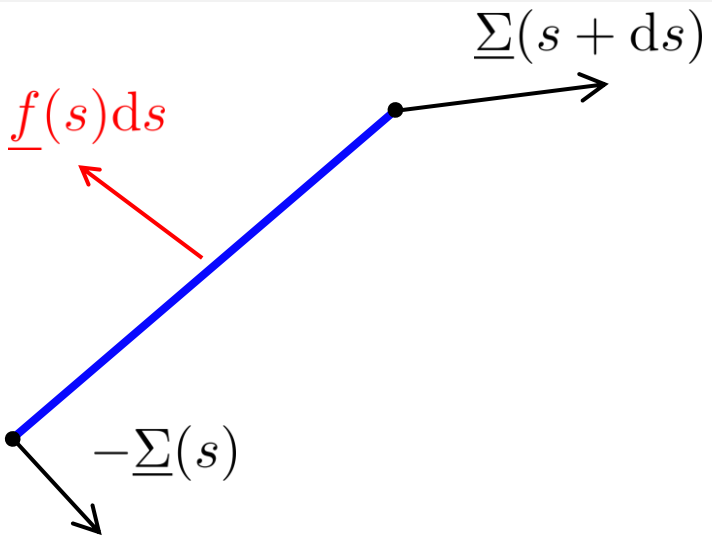
COURS DE MECANIQUE
Milieux curvilignes

Statique (suite), cinématique et Théorème des Puissances Virtuelles (T.P.V)

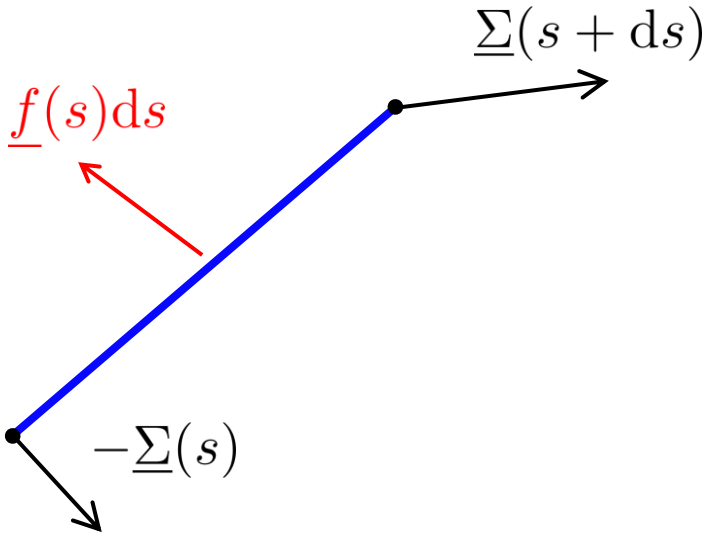
17 septembre 2026

Ecole Nationale des Ponts et Chaussées

Equations **locales** d'équilibre



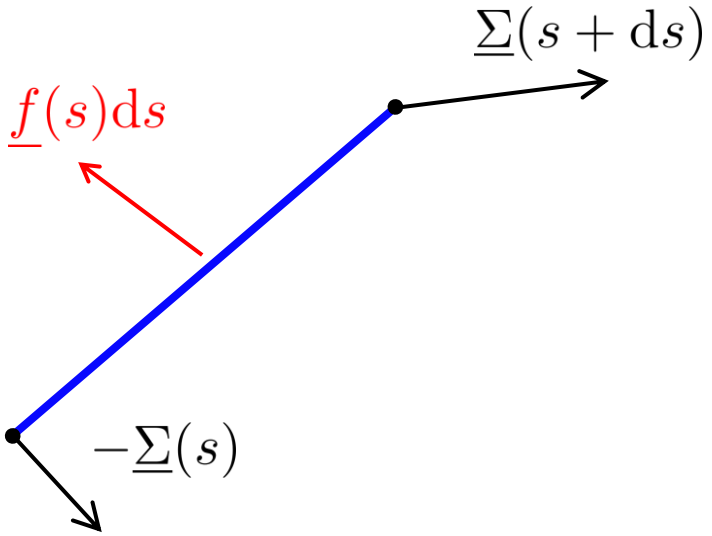
Equations **locales** d'équilibre

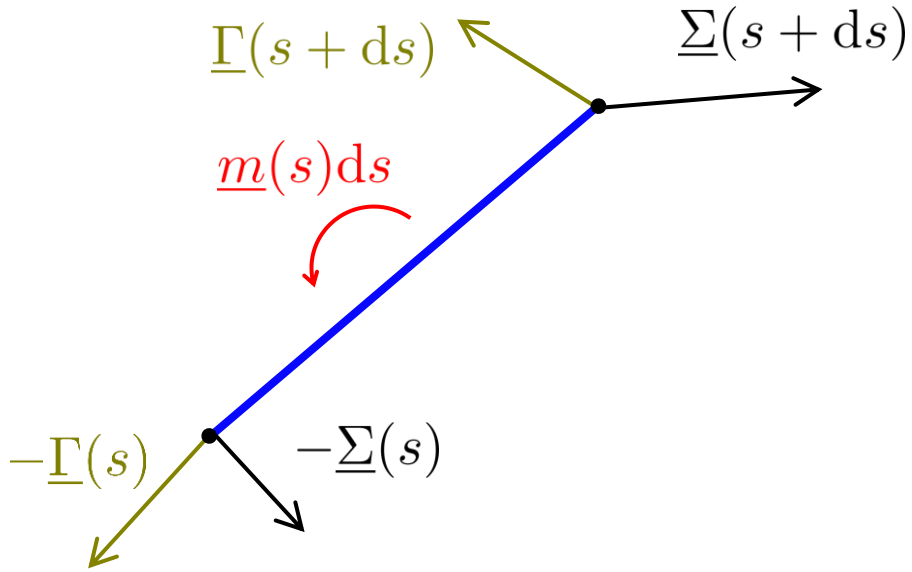


éq. en **résultante** :

$$\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s} + \underline{f} = \underline{0}$$

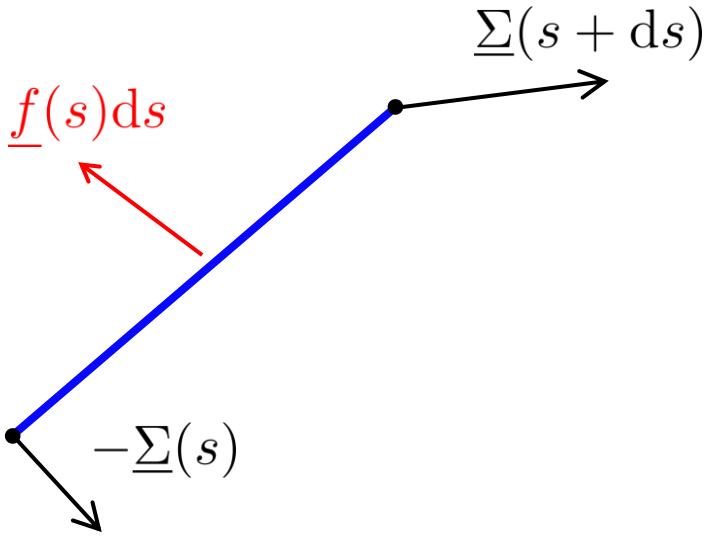
Equations **locales** d'équilibre





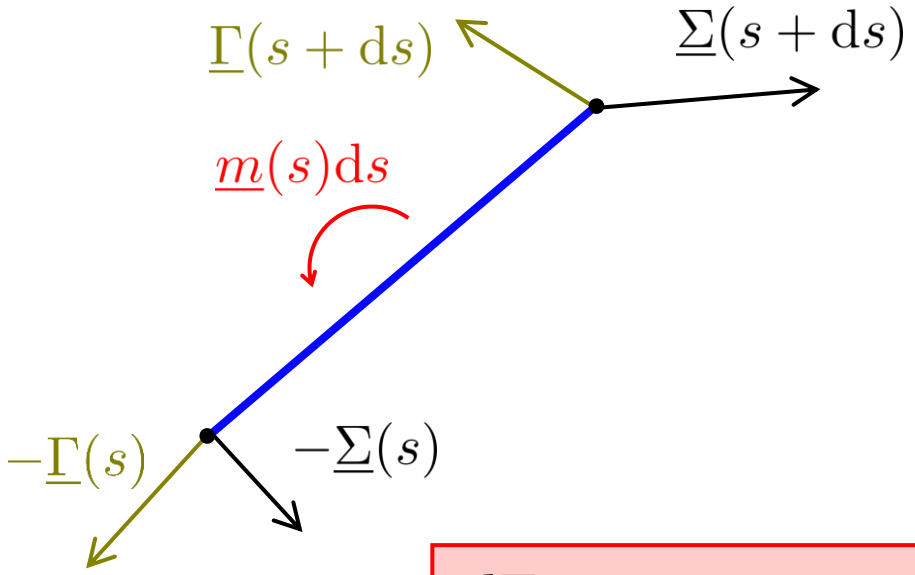
éq. en **résultante** : $\dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s} + \underline{f} = \underline{0}$

Equations **locales** d'équilibre



éq. en **résultante** :

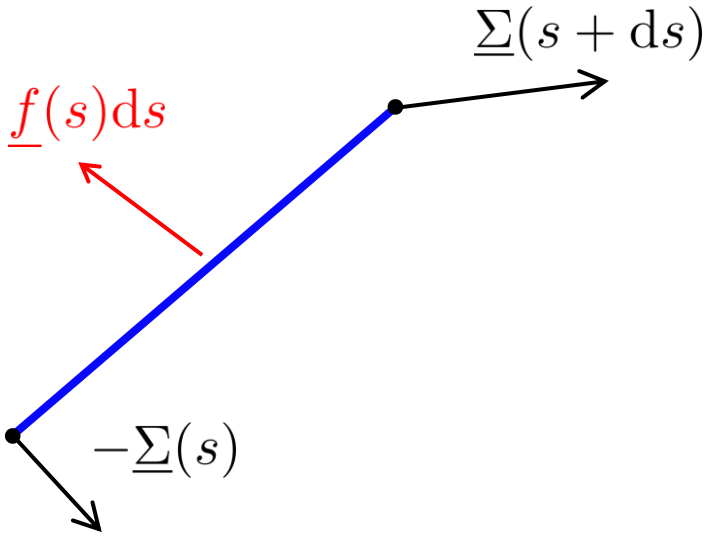
$$\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s} + \underline{f} = \underline{0}$$



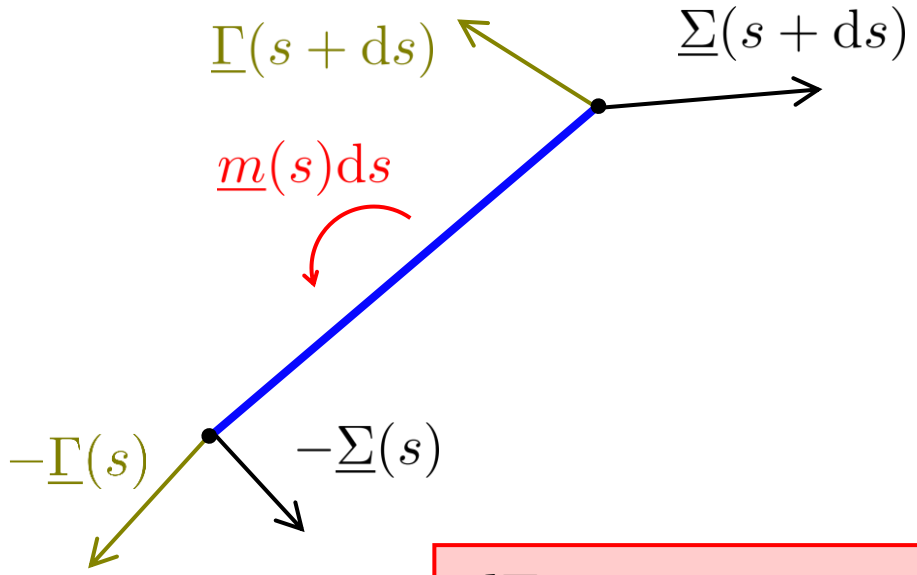
éq. en **moment** :

$$\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s} + \underline{m} + \underline{t} \wedge \underline{\Sigma} = \underline{0}$$

Equations **locales** d'équilibre



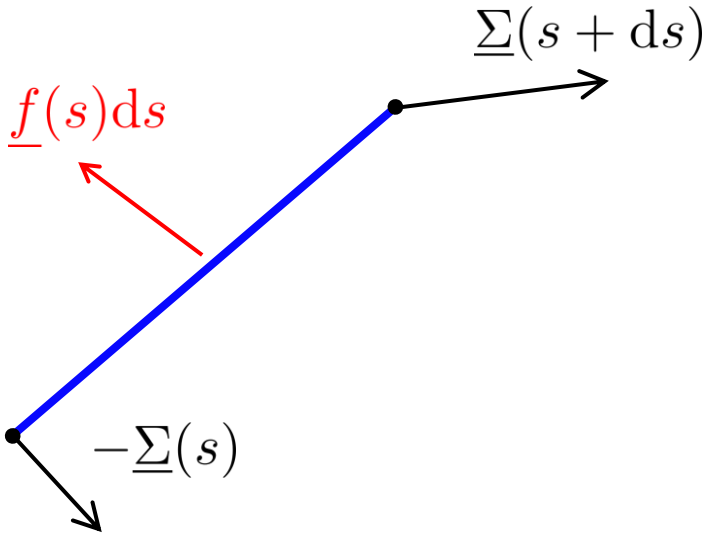
éq. en **résultante** : $\boxed{\dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s} + \underline{f} = \underline{0}}$



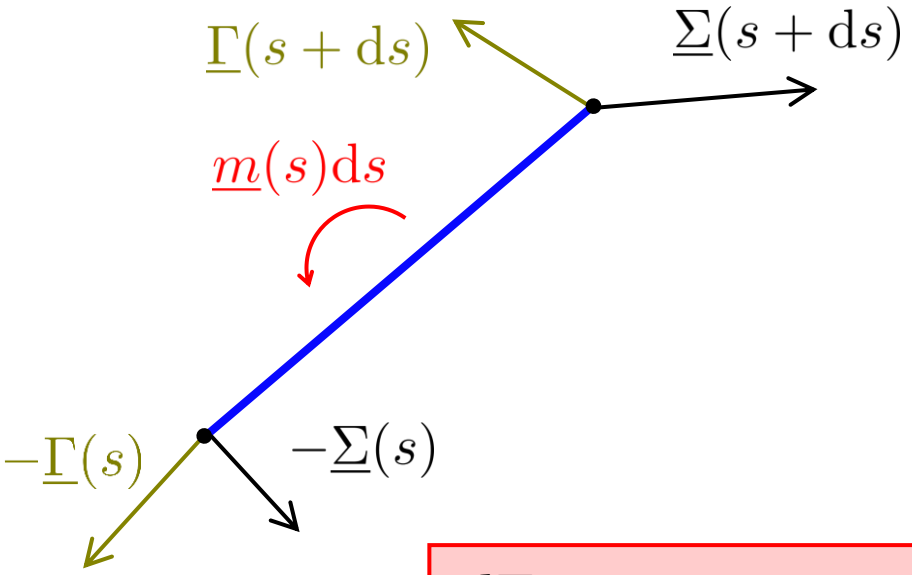
éq. en **moment** : $\boxed{\dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s} + \underline{m} + \underline{t} \wedge \underline{\Sigma} = \underline{0}}$

cas particulier :

Equations **locales** d'équilibre



éq. en **résultante** :
$$\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s} + \underline{f} = \underline{0}$$



éq. en **moment** :
$$\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s} + \underline{m} + \underline{t} \wedge \underline{\Sigma} = \underline{0}$$

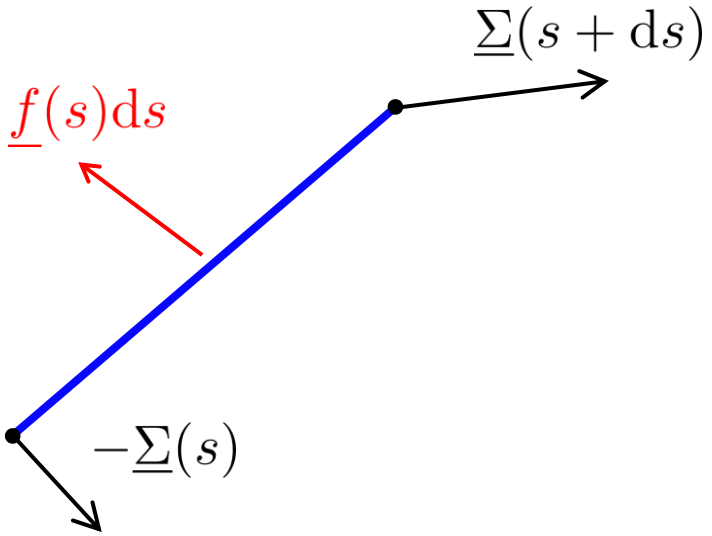
cas particulier :

arc **rectiligne** : $\underline{t}(s) = \underline{t}$

structure **plane**

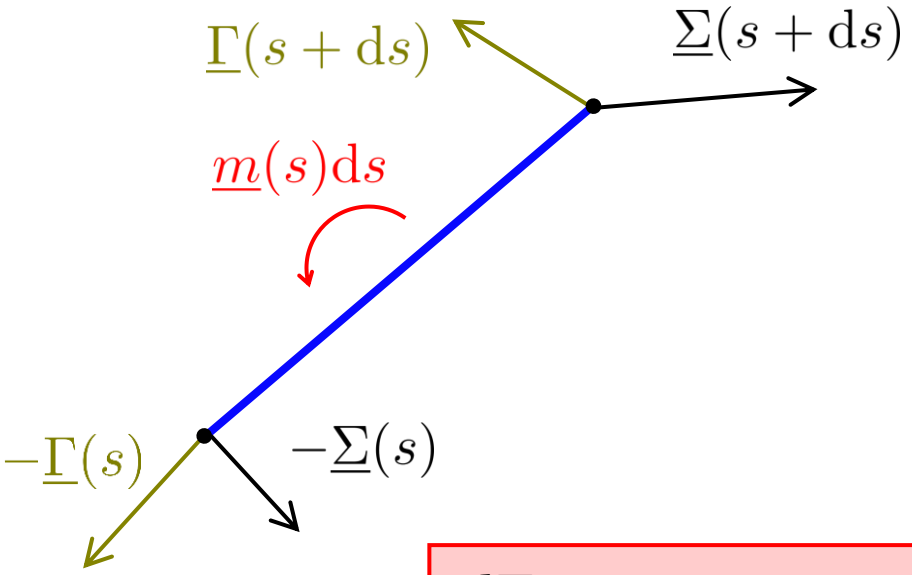
chargement **plan**

Equations **locales** d'équilibre



éq. en **résultante** :

$$\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s} + \underline{f} = \underline{0}$$



éq. en **moment** :

$$\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s} + \underline{m} + \underline{t} \wedge \underline{\Sigma} = \underline{0}$$

cas particulier :

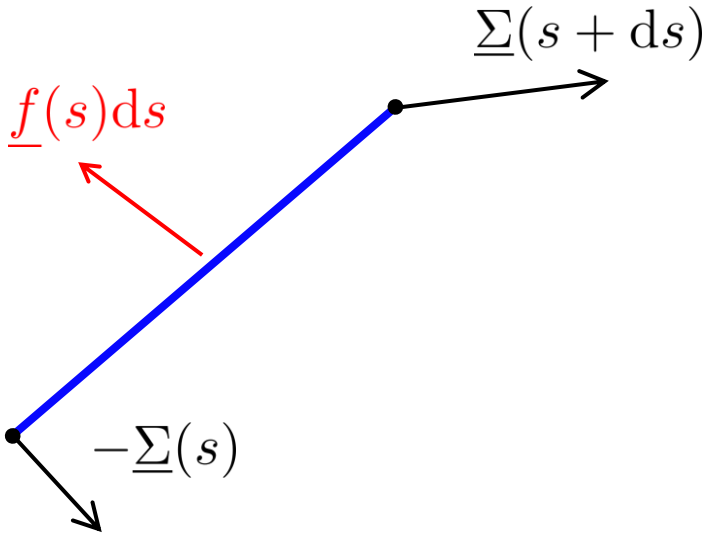
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structure **plane**

chargement **plan**

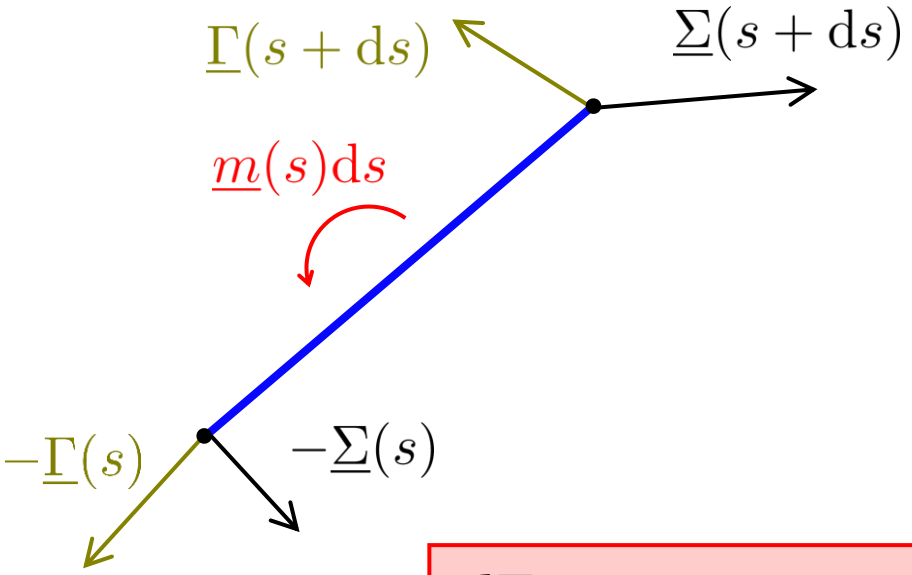
$$\frac{\mathrm{d}N}{\mathrm{d}s} + f_t = 0 \quad ; \quad \frac{\mathrm{d}V}{\mathrm{d}s} + f_n = 0$$

Equations **locales** d'équilibre



éq. en **résultante** :

$$\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s} + \underline{f} = \underline{0}$$



éq. en **moment** :

$$\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s} + \underline{m} + \underline{t} \wedge \underline{\Sigma} = \underline{0}$$

cas particulier :

arc **rectiligne** : $\underline{t}(s) = \underline{t}$

structure **plane**

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$$\frac{\mathrm{d}N}{\mathrm{d}s} + f_t = 0 \quad ; \quad \frac{\mathrm{d}V}{\mathrm{d}s} + f_n = 0$$

$$\frac{\mathrm{d}M}{\mathrm{d}s} + m + V = 0$$

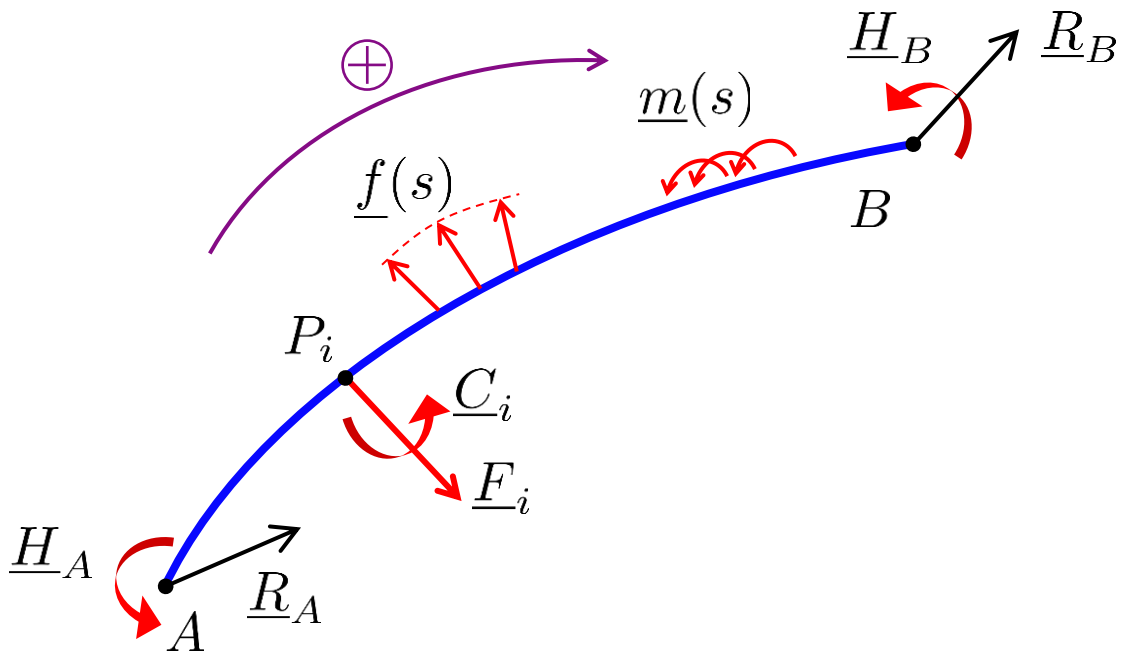
Régularité de $\underline{X}$ et de $\underline{\Gamma}$

$$\text{régularité de} \begin{cases} s \longrightarrow \underline{\Sigma}(s) \\ \\ s \longrightarrow \underline{\Gamma}(s) \end{cases}$$



Régularité de $\underline{X}$ et de $\underline{\Gamma}$

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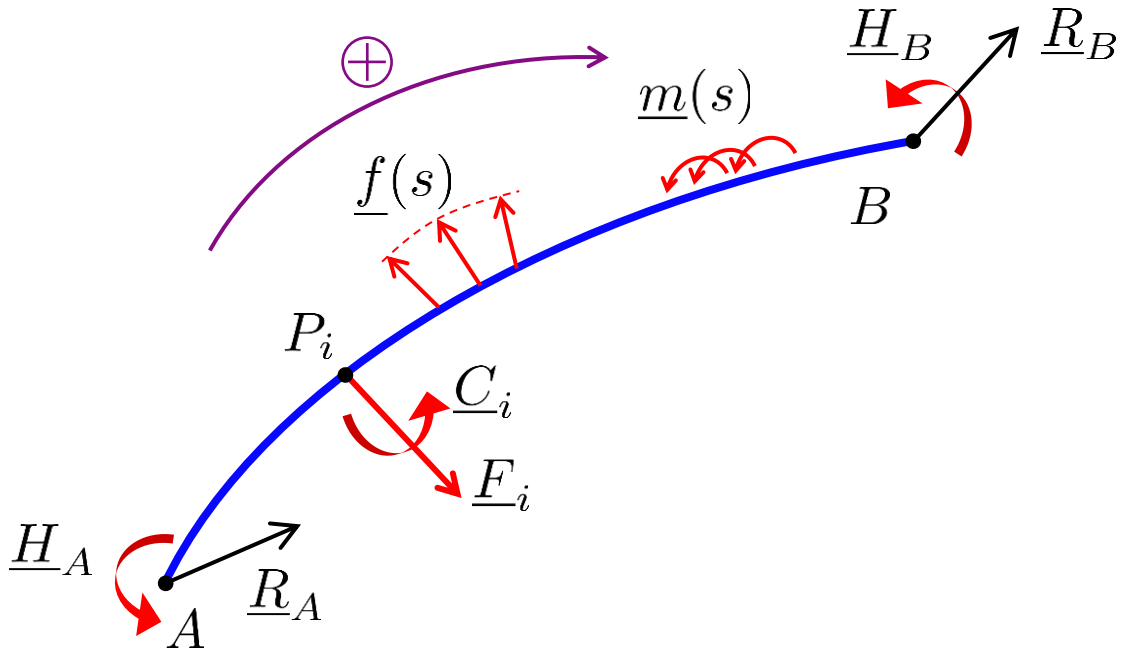


$$\underline{\Sigma}(s_P) = \underline{R}_B + \int_{s_P}^{s_B} \underline{f}(s)\mathrm{d}s + \sum_{s_i > s_P} \underline{F}_i$$

$$\underline{\Gamma}(s_P) = \underline{H}_B + \underline{PB} \wedge \underline{R}_B + \int_{s_P}^{s_B} \underline{m}(s)\mathrm{d}s + \int_{s_P}^{s_B} \underline{PM} \wedge \underline{f}(s_M)\mathrm{d}s_M$$
$$+ \sum_{s_i > s_P} \underline{C}_i + \sum_{s_i > s_P} \underline{PP_i} \wedge \underline{F}_i$$

Régularité de $\underline{X}$ et de $\underline{\Gamma}$

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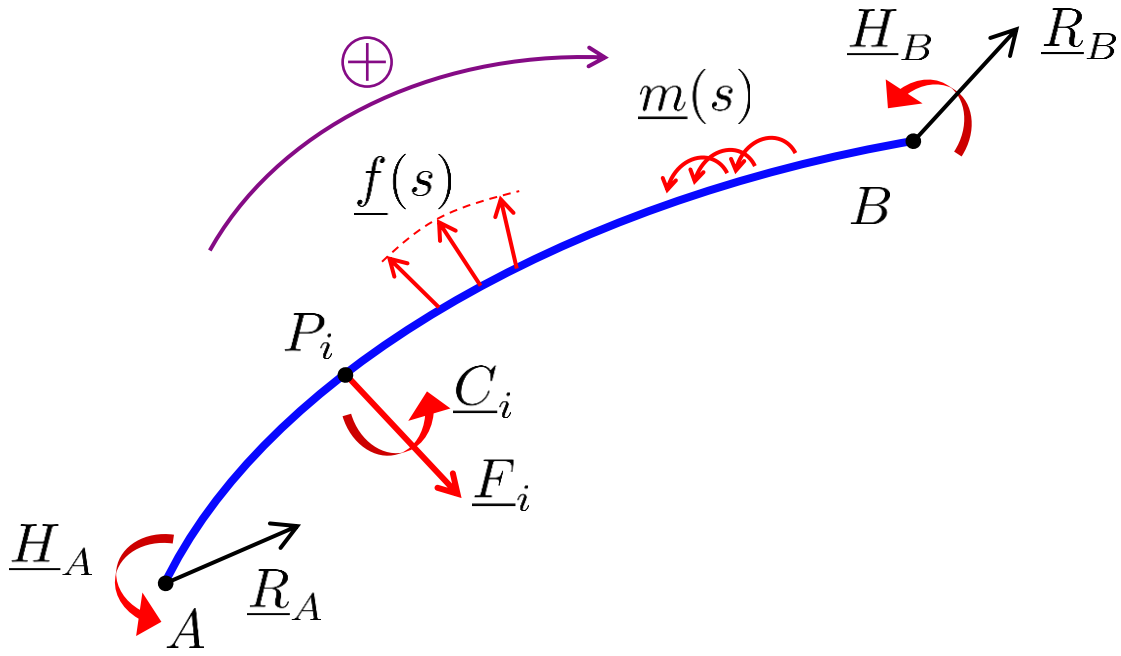


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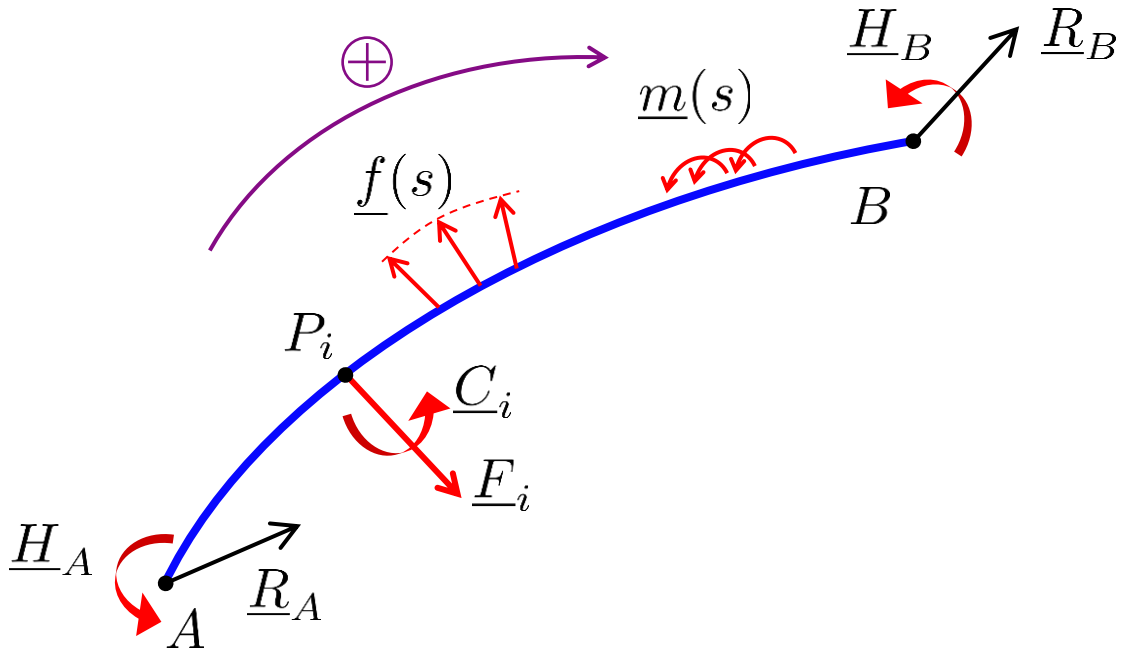


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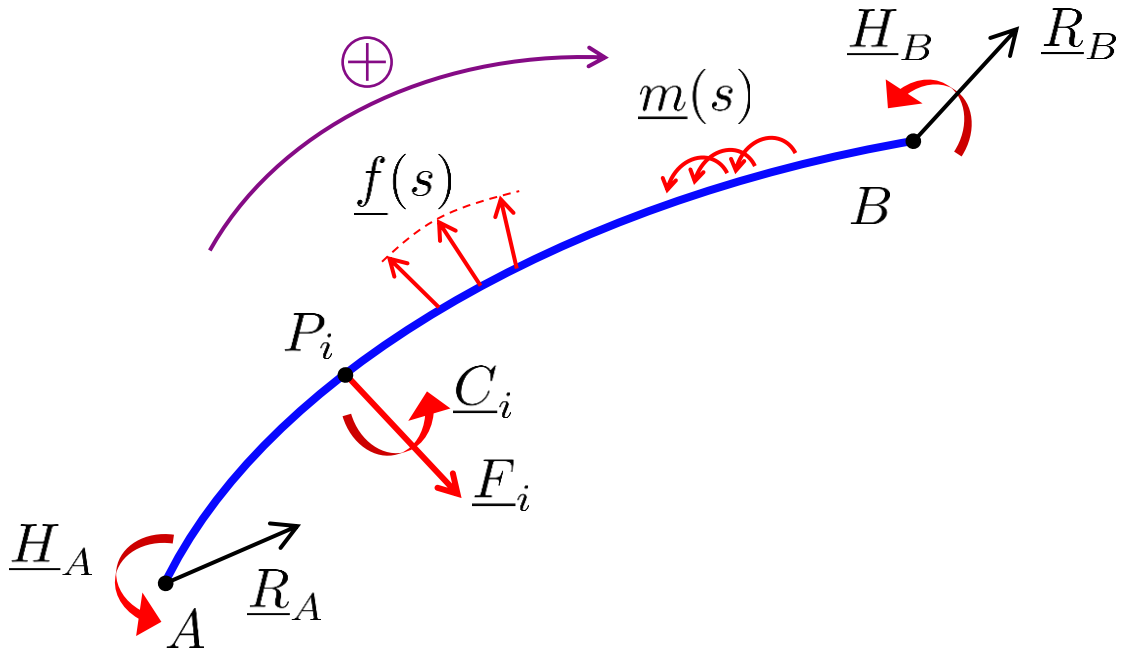


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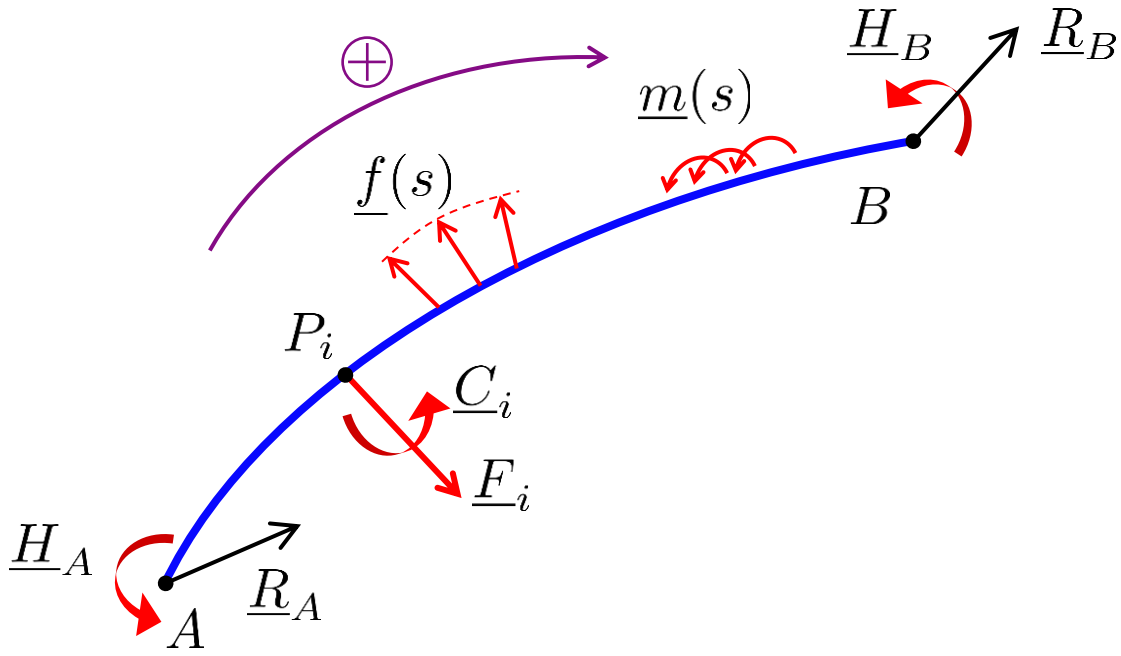


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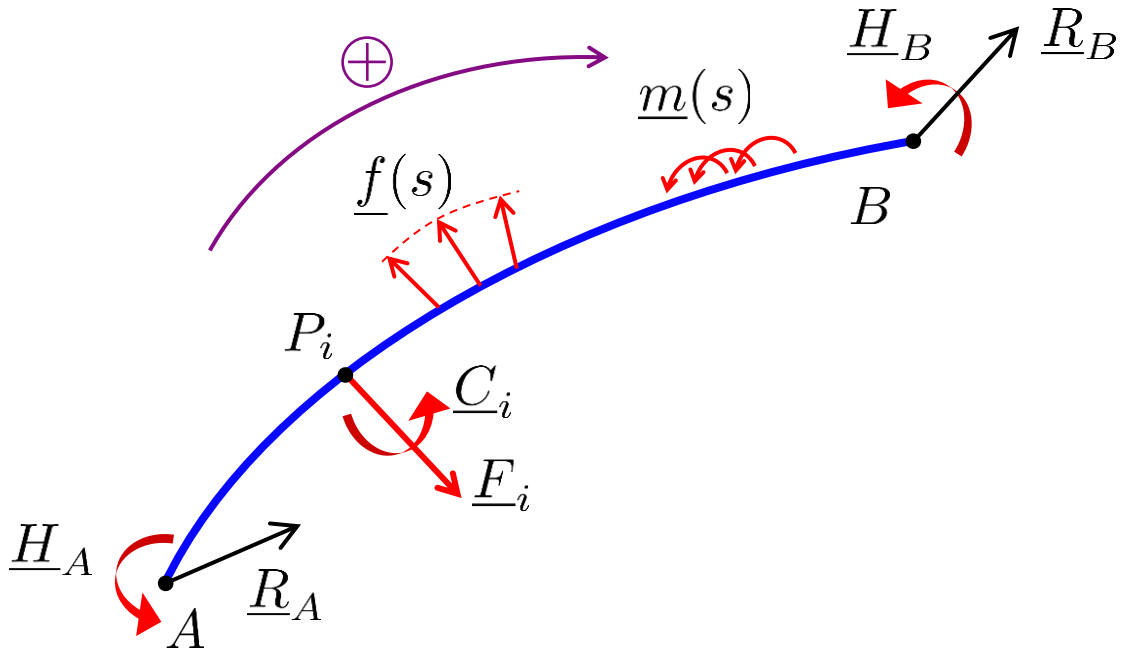


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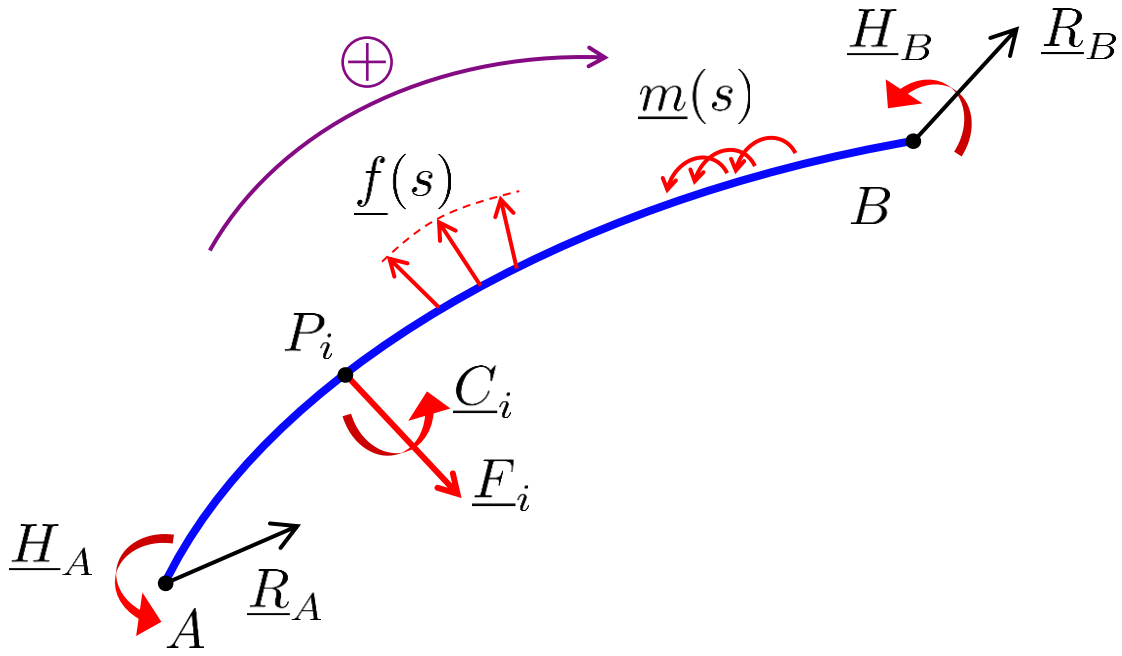


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Régularité de $\underline{X}$ et de $\underline{\Gamma}$

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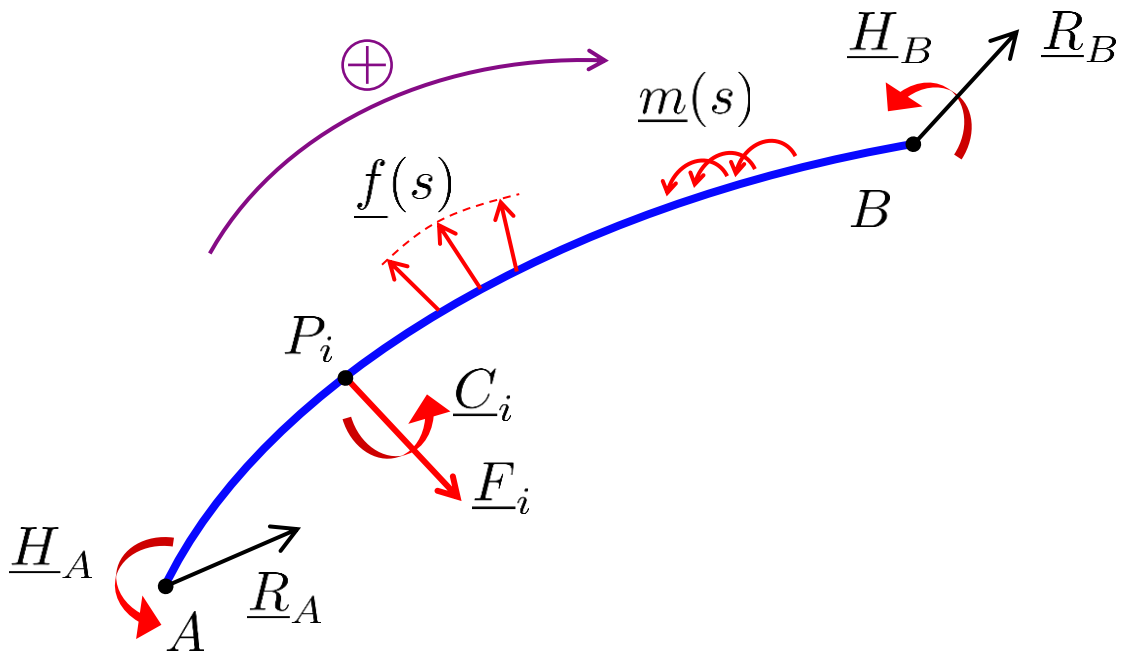


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Régularité de $\underline{X}$ et de $\underline{\Gamma}$

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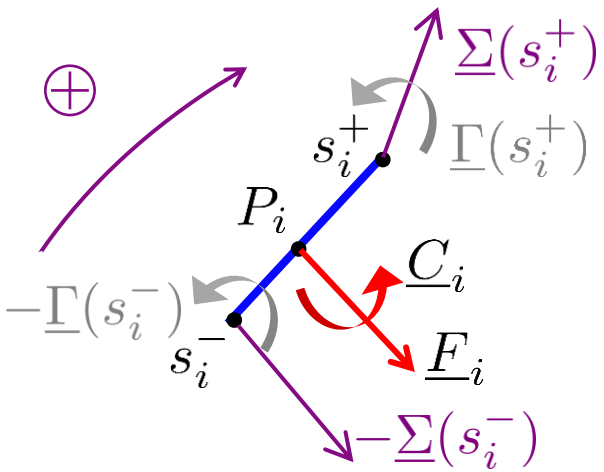
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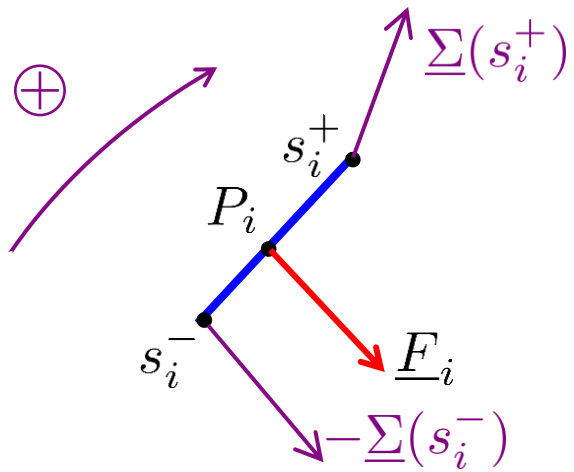
$$+ \sum_{s_i > s_P} \underline{C}_i + \sum_{s_i > s_P} \underline{PP_i} \wedge \underline{F}_i$$

Prise en compte des actions **concentrées** : équations de saut

équilibre



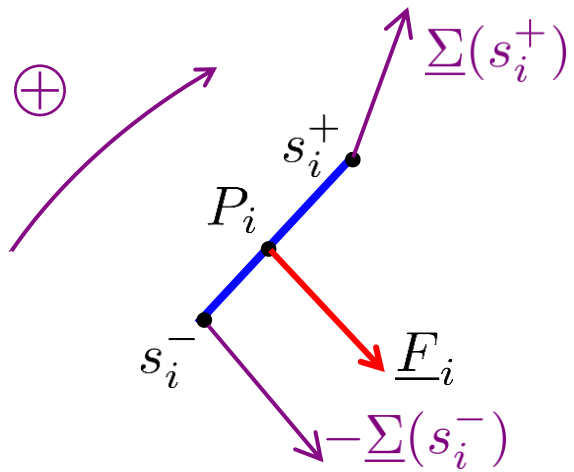
Prise en compte des actions **concentrées** : équations de saut



équilibre

résultante : $\underline{\Sigma}(s_i^+) - \underline{\Sigma}(s_i^-) + \underline{F}_i = \underline{0}$

Prise en compte des actions **concentrées** : équations de saut

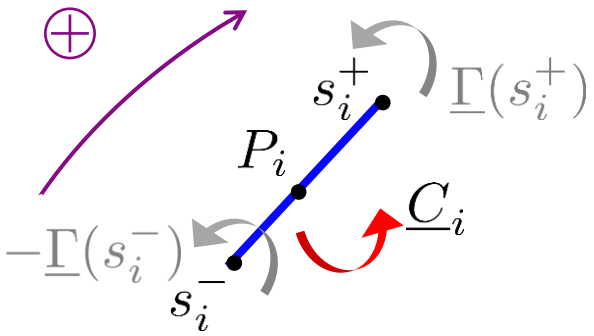


équilibre

$$\overbrace{[\![\underline{\Sigma}]\!]_{P_i}}$$

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Prise en compte des actions **concentrées** : équations de saut



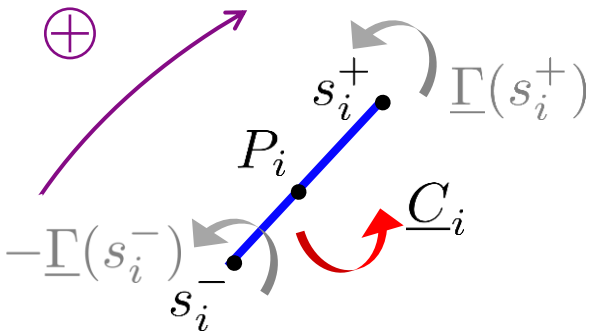
équilibre

$$\overbrace{[\![\underline{\Sigma}]\!]_{P_i}}$$

résultante : $\underbrace{\underline{\Sigma}(s_i^+) - \underline{\Sigma}(s_i^-)} + \underline{F}_i = \underline{0}$

moment :

Prise en compte des actions **concentrées** : équations de saut



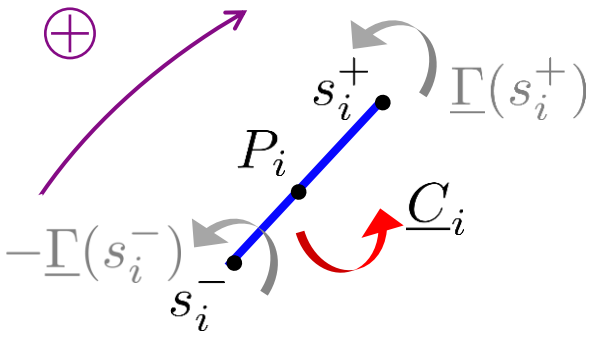
équilibre

$$\overbrace{[\![\underline{\Sigma}]\!]_{P_i}}$$

résultante : $\underline{\Sigma}(s_i^+) - \underline{\Sigma}(s_i^-) + \underline{F}_i = \underline{0}$

moment : $\underline{\Gamma}(s_i^+) - \underline{\Gamma}(s_i^-) + \underline{C}_i = \underline{0}$

Prise en compte des actions **concentrées** : équations de saut



équilibre

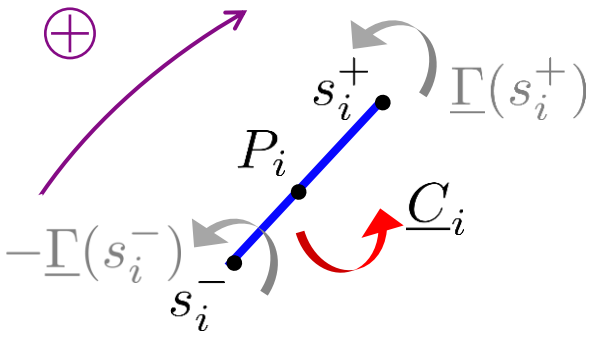
$$\overbrace{[\![\underline{\Sigma}]\!]_{P_i}}$$

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Prise en compte des actions **concentrées** : équations de saut



équilibre

$$\underbrace{[\![\underline{\Sigma}]\!]_{P_i}}$$

résultante : $\overbrace{\underline{\Sigma}(s_i^+) - \underline{\Sigma}(s_i^-)} + \underline{F}_i = \underline{0}$

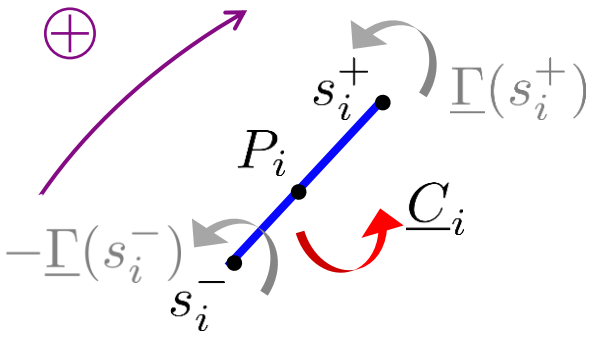
moment : $\underbrace{\underline{\Gamma}(s_i^+) - \underline{\Gamma}(s_i^-)} + \underline{C}_i = \underline{0}$

$$\underbrace{[\![\underline{\Gamma}]\!]_{P_i}}$$

$$[\![\underline{\Sigma}]\!]_{P_i} + \underline{F}_i = \underline{0}$$

$$[\![\underline{\Gamma}]\!]_{P_i} + \underline{C}_i = \underline{0}$$

Prise en compte des actions **concentrées** : équations de saut



équilibre

$$\underbrace{\underline{\Sigma}(s_i^+) - \underline{\Sigma}(s_i^-)}_{[\![\underline{\Sigma}]\!]_{P_i}} + \underline{F}_i = \underline{0} \quad \text{résultante :}$$

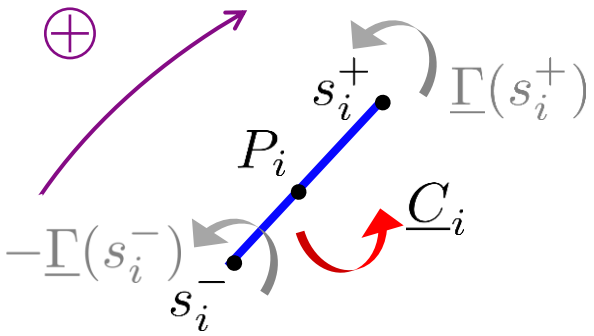
$$\underbrace{\underline{\Gamma}(s_i^+) - \underline{\Gamma}(s_i^-)}_{[\![\underline{\Gamma}]\!]_{P_i}} + \underline{C}_i = \underline{0} \quad \text{moment :}$$

$$[\![\underline{\Sigma}]\!]_{P_i} + \underline{F}_i = \underline{0}$$

$$[\![\underline{\Gamma}]\!]_{P_i} + \underline{C}_i = \underline{0}$$

Attention :

Prise en compte des actions **concentrées** : équations de saut



équilibre

$$\underbrace{[\![\underline{\Sigma}]\!]_{P_i}}$$

résultante : $\overbrace{\underline{\Sigma}(s_i^+) - \underline{\Sigma}(s_i^-)} + \underline{F}_i = \underline{0}$

moment : $\underbrace{\underline{\Gamma}(s_i^+) - \underline{\Gamma}(s_i^-)} + \underline{C}_i = \underline{0}$

$$\underbrace{[\![\underline{\Gamma}]\!]_{P_i}}$$

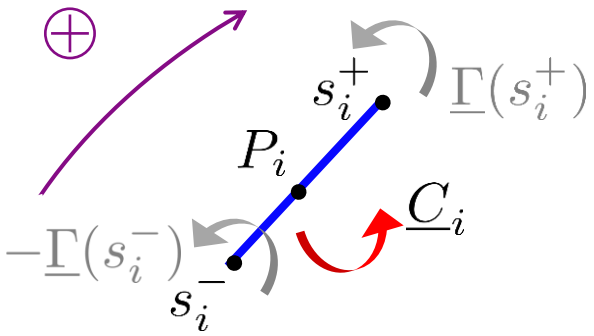
$$[\![\underline{\Sigma}]\!]_{P_i} + \underline{F}_i = \underline{0}$$

$$[\![\underline{\Gamma}]\!]_{P_i} + \underline{C}_i = \underline{0}$$

Attention :

$$[\![\underline{\Sigma}]\!] = \underline{0} \Longleftrightarrow \begin{cases} [\![N]\!] = 0 \\ [\![V]\!] = 0 \end{cases}$$

Prise en compte des actions **concentrées** : équations de saut



équilibre

$$\underbrace{[\![\underline{\Sigma}]\!]_{P_i}}$$

résultante : $\overbrace{\underline{\Sigma}(s_i^+) - \underline{\Sigma}(s_i^-)} + \underline{F}_i = \underline{0}$

moment : $\underbrace{\underline{\Gamma}(s_i^+) - \underline{\Gamma}(s_i^-)} + \underline{C}_i = \underline{0}$

$$\underbrace{[\![\underline{\Gamma}]\!]_{P_i}}$$

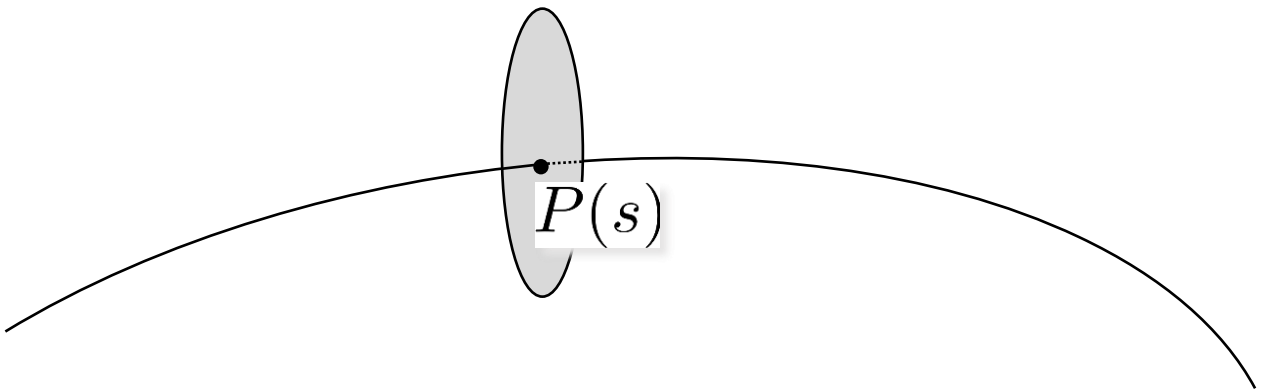
$$[\![\underline{\Sigma}]\!]_{P_i} + \underline{F}_i = \underline{0}$$

$$[\![\underline{\Gamma}]\!]_{P_i} + \underline{C}_i = \underline{0}$$

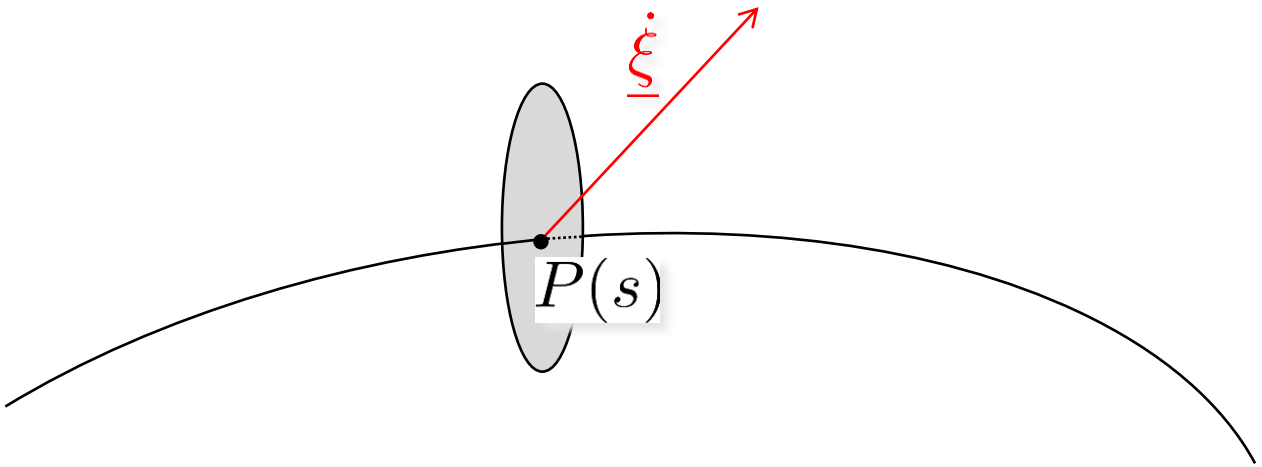
Attention :

$$[\![\underline{\Sigma}]\!] = \underline{0} \not\Longleftrightarrow \begin{cases} [\![N]\!] = 0 \\ [\![V]\!] = 0 \end{cases}$$

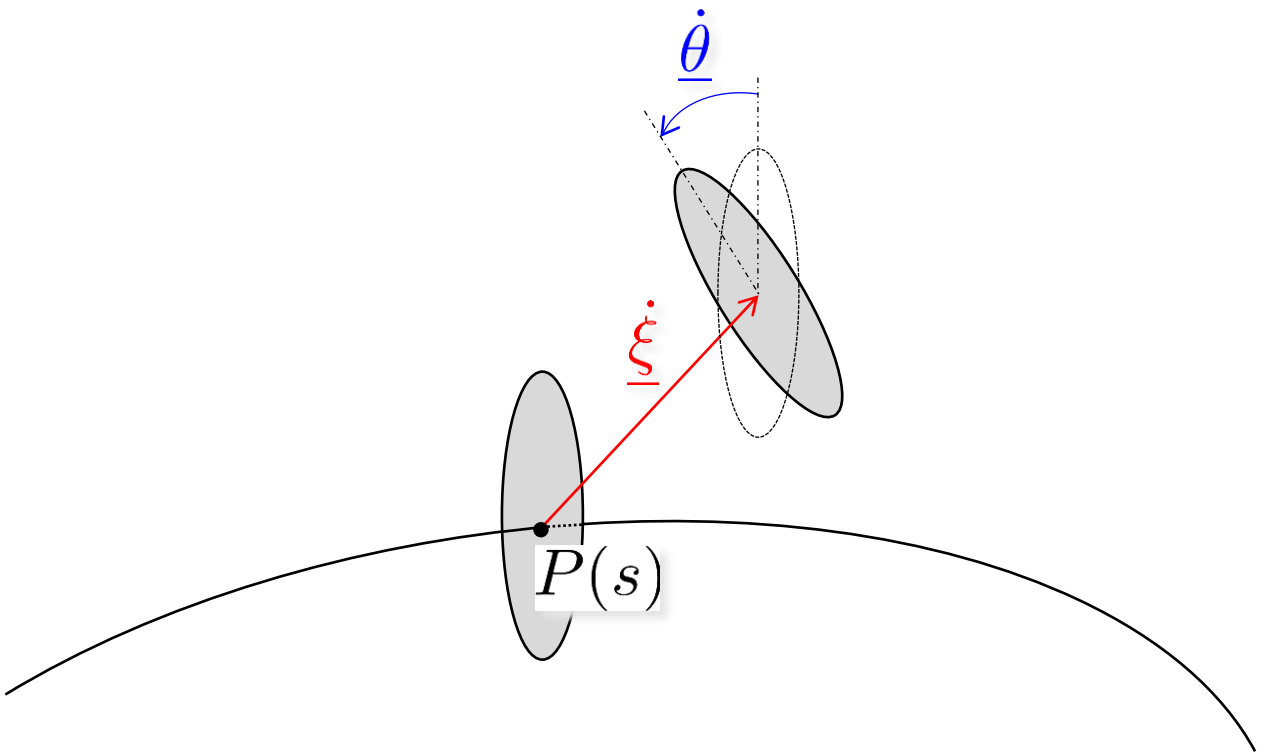
Transformation géométrique d'un arc (cinématique)



Transformation géométrique d'un arc (cinématique)

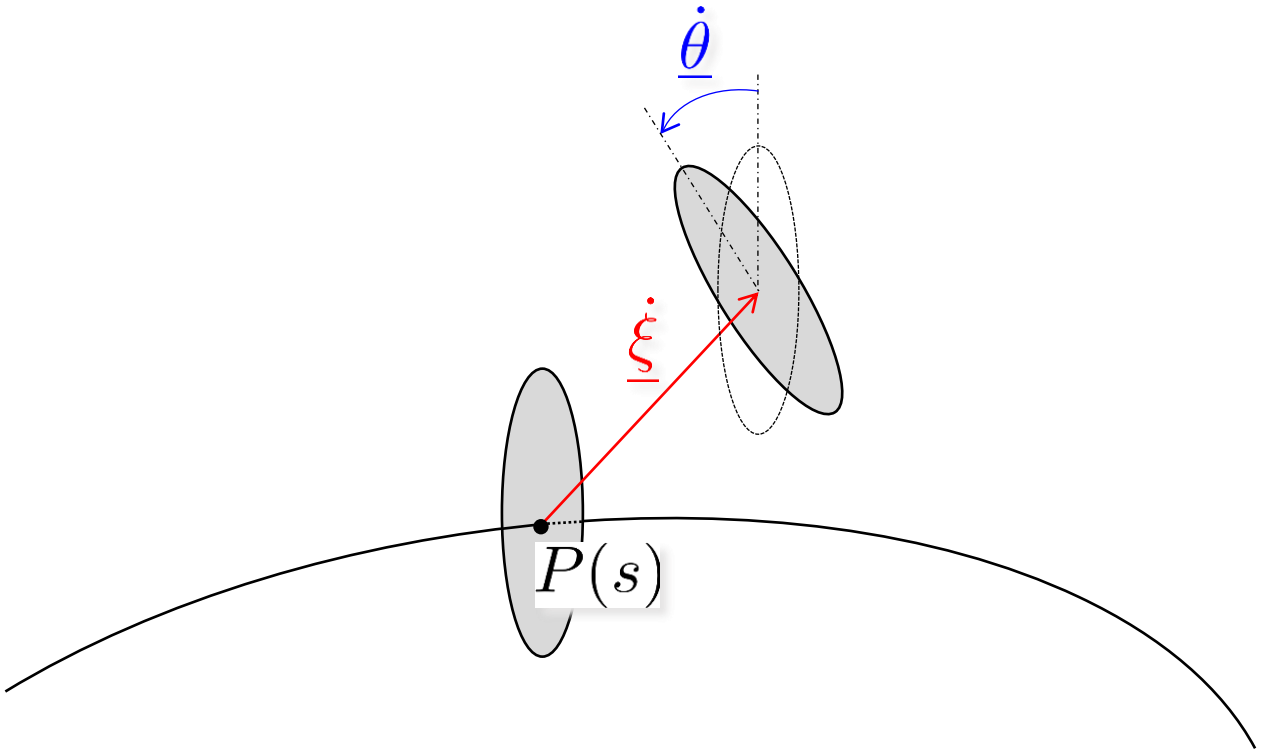


Transformation géométrique d'un arc (cinématique)



Transformation géométrique d'un arc (cinématique)

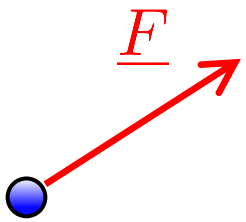
mouvement de la section d'abscisse s à l'instant t : $\begin{cases} \text{vitesse de translation : } \underline{\dot{\xi}}(s,t) \\ \\ \text{vitesse de rotation : } \underline{\dot{\theta}}(s,t) \end{cases}$



Théorème des Puissances Virtuelles (T.P.V) : Cas d'un point matériel

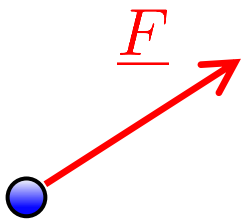


Théorème des Puissances Virtuelles (T.P.V) : Cas d'un point matériel



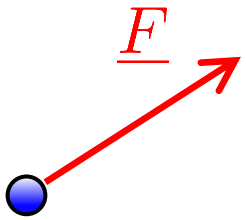
équilibre : $\underline{F} = \underline{0}$

Théorème des Puissances Virtuelles (T.P.V) : Cas d'un point matériel



équilibre : $\underline{F} = \underline{0} \iff \forall\, \underline{\hat{U}} \in \mathbb{R}^3 : \underline{F} \cdot \underline{\hat{U}} = 0$

Théorème des Puissances Virtuelles (T.P.V) : Cas d'un point matériel



équilibre : $\underline{F} = \underline{0} \iff \forall\, \underline{\hat{U}} \in \mathbb{R}^3 : \underline{F} \cdot \underline{\hat{U}} = 0$

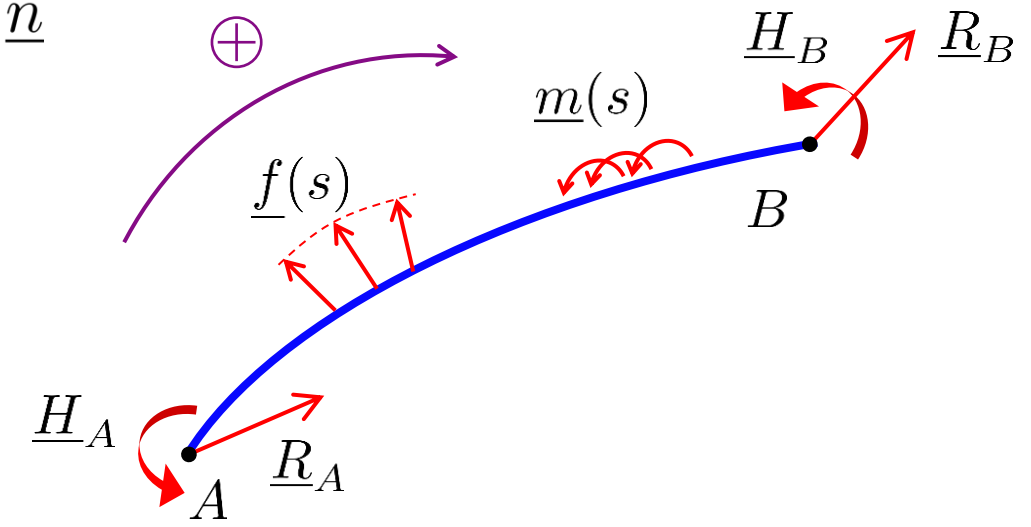
$\underline{\hat{U}}$: vitesse **virtuelle** (aucune restriction)

$\underline{F} \cdot \underline{\hat{U}}$: puissance **virtuelle**

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$

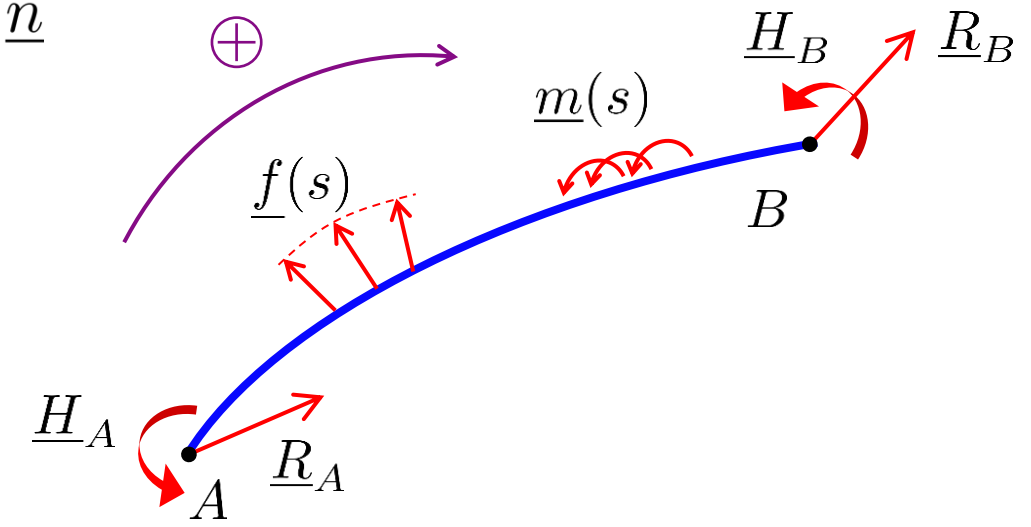
équilibre : $\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$



T.P.V : Absence d'actions concentrées (1)

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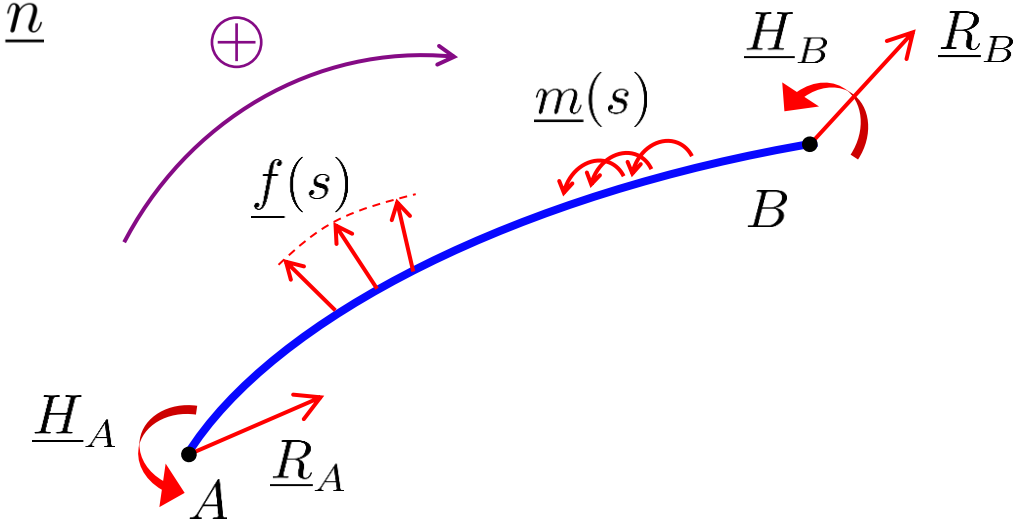
équilibre : $\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$



Dualisation des équations d'équilibre

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



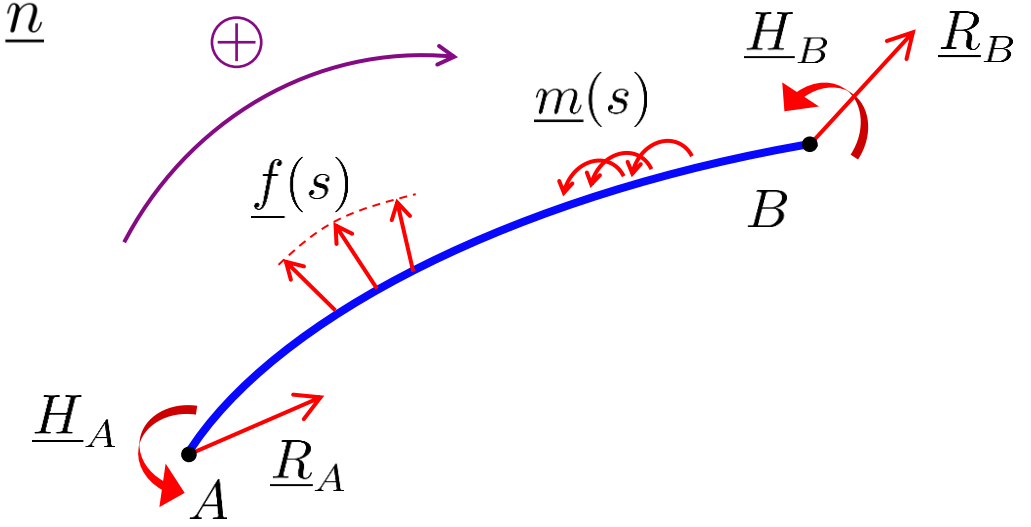
équilibre : $\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$

Dualisation des équations d'équilibre

$$\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right)$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre : $\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$

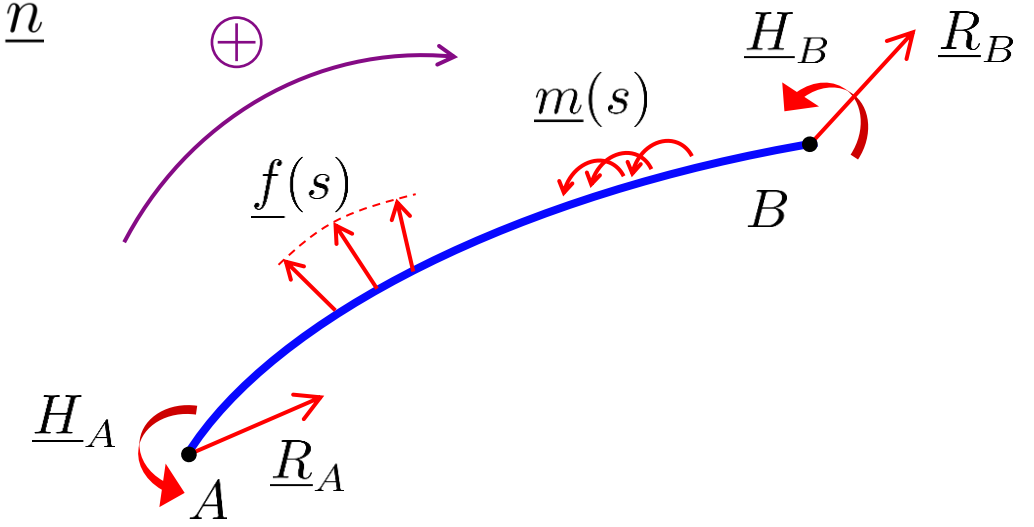
Dualisation des équations d'équilibre

$$\forall s \to \underline{A}(s)$$

$$\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right) \cdot \underline{A}(s)$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre :

$$\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0}$$

$$\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0}$$

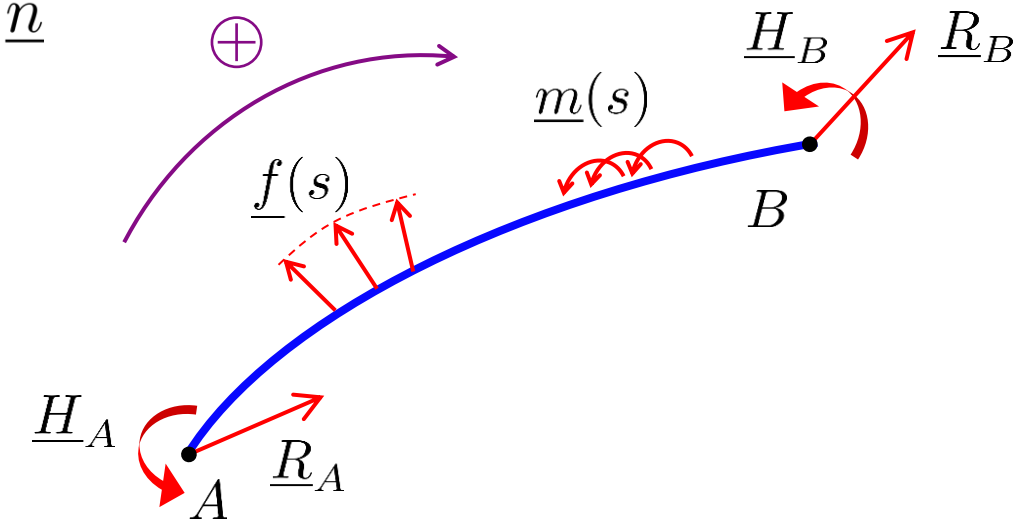
Dualisation des équations d'équilibre

$$\forall s \to \underline{A}(s)$$

$$\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s)\right) \cdot \underline{A}(s) \qquad \left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s)\right)$$

T.P.V : Absence d'actions concentrées (1)

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équilibre :

$$\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$$

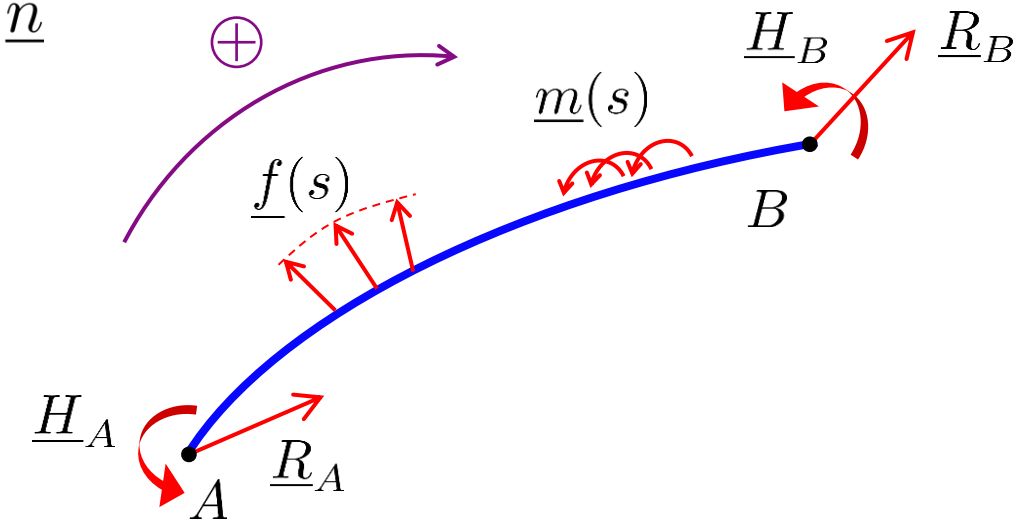
Dualisation des équations d'équilibre

$$\forall s \to \underline{A}(s) \quad ; \forall s \to \underline{B}(s)$$

$$\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s)\right) \cdot \underline{A}(s) \qquad \left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s)\right) \cdot \underline{B}(s)$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre :
$$\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$$

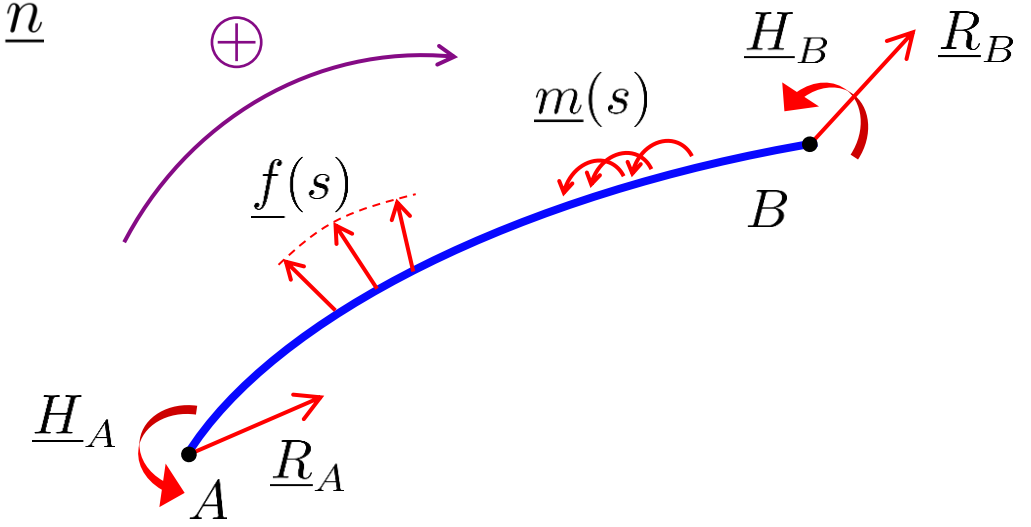
Dualisation des équations d'équilibre

$$\forall s \to \underline{A}(s) \quad ; \forall s \to \underline{B}(s)$$

$$\int_A^B \left[\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right) \cdot \underline{A}(s) + \left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) \right) \cdot \underline{B}(s) \right] \mathrm{d}s = 0$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre :

$$\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$$

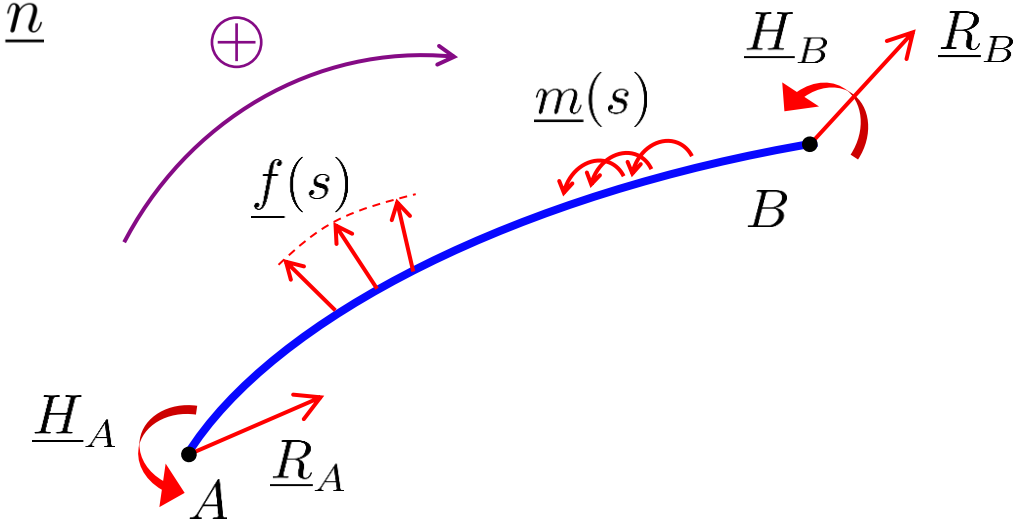
Dualisation des équations d'équilibre

$$\forall s \to \underline{A}(s) \quad ; \forall s \to \underline{B}(s)$$

$$\int_A^B \left[\underbrace{\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right)}_{N/m} \cdot \underline{A}(s) + \left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) \right) \cdot \underline{B}(s) \right] \mathrm{d}s = 0$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre :
$$\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$$

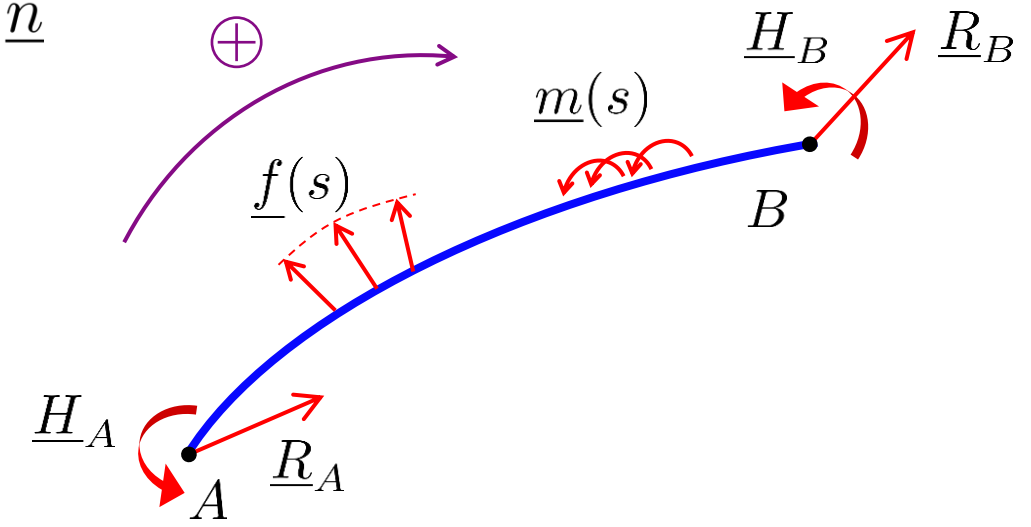
Dualisation des équations d'équilibre

$$\forall s \to \underline{A}(s) \quad ; \forall s \to \underline{B}(s)$$

$$\int_A^B \left[\underbrace{\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right)}_{N/m} \cdot \underline{A}(s) + \underbrace{\left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) \right)}_{N} \cdot \underline{B}(s) \right] \mathrm{d}s = 0$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre :
$$\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$$

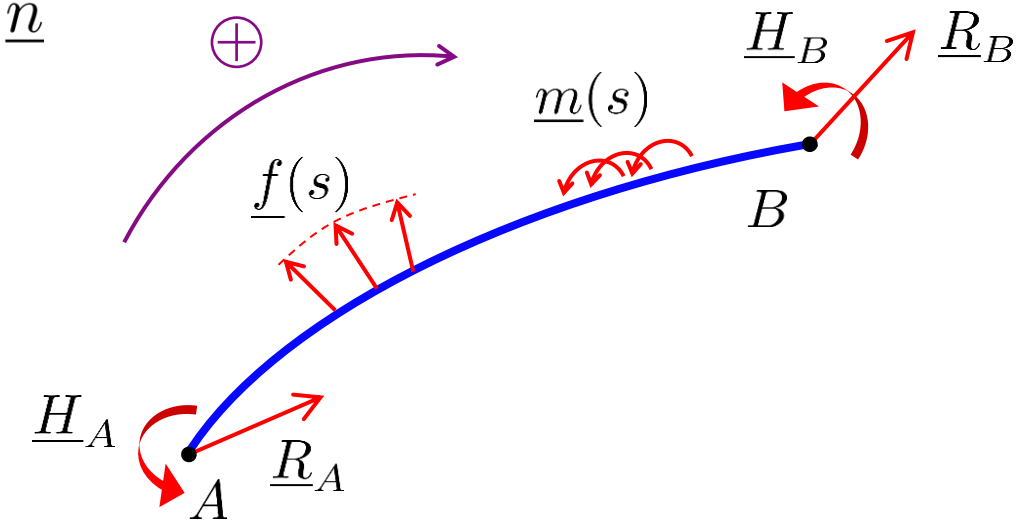
Dualisation des équations d'équilibre

$$\forall s \to \underline{A}(s) \quad ; \forall s \to \underline{B}(s)$$

$$\int_A^B \Big[\underbrace{\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right)}_{N/m} \cdot \underline{A}(s) + \underbrace{\left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) \right)}_{N} \cdot \underline{B}(s) \Big] \underbrace{\mathrm{d}s}_{m} = 0$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre :
$$\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$$

Dualisation des équations d'équilibre

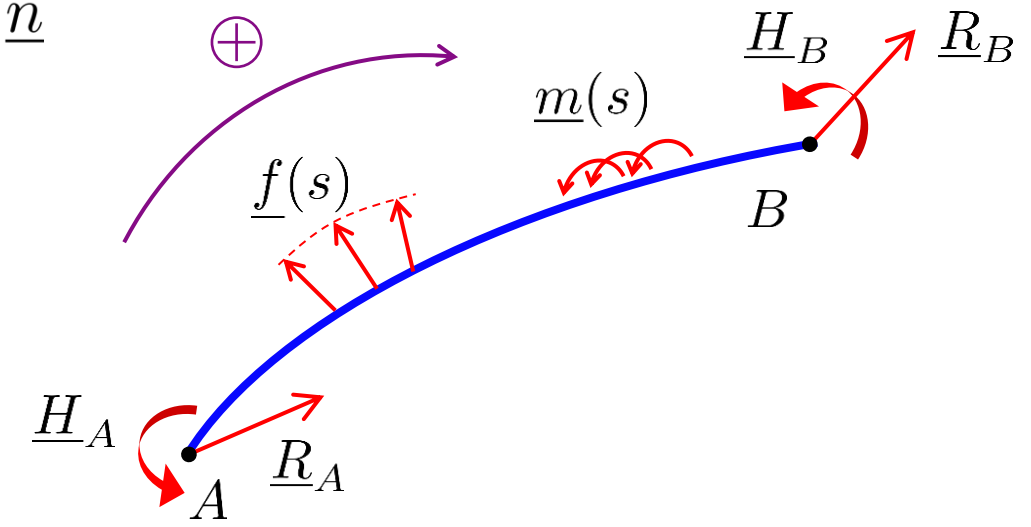
$$\forall s \to \underline{A}(s) \quad ; \forall s \to \underline{B}(s)$$

puissance virtuelle

$$\int_A^B \left[\underbrace{\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right)}_{N/m} \cdot \underline{A}(s) + \underbrace{\left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) \right)}_{N} \cdot \underline{B}(s) \right] \underbrace{\mathrm{d}s}_{m} = 0$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre :
$$\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$$

Dualisation des équations d'équilibre

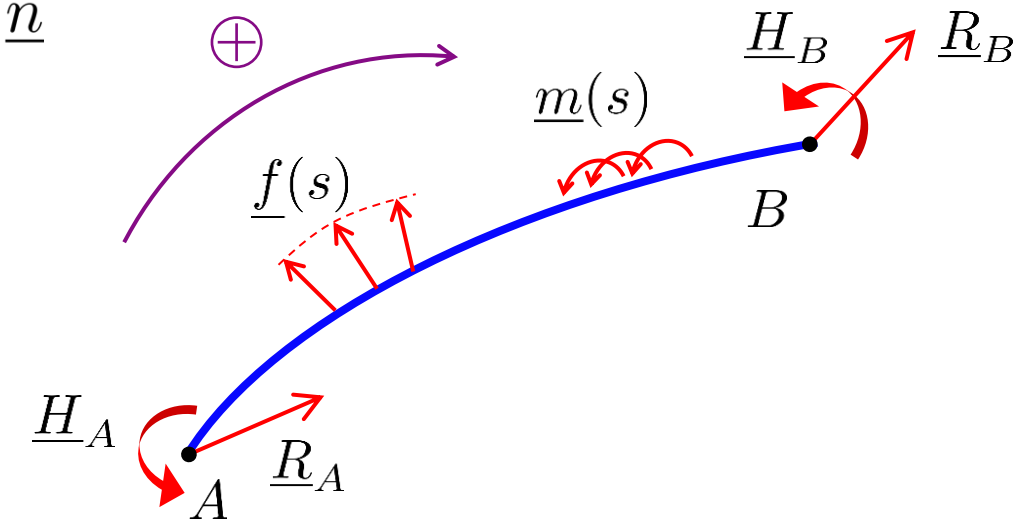
$$\forall s \to \underline{A}(s) \quad ; \forall s \to \underline{B}(s)$$

puissance virtuelle

$$\int_A^B \left[\underbrace{\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right)}_{N/m} \cdot \underbrace{\underline{A}(s)}_{m/s} + \underbrace{\left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) \right)}_{N} \cdot \underbrace{\underline{B}(s)}_{s^{-1}} \right] \underbrace{\mathrm{d}s}_{m} = 0$$

T.P.V : Absence d'actions concentrées (1)

arc plan + chargement plan $\begin{cases} \underline{\Sigma} = N\underline{t} + V\underline{n} \\ \underline{\Gamma} = M\underline{e}_z \end{cases}$



équilibre :
$$\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$$

Dualisation des équations d'équilibre

$$\forall s \to \hat{\underline{\xi}}(s) \quad ; \forall s \to \hat{\underline{\theta}}(s)$$

puissance virtuelle

$$\int_A^B \left[\underbrace{\left(\frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) \right)}_{N/m} \cdot \underbrace{\hat{\underline{\xi}}(s)}_{m/s} + \underbrace{\left(\frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) \right)}_{N} \cdot \underbrace{\hat{\underline{\theta}}(s)}_{s^{-1}} \right] \underbrace{\mathrm{d}s}_{m} = 0$$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

(1) $\displaystyle\int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \left[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\right]_A^B - \int_A^B \underline{\Sigma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \left[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\right]_A^B - \int_A^B \boxed{\underline{\Sigma}(s)}\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Sigma}(s) = N(s)\underline{t} + V(s)\underline{n}$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$N(s)\underline{t} + V(s)\underline{n}$

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \left[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\right]_A^B - \int_A^B \underline{\Sigma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$$= \underline{\Sigma}(s_B)\cdot\hat{\dot{\underline{\xi}}}(s_B) - \underline{\Sigma}(s_A)\cdot\hat{\dot{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$N(s)\underline{t} + V(s)\underline{n}$

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \left[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\right]_A^B - \int_A^B \underline{\Sigma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$$= \underline{\Sigma}(s_B)\cdot\hat{\dot{\underline{\xi}}}(s_B) - \underline{\Sigma}(s_A)\cdot\hat{\dot{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{R}_B$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \left[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\right]_A^B - \int_A^B \underline{\Sigma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Sigma}(s) \to N(s)\underline{t} + V(s)\underline{n}$

$$= \underline{\Sigma}(s_B)\cdot\hat{\dot{\underline{\xi}}}(s_B) - \underline{\Sigma}(s_A)\cdot\hat{\dot{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{\Sigma}(s_B) \to \underline{R}_B$

$\underline{\Sigma}(s_A) \to -\underline{R}_A$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \Big[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\Big]_A^B - \int_A^B \underline{\Sigma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Sigma}(s) = N(s)\underline{t} + V(s)\underline{n}$

$$= \underline{\Sigma}(s_B)\cdot\hat{\dot{\underline{\xi}}}(s_B) - \underline{\Sigma}(s_A)\cdot\hat{\dot{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{\Sigma}(s_B) = \underline{R}_B$; $\underline{\Sigma}(s_A) = -\underline{R}_A$

$$(2)\quad \int_A^B \frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\theta}}}(s)\,\mathrm{d}s = \Big[\underline{\Gamma}(s)\cdot\hat{\dot{\underline{\theta}}}(s)\Big]_A^B - \int_A^B \underline{\Gamma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\theta}}}}{\mathrm{d}s}\,\mathrm{d}s$$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$N(s)\underline{t} + V(s)\underline{n}$

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \Big[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\Big]_A^B - \int_A^B \underline{\Sigma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$$= \underline{\Sigma}(s_B)\cdot\hat{\dot{\underline{\xi}}}(s_B) - \underline{\Sigma}(s_A)\cdot\hat{\dot{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{\Sigma}(s_B) \to \underline{R}_B$; $\underline{\Sigma}(s_A) \to -\underline{R}_A$

$M(s)\underline{e}_z$

$$(2)\quad \int_A^B \frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\theta}}}(s)\,\mathrm{d}s = \Big[\underline{\Gamma}(s)\cdot\hat{\dot{\underline{\theta}}}(s)\Big]_A^B - \int_A^B \underline{\Gamma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\theta}}}}{\mathrm{d}s}\,\mathrm{d}s$$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \Big[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\Big]_A^B - \int_A^B \underline{\Sigma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Sigma}(s) \to N(s)\underline{t} + V(s)\underline{n}$

$$= \underline{\Sigma}(s_B)\cdot\hat{\dot{\underline{\xi}}}(s_B) - \underline{\Sigma}(s_A)\cdot\hat{\dot{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{\Sigma}(s_B) \to \underline{R}_B$; $\underline{\Sigma}(s_A) \to -\underline{R}_A$

$$(2)\quad \int_A^B \frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\theta}}}(s)\,\mathrm{d}s = \Big[\underline{\Gamma}(s)\cdot\hat{\dot{\underline{\theta}}}(s)\Big]_A^B - \int_A^B \underline{\Gamma}(s)\cdot\frac{\mathrm{d}\hat{\dot{\underline{\theta}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Gamma}(s) \to M(s)\underline{e}_z$

$$= \underline{\Gamma}(s_B)\cdot\hat{\dot{\underline{\theta}}}(s_B) - \underline{\Gamma}(s_A)\cdot\hat{\dot{\underline{\theta}}}(s_A) - \int_A^B M(s)\frac{\mathrm{d}\hat{\dot{\theta}}_z}{\mathrm{d}s}(s)\,\mathrm{d}s$$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\,\mathrm{d}s = \Big[\underline{\Sigma}(s)\cdot\hat{\dot{\underline{\xi}}}(s)\Big]_A^B - \int_A^B \boxed{\underline{\Sigma}(s)}\cdot\frac{\mathrm{d}\hat{\dot{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Sigma}(s) = N(s)\underline{t} + V(s)\underline{n}$

$$= \boxed{\underline{\Sigma}(s_B)}\cdot\hat{\dot{\underline{\xi}}}(s_B) - \boxed{\underline{\Sigma}(s_A)}\cdot\hat{\dot{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{\Sigma}(s_B) = \underline{R}_B$; $\underline{\Sigma}(s_A) = -\underline{R}_A$

$$(2)\quad \int_A^B \frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s)\cdot\hat{\dot{\underline{\theta}}}(s)\,\mathrm{d}s = \Big[\underline{\Gamma}(s)\cdot\hat{\dot{\underline{\theta}}}(s)\Big]_A^B - \int_A^B \boxed{\underline{\Gamma}(s)}\cdot\frac{\mathrm{d}\hat{\dot{\underline{\theta}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Gamma}(s) = M(s)\underline{e}_z$

$$= \boxed{\underline{\Gamma}(s_B)}\cdot\hat{\dot{\underline{\theta}}}(s_B) - \underline{\Gamma}(s_A)\cdot\hat{\dot{\underline{\theta}}}(s_A) - \int_A^B M(s)\frac{\mathrm{d}\hat{\dot{\theta}}_z}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{\Gamma}(s_B) = \underline{H}_B = M(s_B)\underline{e}_z$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\dot{\hat{\underline{\xi}}}(s)\,\mathrm{d}s = \Big[\underline{\Sigma}(s)\cdot\dot{\hat{\underline{\xi}}}(s)\Big]_A^B - \int_A^B \boxed{\underline{\Sigma}(s)}\cdot\frac{\mathrm{d}\dot{\hat{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Sigma}(s) = N(s)\underline{t} + V(s)\underline{n}$

$$= \boxed{\underline{\Sigma}(s_B)}\cdot\dot{\hat{\underline{\xi}}}(s_B) - \boxed{\underline{\Sigma}(s_A)}\cdot\dot{\hat{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\dot{\hat{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{\Sigma}(s_B) = \underline{R}_B$; $\underline{\Sigma}(s_A) = -\underline{R}_A$

$$(2)\quad \int_A^B \frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s)\cdot\dot{\hat{\underline{\theta}}}(s)\,\mathrm{d}s = \Big[\underline{\Gamma}(s)\cdot\dot{\hat{\underline{\theta}}}(s)\Big]_A^B - \int_A^B \boxed{\underline{\Gamma}(s)}\cdot\frac{\mathrm{d}\dot{\hat{\underline{\theta}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$\underline{\Gamma}(s) = M(s)\underline{e}_z$

$$= \boxed{\underline{\Gamma}(s_B)}\cdot\dot{\hat{\underline{\theta}}}(s_B) - \boxed{\underline{\Gamma}(s_A)}\cdot\dot{\hat{\underline{\theta}}}(s_A) - \int_A^B M(s)\frac{\mathrm{d}\dot{\hat{\theta}}_z}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\underline{\Gamma}(s_B) = \underline{H}_B = M(s_B)\underline{e}_z$; $-\underline{\Gamma}(s_A) = -\underline{H}_A = M(s_A)\underline{e}_z$

T.P.V : Absence d'actions concentrées (2)

arc plan + chargement plan

$\to N(s)\underline{t} + V(s)\underline{n}$

$$(1)\quad \int_A^B \frac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s)\cdot\dot{\hat{\underline{\xi}}}(s)\,\mathrm{d}s = \left[\underline{\Sigma}(s)\cdot\dot{\hat{\underline{\xi}}}(s)\right]_A^B - \int_A^B \underline{\Sigma}(s)\cdot\frac{\mathrm{d}\dot{\hat{\underline{\xi}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$$= \underline{\Sigma}(s_B)\cdot\dot{\hat{\underline{\xi}}}(s_B) - \underline{\Sigma}(s_A)\cdot\dot{\hat{\underline{\xi}}}(s_A) - \int_A^B N(s)\frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}(s) + V(s)\frac{\mathrm{d}\dot{\hat{\xi}}_n}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\to \underline{R}_B$ $\quad \to -\underline{R}_A$

$\to M(s)\underline{e}_z$

$$(2)\quad \int_A^B \frac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s)\cdot\dot{\hat{\underline{\theta}}}(s)\,\mathrm{d}s = \left[\underline{\Gamma}(s)\cdot\dot{\hat{\underline{\theta}}}(s)\right]_A^B - \int_A^B \underline{\Gamma}(s)\cdot\frac{\mathrm{d}\dot{\hat{\underline{\theta}}}}{\mathrm{d}s}\,\mathrm{d}s$$

$$= \underline{\Gamma}(s_B)\cdot\dot{\hat{\underline{\theta}}}(s_B) - \underline{\Gamma}(s_A)\cdot\dot{\hat{\underline{\theta}}}(s_A) - \int_A^B M(s)\frac{\mathrm{d}\dot{\hat{\theta}}_z}{\mathrm{d}s}(s)\,\mathrm{d}s$$

$\to \underline{H}_B = M(s_B)\underline{e}_z$ $\quad \to -\underline{H}_A = M(s_A)\underline{e}_z$

$$(3)\quad (\underline{t}\wedge\underline{\Sigma})\cdot\dot{\hat{\underline{\theta}}} = \underline{t}\wedge(N\underline{t}+V\underline{n})\cdot\dot{\hat{\underline{\theta}}} = V\dot{\hat{\theta}}_z$$

T.P.V : Absence d'actions concentrées (3)

arc plan + chargement plan

Equilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\dot{\underline{\xi}}}(s) \ ; \ \forall s \to \hat{\dot{\underline{\theta}}}(s)$$

$$\underline{R}_A \cdot \hat{\dot{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \hat{\dot{\underline{\xi}}}(s_B) + M(s_A)\hat{\dot{\theta}}_z(s_A) + M(s_B)\hat{\dot{\theta}}_z(s_B) + \int_A^B \underline{f}(s) \cdot \hat{\dot{\underline{\xi}}} + m(s)\hat{\dot{\theta}}_z \, \mathrm{d}s$$

$$+$$

$$- \left(\int_A^B N(s) \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}}_z \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}_z}{\mathrm{d}s}(s) \, \mathrm{d}s \right)$$

$$= 0$$

T.P.V : Absence d'actions concentrées (3)

arc plan + chargement plan

Equilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\dot{\underline{\xi}}}(s) \ ; \ \forall s \to \hat{\dot{\theta}}_z(s)$$

$$\underline{R}_A \cdot \hat{\dot{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \hat{\dot{\underline{\xi}}}(s_B) + M(s_A)\hat{\dot{\theta}}_z(s_A) + M(s_B)\hat{\dot{\theta}}_z(s_B) + \int_A^B \underline{f}(s) \cdot \hat{\dot{\underline{\xi}}} + m(s)\hat{\dot{\theta}}_z \, \mathrm{d}s$$

$$+$$

$$- \left(\int_A^B N(s) \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}}_z \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}_z}{\mathrm{d}s}(s) \, \mathrm{d}s \right)$$

$$= 0$$

T.P.V : Absence d'actions concentrées (3)

arc plan + chargement plan

Equilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\underline{\xi}}(s) \ ; \ \forall s \to \ \hat{\dot{\theta}}_z(s) = \hat{\dot{\theta}}(s)$$

$$\underline{R}_A \cdot \hat{\underline{\xi}}(s_A) + \underline{R}_B \cdot \hat{\underline{\xi}}(s_B) + M(s_A)\hat{\dot{\theta}}_z(s_A) + M(s_B)\hat{\dot{\theta}}_z(s_B) + \int_A^B \underline{f}(s) \cdot \hat{\underline{\xi}} + m(s)\hat{\dot{\theta}}_z \, \mathrm{d}s$$

$$+$$

$$- \left(\int_A^B N(s) \frac{\mathrm{d}\hat{\xi}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\xi}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}}_z \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}_z}{\mathrm{d}s}(s) \, \mathrm{d}s \right)$$

$$= 0$$

T.P.V : Absence d'actions concentrées (3)

arc plan + chargement plan

Equilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\underline{\xi}}(s) \ ; \ \forall s \to \ \hat{\dot{\theta}}_z(s) = \hat{\dot{\theta}}(s)$$

$$\underline{R}_A \cdot \hat{\underline{\xi}}(s_A) + \underline{R}_B \cdot \hat{\underline{\xi}}(s_B) + M(s_A)\hat{\dot{\theta}}(s_A) + M(s_B)\hat{\dot{\theta}}(s_B) + \int_A^B \underline{f}(s) \cdot \hat{\underline{\xi}} + m(s)\hat{\dot{\theta}} \, \mathrm{d}s$$

$$+$$

$$- \left(\int_A^B N(s) \frac{\mathrm{d}\hat{\xi}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\xi}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}}_z \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}_z}{\mathrm{d}s}(s) \, \mathrm{d}s \right)$$

$$= 0$$

T.P.V : Absence d'actions concentrées (3)

arc plan + chargement plan

Equilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\underline{\dot{\xi}}}(s) \ ; \ \forall s \to \hat{\dot{\theta}}_z(s) = \hat{\dot{\theta}}(s)$$

$$\underline{R}_A \cdot \hat{\underline{\dot{\xi}}}(s_A) + \underline{R}_B \cdot \hat{\underline{\dot{\xi}}}(s_B) + M(s_A)\hat{\dot{\theta}}(s_A) + M(s_B)\hat{\dot{\theta}}(s_B) + \int_A^B \underline{f}(s) \cdot \hat{\underline{\dot{\xi}}} + m(s)\hat{\dot{\theta}} \, \mathrm{d}s$$

$$+$$

$$- \int_A^B N(s) \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}} \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

$$= 0$$

T.P.V : Absence d'actions concentrées (3)

arc plan + chargement plan

Equilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\dot{\underline{\xi}}}(s) \ ; \ \forall s \to \hat{\dot{\theta}}_z(s) = \hat{\dot{\theta}}(s)$$

$P_{ext}(\hat{\dot{\underline{\xi}}}, \hat{\dot{\theta}})$

$$\underline{R}_A \cdot \hat{\dot{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \hat{\dot{\underline{\xi}}}(s_B) + M(s_A)\hat{\dot{\theta}}(s_A) + M(s_B)\hat{\dot{\theta}}(s_B) + \int_A^B \underline{f}(s) \cdot \hat{\dot{\underline{\xi}}} + m(s)\hat{\dot{\theta}} \, \mathrm{d}s$$

$$+$$

$$- \int_A^B N(s) \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}} \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

$$= 0$$

T.P.V : Absence d'actions concentrées (3)

arc plan + chargement plan

Equilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\dot{\underline{\xi}}}(s) \ ; \ \forall s \to \hat{\dot{\theta}}_z(s) = \hat{\dot{\theta}}(s)$$

$P_{ext}(\hat{\dot{\underline{\xi}}}, \hat{\dot{\theta}})$

$$\underline{R}_A \cdot \hat{\dot{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \hat{\dot{\underline{\xi}}}(s_B) + M(s_A)\hat{\dot{\theta}}(s_A) + M(s_B)\hat{\dot{\theta}}(s_B) + \int_A^B \underline{f}(s) \cdot \hat{\dot{\underline{\xi}}} + m(s)\hat{\dot{\theta}} \, \mathrm{d}s$$

$+$

$P_{int}(\hat{\dot{\underline{\xi}}}, \hat{\dot{\theta}})$

$$-\int_A^B N(s)\frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s)\left(\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}}\right) + M(s)\frac{\mathrm{d}\hat{\dot{\theta}}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

$$= 0$$

T.P.V : Absence d'actions concentrées (3)

arc plan + chargement plan

Equilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\dot{\underline{\xi}}}(s) \ ; \ \forall s \to \hat{\dot{\theta}}_z(s) = \hat{\dot{\theta}}(s) \quad : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

$P_{ext}(\hat{\dot{\underline{\xi}}}, \hat{\dot{\theta}})$

$$\underline{R}_A \cdot \hat{\dot{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \hat{\dot{\underline{\xi}}}(s_B) + M(s_A)\hat{\dot{\theta}}(s_A) + M(s_B)\hat{\dot{\theta}}(s_B) + \int_A^B \underline{f}(s) \cdot \hat{\dot{\underline{\xi}}} + m(s)\hat{\dot{\theta}} \, \mathrm{d}s$$

$+$

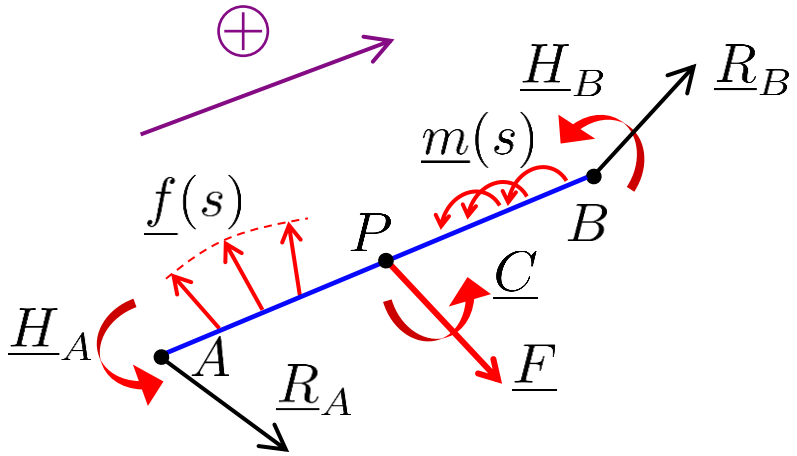
$P_{int}(\hat{\dot{\underline{\xi}}}, \hat{\dot{\theta}})$

$$- \int_A^B N(s) \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}} \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

$$= 0$$

T.P.V : Prise en compte des actions concentrées (1)

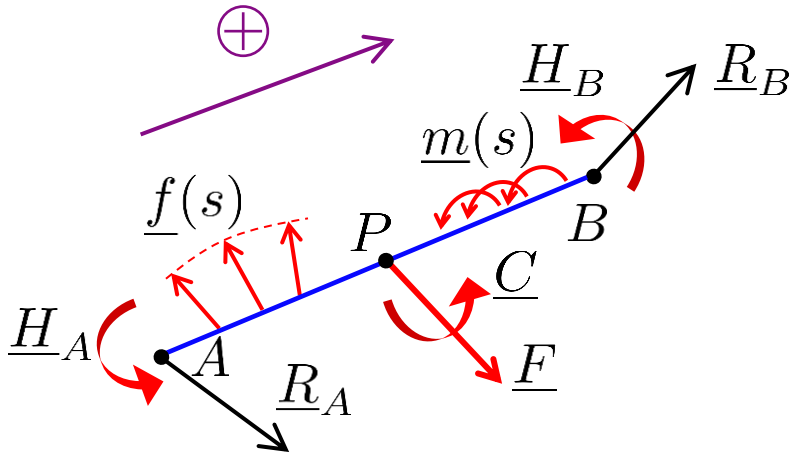
arc plan + chargement plan



équilibre : $\begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases}$ sur $[AC[\cup]CB]$ et $\begin{cases} [\![\underline{\Sigma}]\!]_P + \underline{F} = \underline{0} \\ [\![\underline{\Gamma}]\!]_P + \underline{C} = \underline{0} \end{cases}$

T.P.V : Prise en compte des actions concentrées (1)

arc plan + chargement plan



$$\text{équilibre : } \begin{cases} \dfrac{\mathrm{d}\underline{\Sigma}}{\mathrm{d}s}(s) + \underline{f}(s) = \underline{0} \\ \\ \dfrac{\mathrm{d}\underline{\Gamma}}{\mathrm{d}s}(s) + \underline{m}(s) + \underline{t} \wedge \underline{\Sigma}(s) = \underline{0} \end{cases} \quad \text{sur } [AC[\cup]CB] \text{ et } \begin{cases} [\![\underline{\Sigma}]\!]_P + \underline{F} = \underline{0} \\ \\ [\![\underline{\Gamma}]\!]_P + \underline{C} = \underline{0} \end{cases}$$

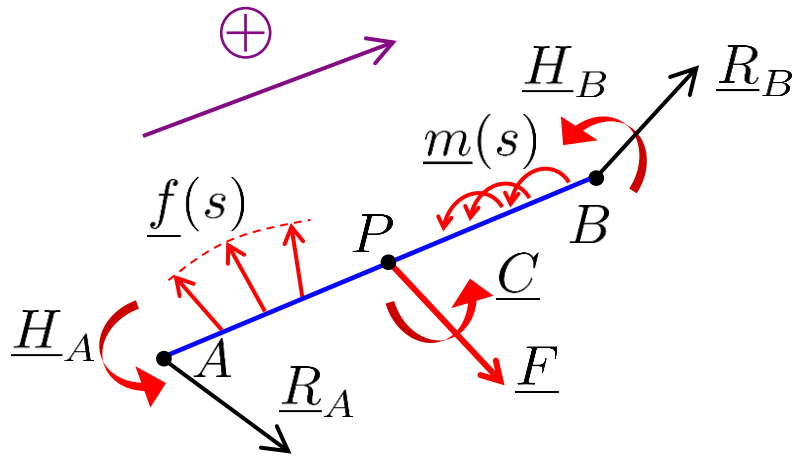
$$\Longleftrightarrow$$

$$\forall s \to \hat{\dot{\underline{\xi}}}(s) \ ; \ \forall s \to \hat{\dot{\underline{\theta}}}(s)$$

$$\int_A^P \left[(...) \cdot \hat{\dot{\underline{\xi}}}(s) + (...) \cdot \hat{\dot{\underline{\theta}}}(s) \right] \mathrm{d}s + \int_P^B \left[(...) \cdot \hat{\dot{\underline{\xi}}}(s) + (...) \cdot \hat{\dot{\underline{\theta}}}(s) \right] \mathrm{d}s = 0$$

T.P.V : Prise en compte des actions concentrées (2)

arc plan + chargement plan



équilibre

$$\Longleftrightarrow$$

$$\forall s \to \underline{\hat{\dot{\xi}}}(s) \ ; \ \forall s \to \underline{\hat{\dot{\theta}}}(s)$$

$$\underline{R}_A \cdot \underline{\hat{\dot{\xi}}}(s_A) + \underline{R}_B \cdot \underline{\hat{\dot{\xi}}}(s_B) + M(s_A)\hat{\dot{\theta}}_z(s_A) + M(s_B)\hat{\dot{\theta}}_z(s_B) + \int_A^B \underline{f}(s) \cdot \underline{\hat{\dot{\xi}}} + m(s)\hat{\dot{\theta}}_z \, \mathrm{d}s$$

$$+\underline{\Sigma}(s_P^-) \cdot \underline{\hat{\dot{\xi}}}(s_P) - \underline{\Sigma}(s_P^+) \cdot \underline{\hat{\dot{\xi}}}(s_P) + \underline{\Gamma}(s_P^-) \cdot \underline{\hat{\dot{\theta}}}(s_P) - \underline{\Gamma}(s_P^+) \cdot \underline{\hat{\dot{\theta}}}(s_P)$$

$$+$$

$$- \left(\int_A^B N(s) \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\dot{\xi}}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}}_z \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}_z}{\mathrm{d}s}(s) \, \mathrm{d}s \right)$$

$$= 0$$

T.P.V : Prise en compte des actions concentrées (2)

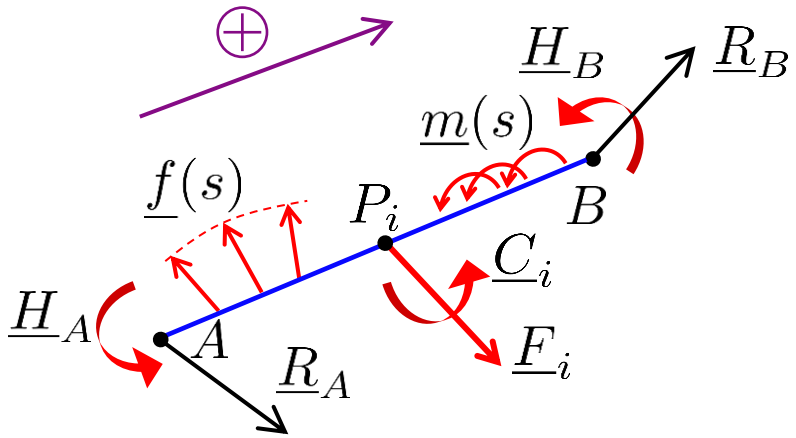
arc plan + chargement plan

équilibre

$$\Longleftrightarrow$$

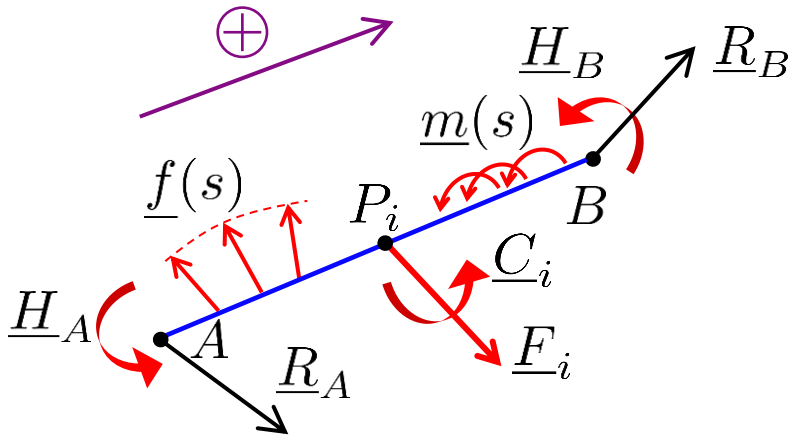
$$\forall s \to \underline{\hat{\dot{\xi}}}(s) \ ; \ \forall s \to \hat{\dot{\theta}}(s) \ : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

$$P_{ext}(\underline{\hat{\dot{\xi}}}, \hat{\dot{\theta}}) + P_{int}(\underline{\hat{\dot{\xi}}}, \hat{\dot{\theta}}) = 0$$



T.P.V : Prise en compte des actions concentrées (2)

arc plan + chargement plan



équilibre

$$\Longleftrightarrow$$

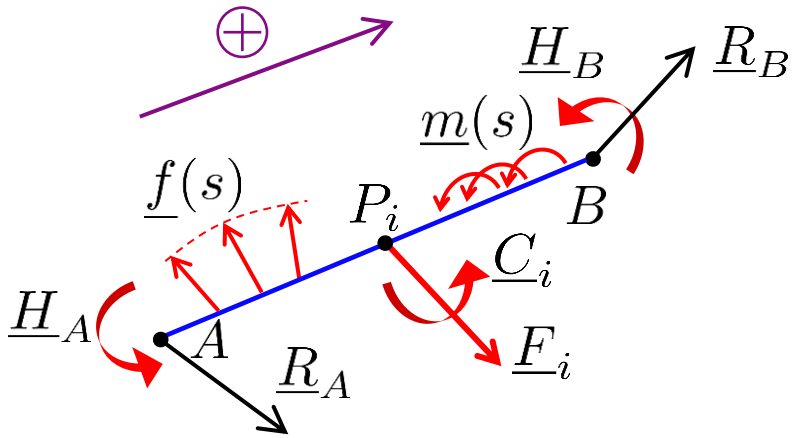
$$\forall s \to \hat{\underline{\xi}}(s) \ ; \ \forall s \to \hat{\theta}(s) \ : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

$$P_{ext}(\hat{\underline{\xi}}, \hat{\theta}) + P_{int}(\hat{\underline{\xi}}, \hat{\theta}) = 0$$

$$P_{ext}(\hat{\underline{\xi}}, \hat{\theta}) = \underline{R}_A \cdot \hat{\underline{\xi}}(s_A) + \underline{R}_B \cdot \hat{\underline{\xi}}(s_B) + M(s_A)\hat{\theta}(s_A) + M(s_B)\hat{\theta}(s_B)$$
$$+ \sum_i \left(\underline{F}_i \cdot \hat{\underline{\xi}}(s_{P_i}) + C_i \hat{\theta}(s_{P_i}) \right) + \int_A^B \underline{f}(s) \cdot \hat{\underline{\xi}} + m(s)\hat{\theta} \, \mathrm{d}s$$

T.P.V : Prise en compte des actions concentrées (2)

arc plan + chargement plan



équilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\underline{\xi}}(s) \ ; \ \forall s \to \hat{\dot{\theta}}(s) \ : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

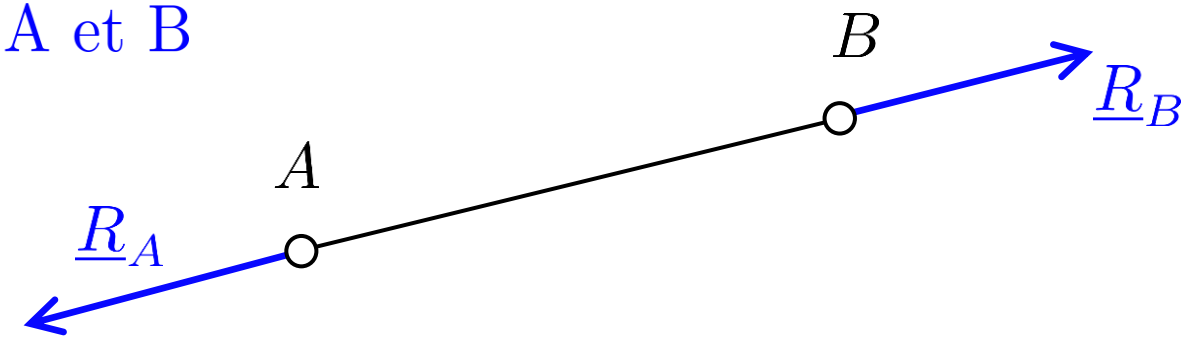
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$$P_{int}(\hat{\underline{\xi}}, \hat{\dot{\theta}}) = -\int_A^B N(s) \frac{\mathrm{d}\hat{\xi}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\xi}_n}{\mathrm{d}s}(s) - \hat{\dot{\theta}} \right) + M(s) \frac{\mathrm{d}\hat{\dot{\theta}}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

T.P.V : Cas d'une poutre droite bi-articulée (1)

Aucune action extérieure entre les extrémités A et B



équilibre

$$\Longleftrightarrow$$

$$\forall s \to \hat{\underline{\xi}}(s) \ ; \ \forall s \to \hat{\theta}(s) \ : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

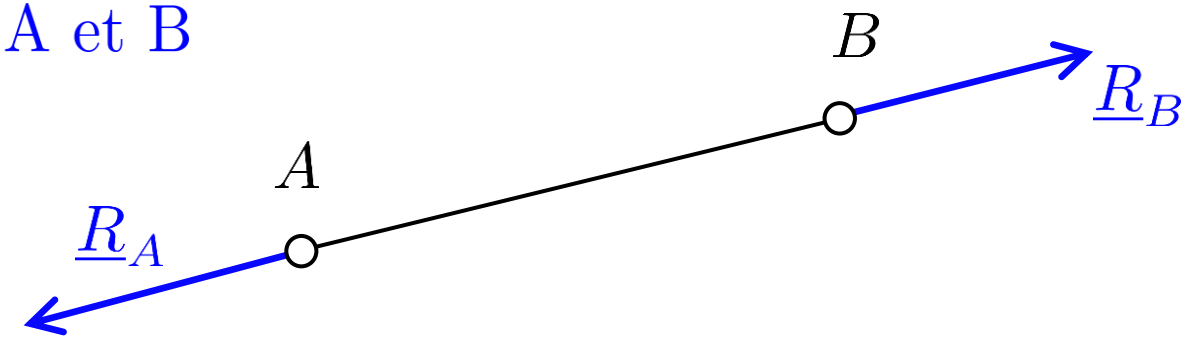
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$$P_{int}(\hat{\underline{\xi}}, \hat{\theta}) = - \int_A^B N(s) \frac{\mathrm{d}\hat{\xi}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\xi}_n}{\mathrm{d}s}(s) - \hat{\theta} \right) + M(s) \frac{\mathrm{d}\hat{\theta}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

T.P.V : Cas d'une poutre droite bi-articulée (1)

Aucune action extérieure entre les extrémités A et B



équilibre

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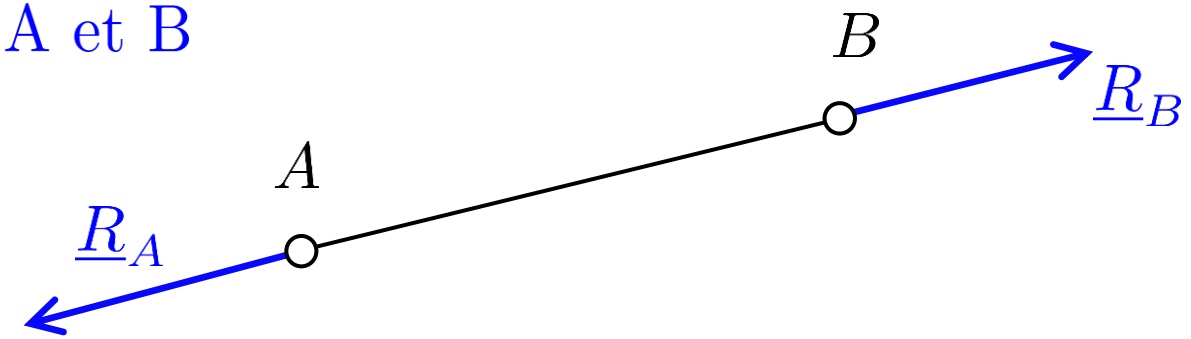
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T.P.V : Cas d'une poutre droite bi-articulée (1)

Aucune action extérieure entre les extrémités A et B



équilibre

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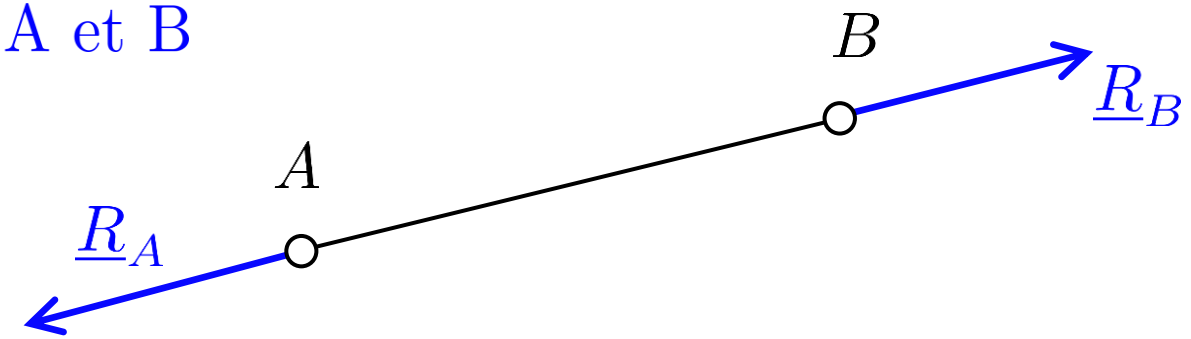
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T.P.V : Cas d'une poutre droite bi-articulée (1)

Aucune action extérieure entre les extrémités A et B



équilibre

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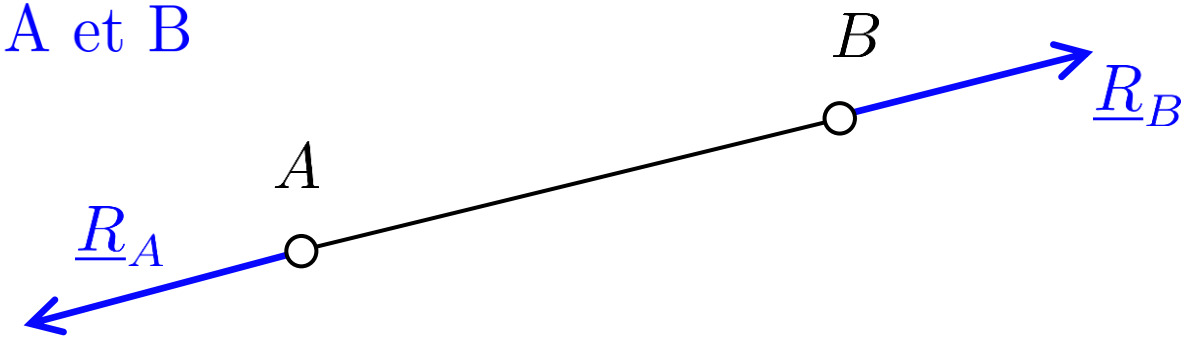
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T.P.V : Cas d'une poutre droite bi-articulée (1)

Aucune action extérieure entre les extrémités A et B



équilibre

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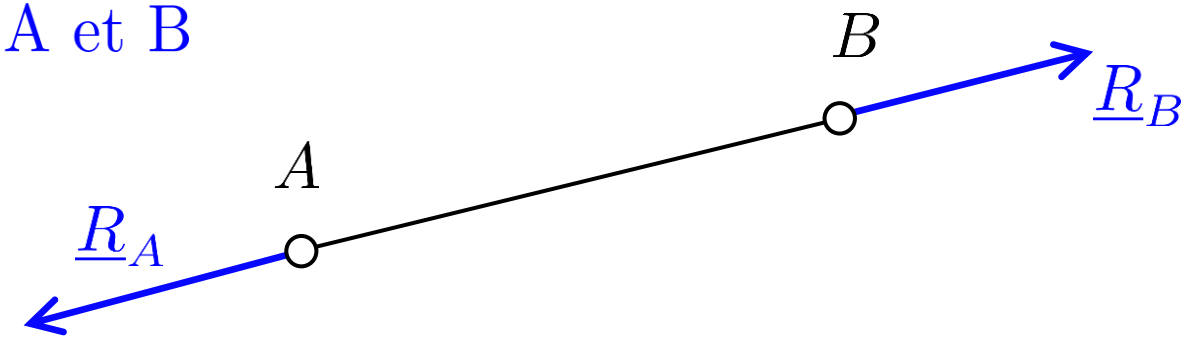
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$$P_{int}(\hat{\underline{\xi}}, \hat{\theta}) = - \int_A^B N(s) \frac{\mathrm{d}\hat{\xi}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\hat{\xi}_n}{\mathrm{d}s}(s) - \hat{\theta} \right) + M(s) \frac{\mathrm{d}\hat{\theta}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

T.P.V : Cas d'une poutre droite bi-articulée (1)

Aucune action extérieure entre les extrémités A et B



équilibre

$$\Longleftrightarrow$$

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T.P.V : Cas d'une poutre bi-articulée

Aucune action extérieure entre les extrémités A et B

équilibre $\Longleftrightarrow$

$$\forall s \to \hat{\dot{\underline{\xi}}}(s) : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

$$\underline{R}_A \cdot \hat{\dot{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \hat{\dot{\underline{\xi}}}(s_B) = \int_A^B N \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}\, \mathrm{d}s$$

T.P.V : Cas d'une poutre bi-articulée

Aucune action extérieure entre les extrémités A et B

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$$\forall s \to \hat{\dot{\underline{\xi}}}(s) : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

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T.P.V : Cas d'une poutre bi-articulée

Aucune action extérieure entre les extrémités A et B

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Remarque :

orientation : $A \to B$

équilibre : $\begin{cases} \dfrac{\mathrm{d}N}{\mathrm{d}s} = 0 \\ \\ \underline{R}_B = N\underline{t}\ ;\ \underline{R}_A = -N\underline{t} \end{cases}$

T.P.V : Cas d'une poutre bi-articulée

Aucune action extérieure entre les extrémités A et B

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T.P.V : Cas d'une poutre bi-articulée

Aucune action extérieure entre les extrémités A et B

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$$\forall s \to \hat{\dot{\underline{\xi}}}(s) : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

$$\underline{R}_A \cdot \hat{\dot{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \hat{\dot{\underline{\xi}}}(s_B) = \int_A^B N \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}\, \mathrm{d}s \; = N(\hat{\dot{\xi}}_t(s_B) - \hat{\dot{\xi}}_t(s_A))$$

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$$\int_A^B \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}\, \mathrm{d}s = \hat{\dot{\xi}}_t(s_B) - \hat{\dot{\xi}}_t(s_A)$$
$$= \underline{t} \cdot \hat{\dot{\underline{\xi}}}(s_B) - \underline{t} \cdot \hat{\dot{\underline{\xi}}}(s_A)$$

T.P.V : Cas d'une poutre bi-articulée

Aucune action extérieure entre les extrémités A et B

équilibre $\iff$

$$\forall s \to \dot{\hat{\underline{\xi}}}(s) : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm}$$

$$\underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \dot{\hat{\underline{\xi}}}(s_B) = \int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\,\mathrm{d}s = N(\dot{\hat{\xi}}_t(s_B) - \dot{\hat{\xi}}_t(s_A))$$

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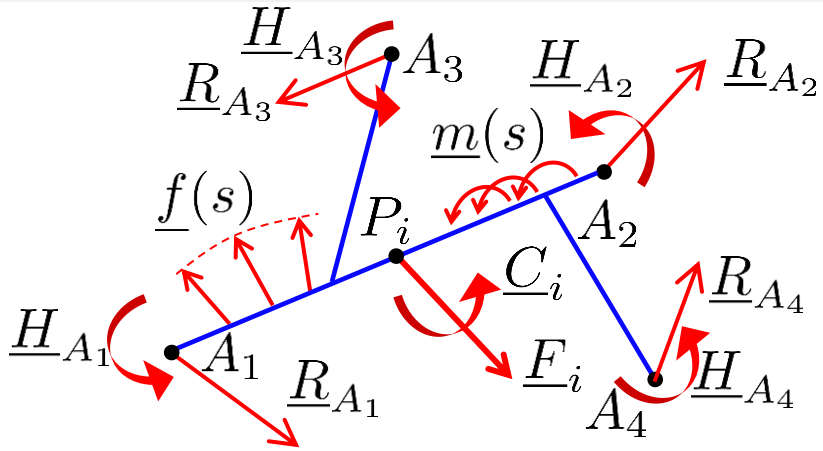
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$$\int_A^B \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\,\mathrm{d}s = \dot{\hat{\xi}}_t(s_B) - \dot{\hat{\xi}}_t(s_A) = \underline{t} \cdot \dot{\hat{\underline{\xi}}}(s_B) - \underline{t} \cdot \dot{\hat{\underline{\xi}}}(s_A)$$

$$\forall s \to \dot{\hat{\underline{\xi}}}(s) \quad \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} : \quad \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(s_A) + \underline{R}_B \cdot \dot{\hat{\underline{\xi}}}(s_B) = \int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\,\mathrm{d}s$$

Enoncé du **T**héorème des **P**uissances **V**irtuelles (1)

$$\text{équilibre} \iff \begin{cases} \forall \text{ mvt. virtuel} \\ \\ P_{ext} + P_{int} = 0 \end{cases}$$



structure plane + chargement plan

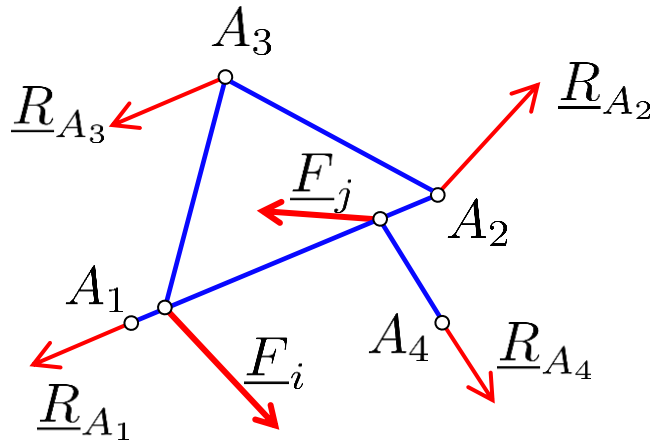
$$P_{ext}(\dot{\hat{\underline{\xi}}}, \dot{\hat{\theta}}) = \sum_i \left(\underline{R}_{A_i} \cdot \dot{\hat{\underline{\xi}}}(s_{A_i}) + M(s_{A_i}) \dot{\hat{\theta}}(s_{A_i}) \right) + \sum_i \left(\underline{F}_i \cdot \dot{\hat{\underline{\xi}}}(s_{P_i}) + C_i \dot{\hat{\theta}}(s_{P_i}) \right)$$
$$+ \int_{struc.} \underline{f}(s) \cdot \dot{\hat{\underline{\xi}}} + m(s) \dot{\hat{\theta}} \, \mathrm{d}s$$

$$P_{int}(\dot{\hat{\underline{\xi}}}, \dot{\hat{\theta}}) = - \int_{struc.} N(s) \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\dot{\hat{\xi}}_n}{\mathrm{d}s}(s) - \dot{\hat{\theta}} \right) + M(s) \frac{\mathrm{d}\dot{\hat{\theta}}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

Enoncé du Théorème des Puissances virtuelles (2)

$$\text{équilibre} \iff \begin{cases} \forall \text{ mvt. virtuel} \\ \\ P_{ext} + P_{int} = 0 \end{cases}$$

structure réticulée

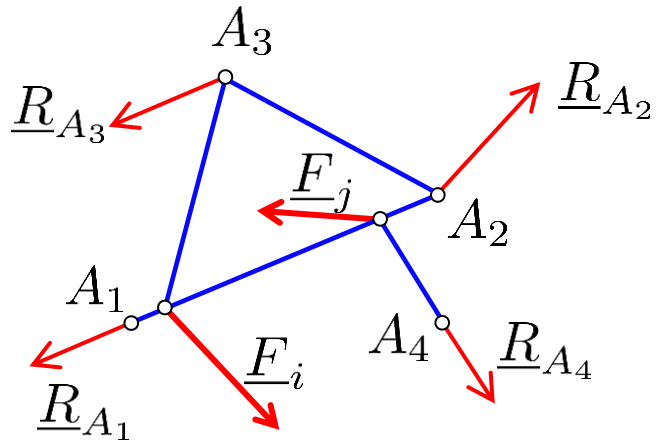


Enoncé du Théorème des Puissances virtuelles (2)

$$\text{équilibre} \iff \begin{cases} \forall \text{ mvt. virtuel} \\ \\ P_{ext} + P_{int} = 0 \end{cases}$$

structure réticulée

$$P_{ext}(\hat{\underline{\xi}}) = \sum_i \underline{R}_{A_i} \cdot \hat{\underline{\xi}}(s_{A_i}) + \sum_i \underline{F}_i \cdot \hat{\underline{\xi}}(s_{P_i})$$



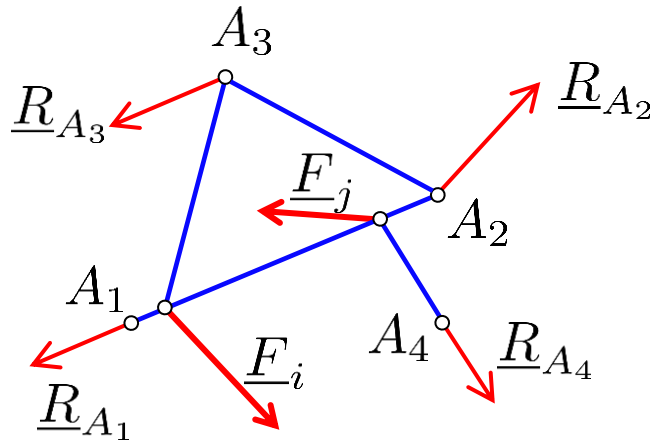
Enoncé du Théorème des Puissances virtuelles (2)

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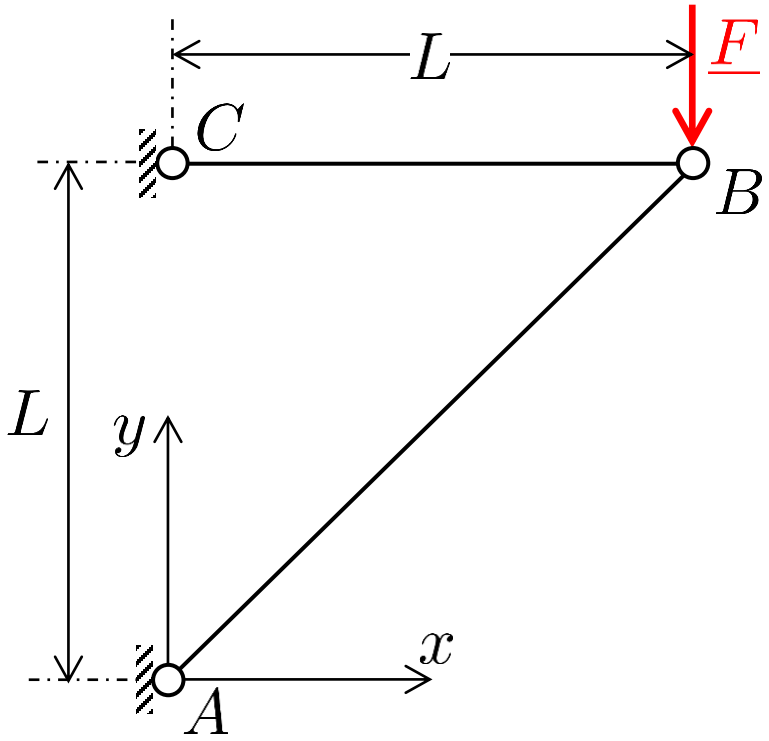
structure réticulée

$$P_{ext}(\hat{\dot{\underline{\xi}}}) = \sum_i \underline{R}_{A_i} \cdot \hat{\dot{\underline{\xi}}}(s_{A_i}) + \sum_i \underline{F}_i \cdot \hat{\dot{\underline{\xi}}}(s_{P_i})$$

$$P_{int}(\hat{\dot{\underline{\xi}}}) = -\int_{struc.} N \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}(s)\, \mathrm{d}s$$

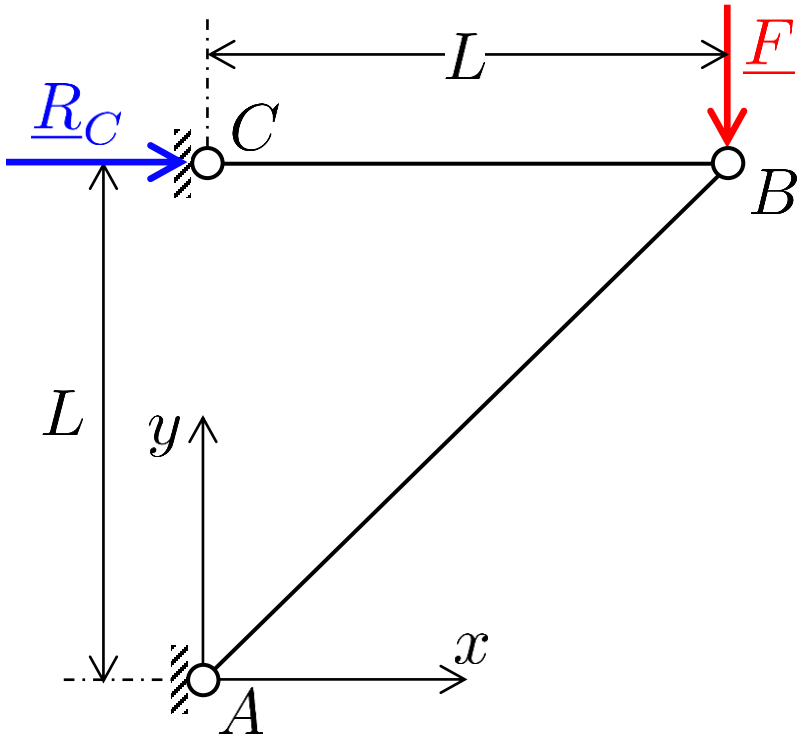


Exemple d'application (1)



$\underline{R}_A = ?\ ;\ \underline{R}_C = ?$

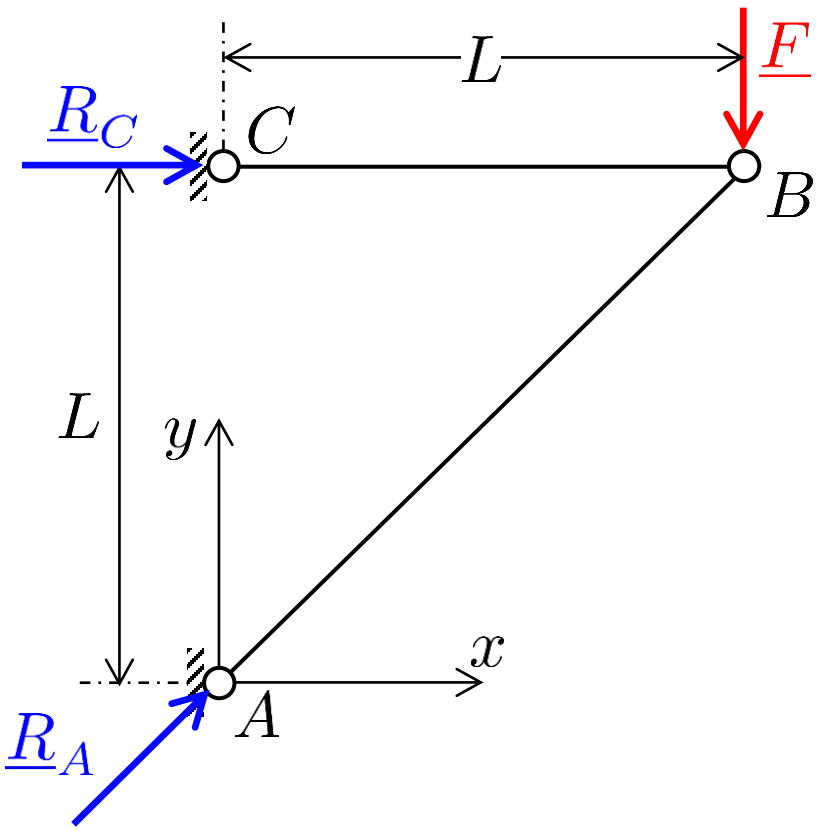
Exemple d'application (1)



$\underline{R}_A = ?$; $\underline{R}_C = ?$

$$\underline{R}_C = R_C \, \underline{e}_x$$

Exemple d'application (1)

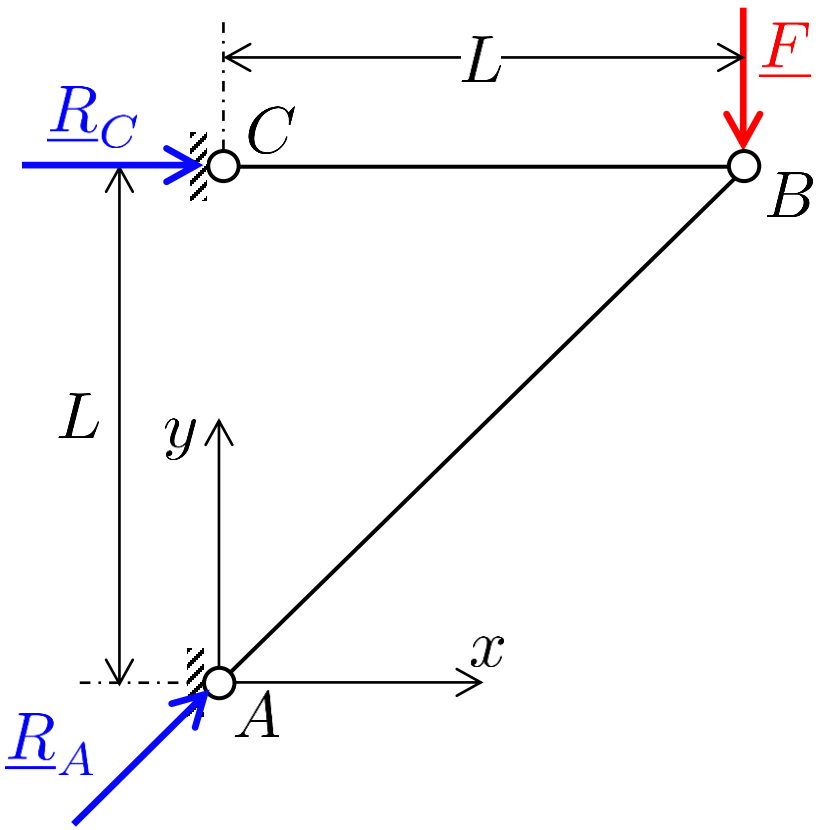


$\underline{R}_A = ?$; $\underline{R}_C = ?$

$$\underline{R}_C = R_C \, \underline{e}_x$$

$$\underline{R}_A = R_A \, \underline{t}_{AB} = R_A \, \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

Exemple d'application (1)



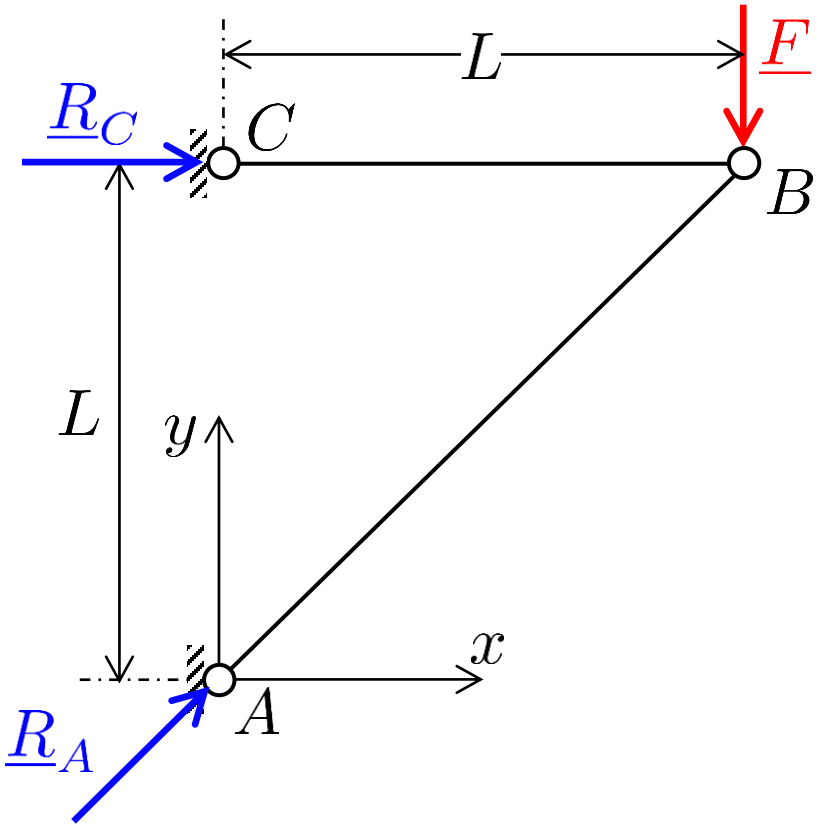
$\underline{R}_A = ?$; $\underline{R}_C = ?$

$$\underline{R}_C = R_C \, \underline{e}_x$$

$$\underline{R}_A = R_A \, \underline{t}_{AB} = R_A \, \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

équilibre : $\underline{R}_A + \underline{R}_C - F\underline{e}_y = \underline{0}$

Exemple d'application (1)



$$\underline{R}_A = ? \; ; \; \underline{R}_C = ?$$

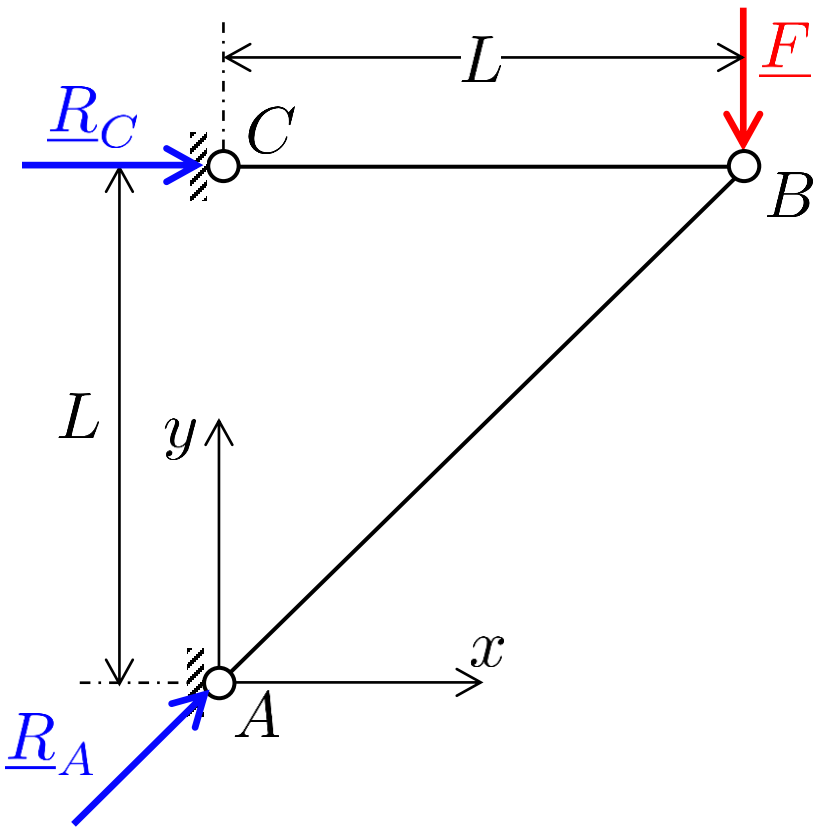
$$\underline{R}_C = R_C \, \underline{e}_x$$

$$\underline{R}_A = R_A \, \underline{t}_{AB} = R_A \, \frac{\sqrt{2}}{2} (\underline{e}_x + \underline{e}_y)$$

équilibre : $\underline{R}_A + \underline{R}_C - F\underline{e}_y = \underline{0}$

$$\Longrightarrow \boxed{\underline{R}_A = F \, (\underline{e}_x + \underline{e}_y) \; ; \; \underline{R}_C = -F\underline{e}_x}$$

Exemple d'application (1)



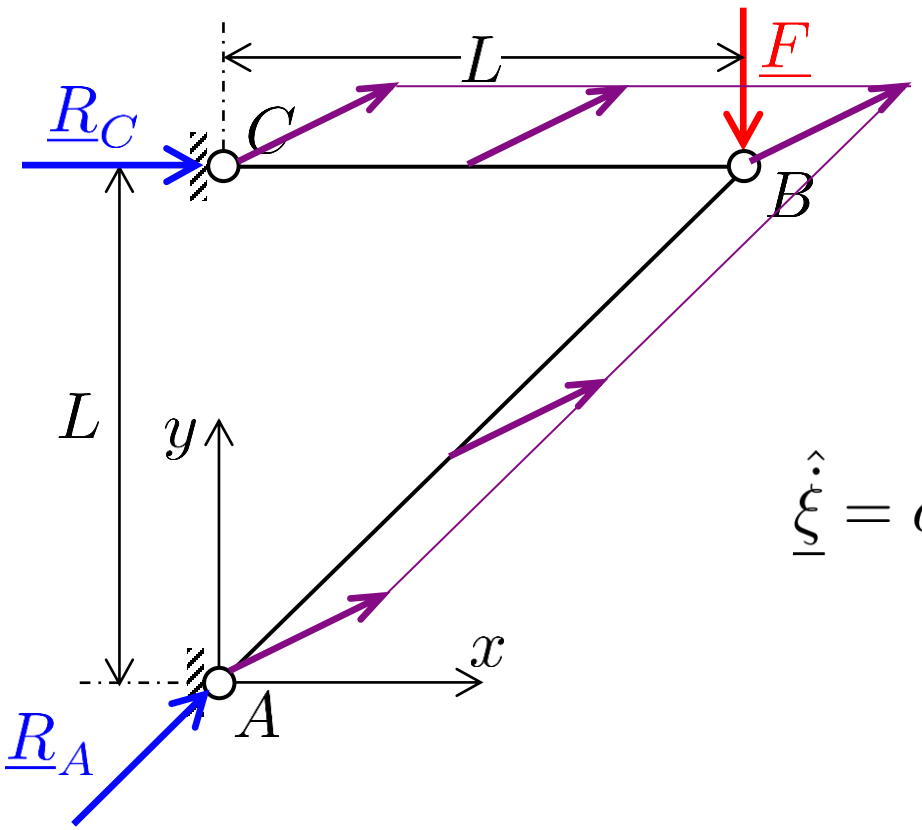
$\underline{R}_A = ?$; $\underline{R}_C = ?$

$$\underline{R}_C = R_C \, \underline{e}_x$$

$$\underline{R}_A = R_A \, \underline{t}_{AB} = R_A \, \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

Mise en œuvre du T.P.V

Exemple d'application (1)



$\underline{R}_A = ?$; $\underline{R}_C = ?$

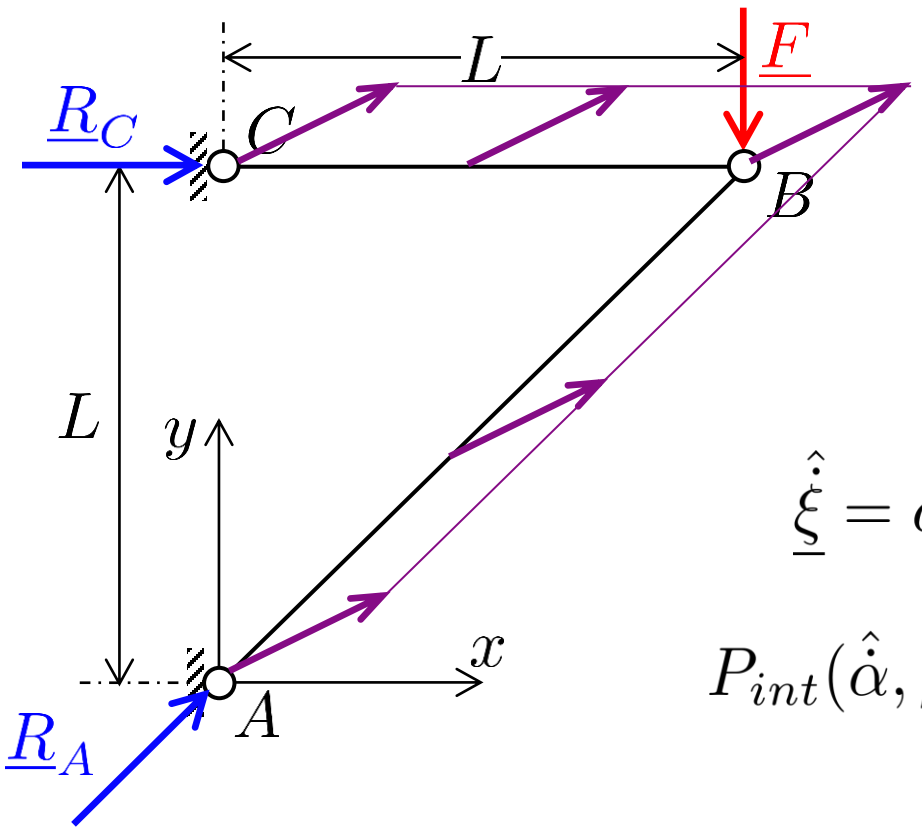
$$\underline{R}_C = R_C\, \underline{e}_x$$

$$\underline{R}_A = R_A\, \underline{t}_{AB} = R_A\, \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

Mise en œuvre du T.P.V

$$\underline{\hat{\xi}} = \hat{\alpha}\underline{e}_x + \hat{\beta}\underline{e}_y \text{ sur } [AB] \text{ et } [CB]$$

Exemple d'application (1)



$\underline{R}_A = ?$; $\underline{R}_C = ?$

$$\underline{R}_C = R_C\, \underline{e}_x$$

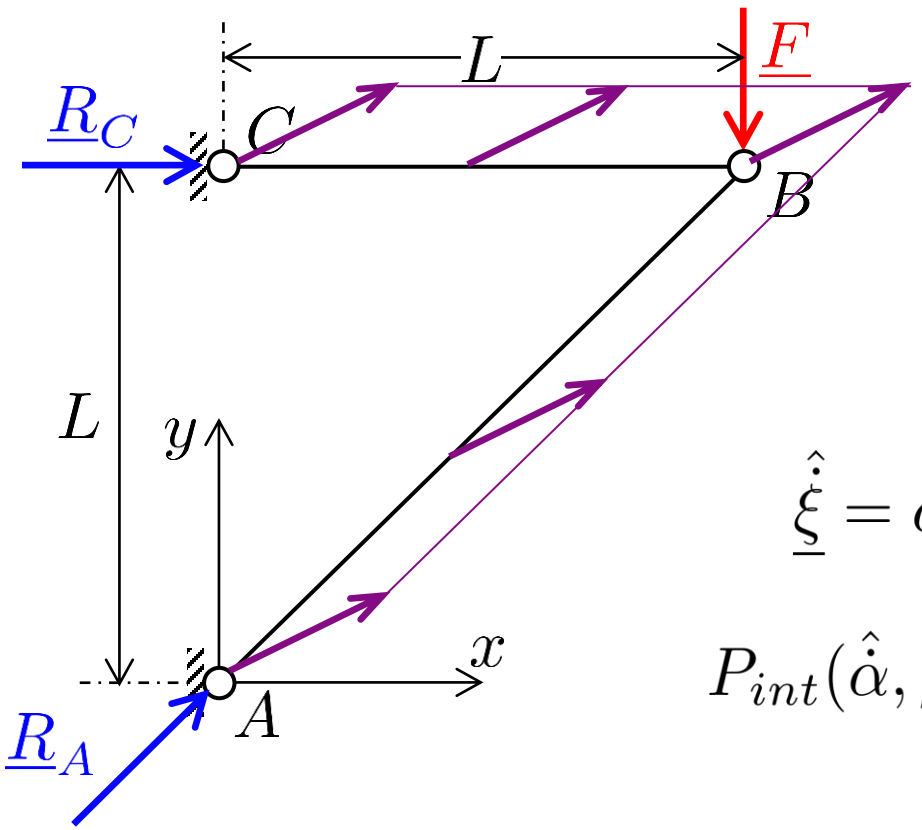
$$\underline{R}_A = R_A\, \underline{t}_{AB} = R_A\, \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

Mise en œuvre du T.P.V

$$\hat{\dot{\underline{\xi}}} = \hat{\dot{\alpha}}\underline{e}_x + \hat{\dot{\beta}}\underline{e}_y \text{ sur } [AB] \text{ et } [CB]$$

$$P_{int}(\hat{\dot{\alpha}}, \hat{\dot{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}\, \mathrm{d}s$$

Exemple d'application (1)



$\underline{R}_A = ?\ ;\ \underline{R}_C = ?$

$$\underline{R}_C = R_C\ \underline{e}_x$$

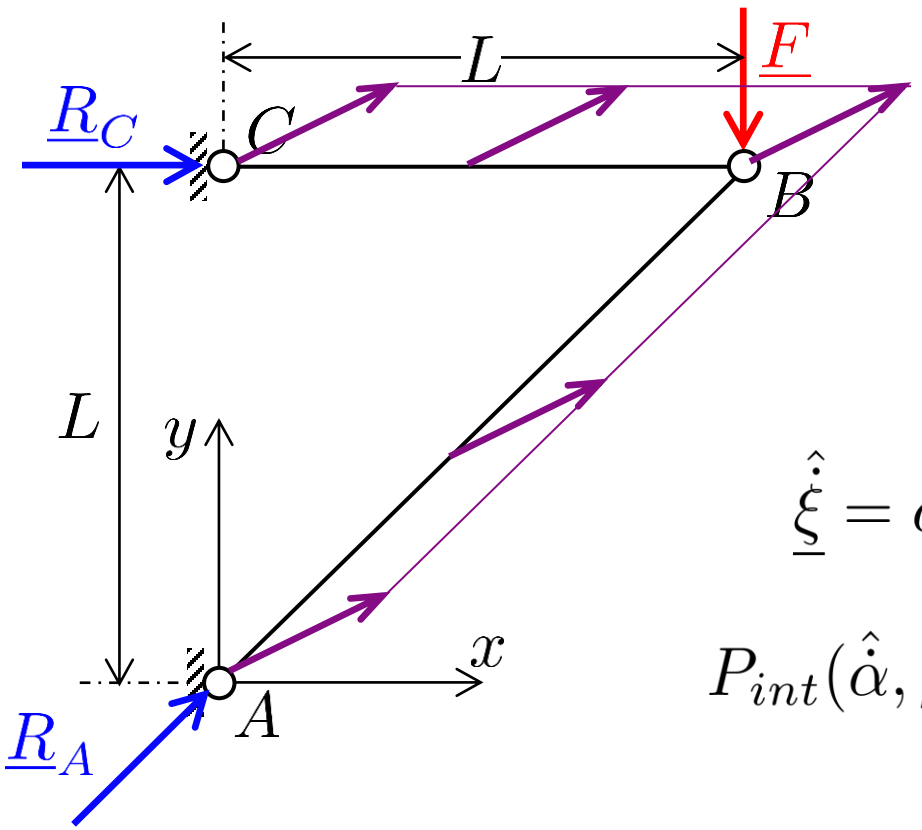
$$\underline{R}_A = R_A\ \underline{t}_{AB} = R_A\ \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

Mise en œuvre du T.P.V

$$\hat{\dot{\underline{\xi}}} = \hat{\dot{\alpha}}\underline{e}_x + \hat{\dot{\beta}}\underline{e}_y \text{ sur } [AB] \text{ et } [CB]$$

$$P_{int}(\hat{\dot{\alpha}}, \hat{\dot{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\hat{\dot{\xi}}_t}{\mathrm{d}s}\ \mathrm{d}s\ = 0$$

Exemple d'application (1)



$\underline{R}_A = ?\ ;\ \underline{R}_C = ?$

$$\underline{R}_C = R_C\ \underline{e}_x$$

$$\underline{R}_A = R_A\ \underline{t}_{AB} = R_A\ \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

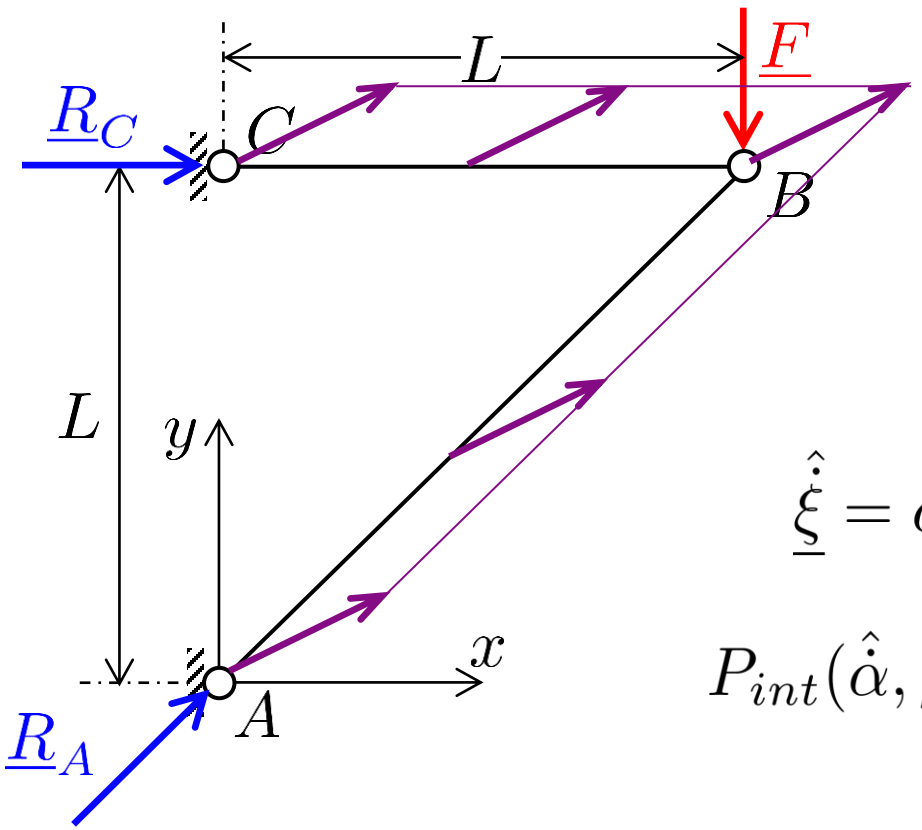
Mise en œuvre du T.P.V

$$\dot{\hat{\underline{\xi}}} = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \text{ sur } [AB] \text{ et } [CB]$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\ \mathrm{d}s \quad = 0$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B)$$

Exemple d'application (1)



$\underline{R}_A = ?\ ;\ \underline{R}_C = ?$

$$\underline{R}_C = R_C\,\underline{e}_x$$

$$\underline{R}_A = R_A\,\underline{t}_{AB} = R_A\,\frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

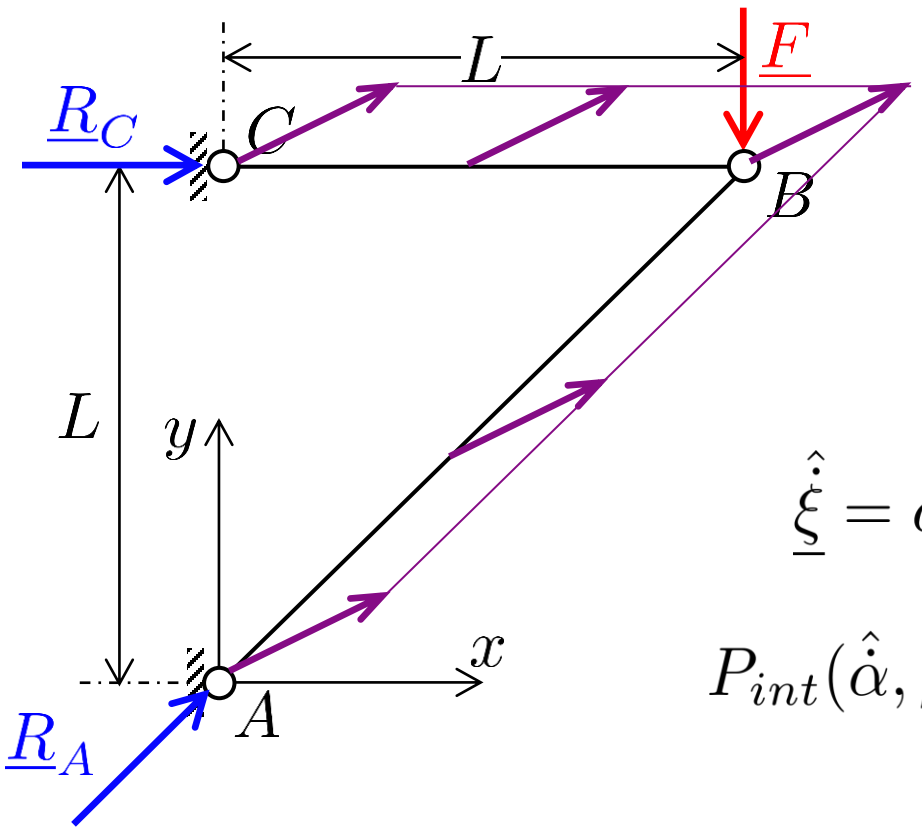
Mise en œuvre du T.P.V

$$\dot{\hat{\underline{\xi}}} = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \text{ sur } [AB] \text{ et } [CB]$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N\frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\,\mathrm{d}s = 0$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = (R_A/\sqrt{2} + R_C)\dot{\hat{\alpha}} + (R_A/\sqrt{2} - F)\dot{\hat{\beta}}$$

Exemple d'application (1)



$\underline{R}_A = ?\ ;\ \underline{R}_C = ?$

$$\underline{R}_C = R_C\ \underline{e}_x$$

$$\underline{R}_A = R_A\ \underline{t}_{AB} = R_A\ \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

Mise en œuvre du T.P.V

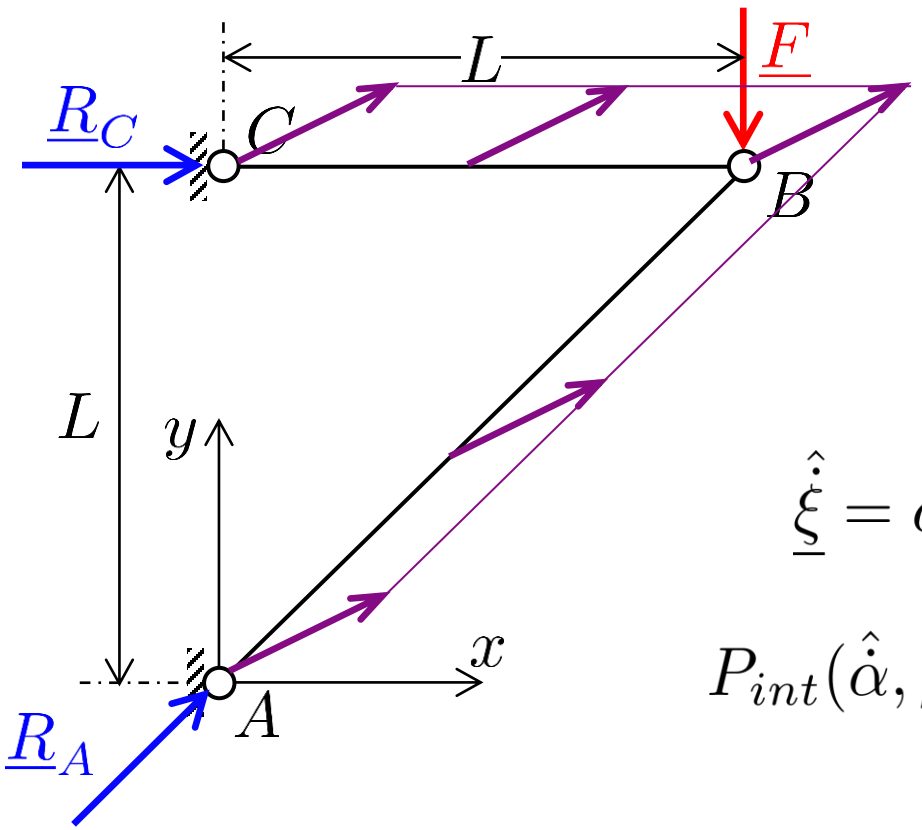
$$\dot{\hat{\underline{\xi}}} = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \text{ sur } [AB] \text{ et } [CB]$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\ \mathrm{d}s \ = 0$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = (R_A/\sqrt{2} + R_C)\dot{\hat{\alpha}} + (R_A/\sqrt{2} - F)\dot{\hat{\beta}}$$

$$\forall(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) : P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) + P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = 0$$

Exemple d'application (1)



$\underline{R}_A = ?\ ;\ \underline{R}_C = ?$

$$\underline{R}_C = R_C\,\underline{e}_x$$

$$\underline{R}_A = R_A\,\underline{t}_{AB} = R_A\,\frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

Mise en œuvre du T.P.V

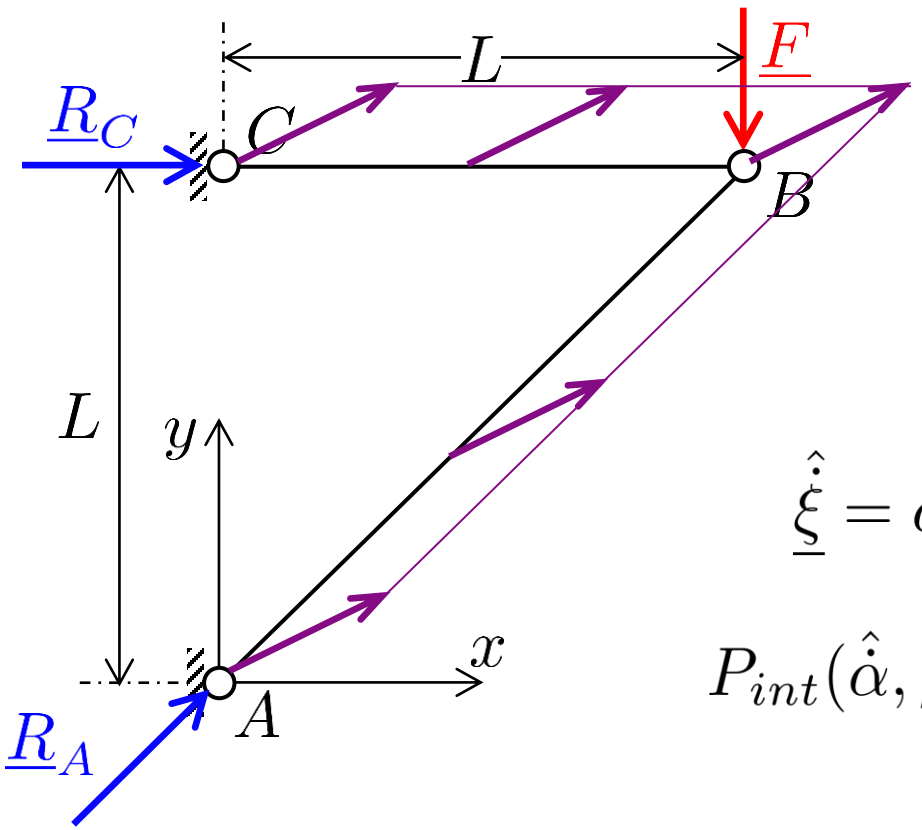
$$\dot{\hat{\underline{\xi}}} = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \text{ sur } [AB] \text{ et } [CB]$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N\frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\,\mathrm{d}s = 0$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = (R_A/\sqrt{2} + R_C)\dot{\hat{\alpha}} + (R_A/\sqrt{2} - F)\dot{\hat{\beta}}$$

$$\forall(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) : P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) + P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = 0 \implies \begin{cases} R_A = \sqrt{2}F \\ R_C = -F \end{cases}$$

Exemple d'application (1)



$\underline{R}_A = ?\ ;\ \underline{R}_C = ?$

$$\underline{R}_C = R_C\ \underline{e}_x$$

$$\underline{R}_A = R_A\ \underline{t}_{AB} = R_A\ \frac{\sqrt{2}}{2}(\underline{e}_x + \underline{e}_y)$$

Mise en œuvre du T.P.V

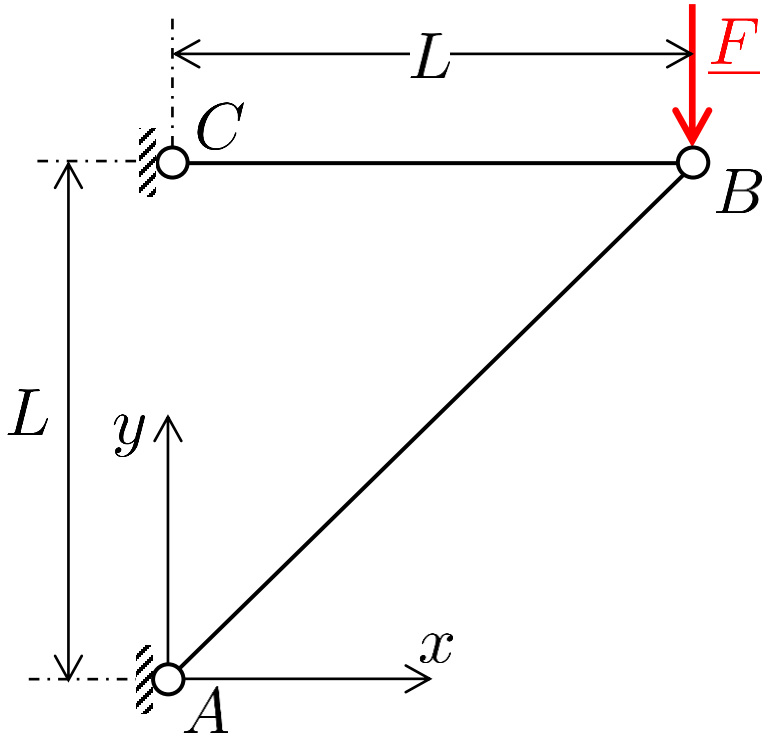
$$\dot{\hat{\underline{\xi}}} = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \text{ sur } [AB] \text{ et } [CB]$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\ \mathrm{d}s\ = 0$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = (R_A/\sqrt{2} + R_C)\dot{\hat{\alpha}} + (R_A/\sqrt{2} - F)\dot{\hat{\beta}}$$

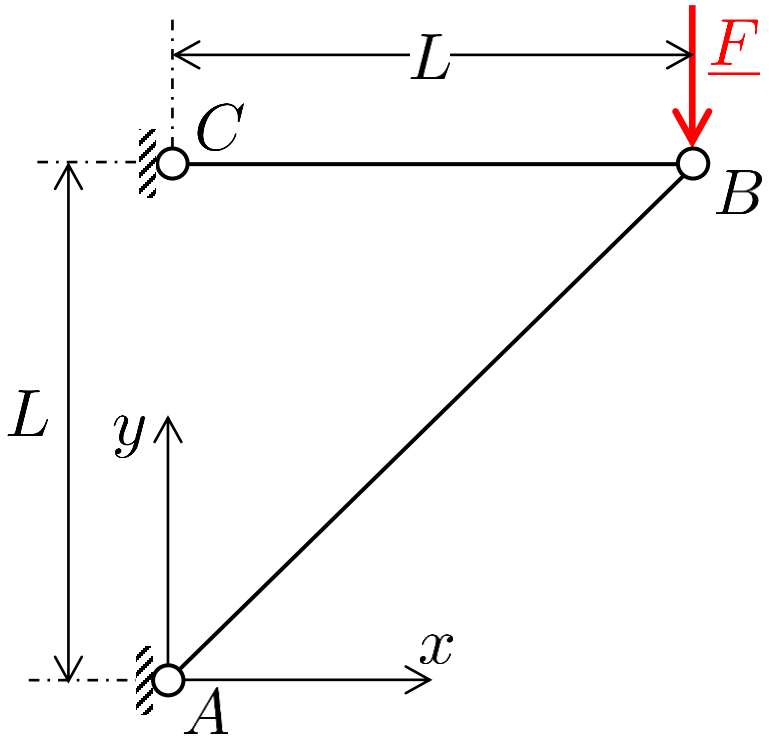
$$\forall(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) : P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) + P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = 0 \quad \Longrightarrow \quad \boxed{\underline{R}_A = F\ (\underline{e}_x + \underline{e}_y)\ ;\ \underline{R}_C = -F\underline{e}_x}$$

Exemple d'application (2)



$N_{AB} = ?\ ;\ N_{BC} = ?$

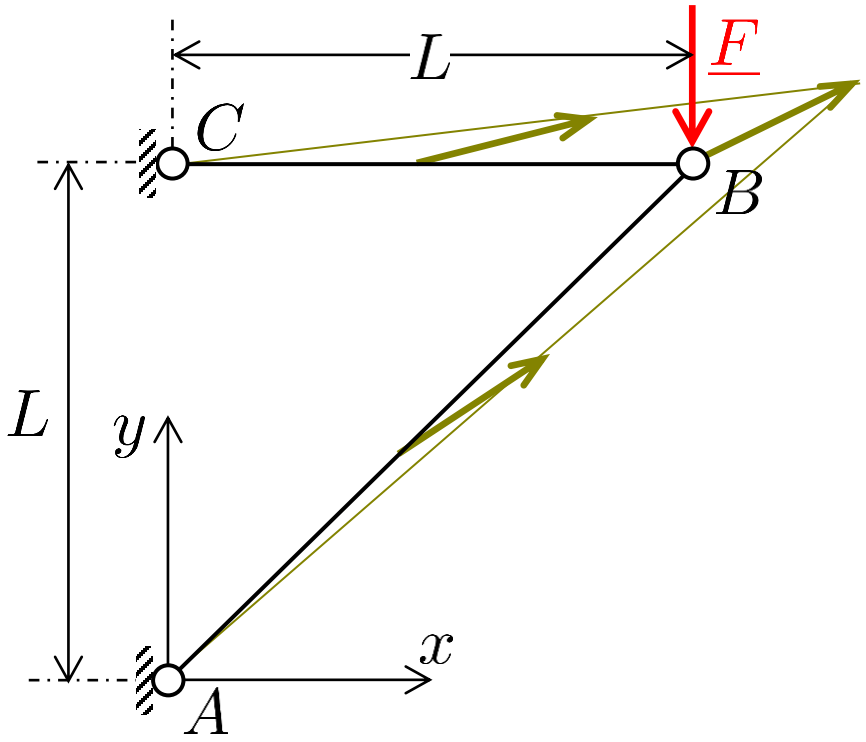
Exemple d'application (2)



$N_{AB} = ?\ ;\ N_{BC} = ?$

$$\begin{cases} \underline{\hat{\dot{\xi}}}(A) = \underline{\hat{\dot{\xi}}}(C) = \underline{0}\ ;\ \underline{\hat{\dot{\xi}}}(B) = \hat{\dot{\alpha}}\underline{e}_x + \hat{\dot{\beta}}\underline{e}_y \\ \\ \underline{\hat{\dot{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

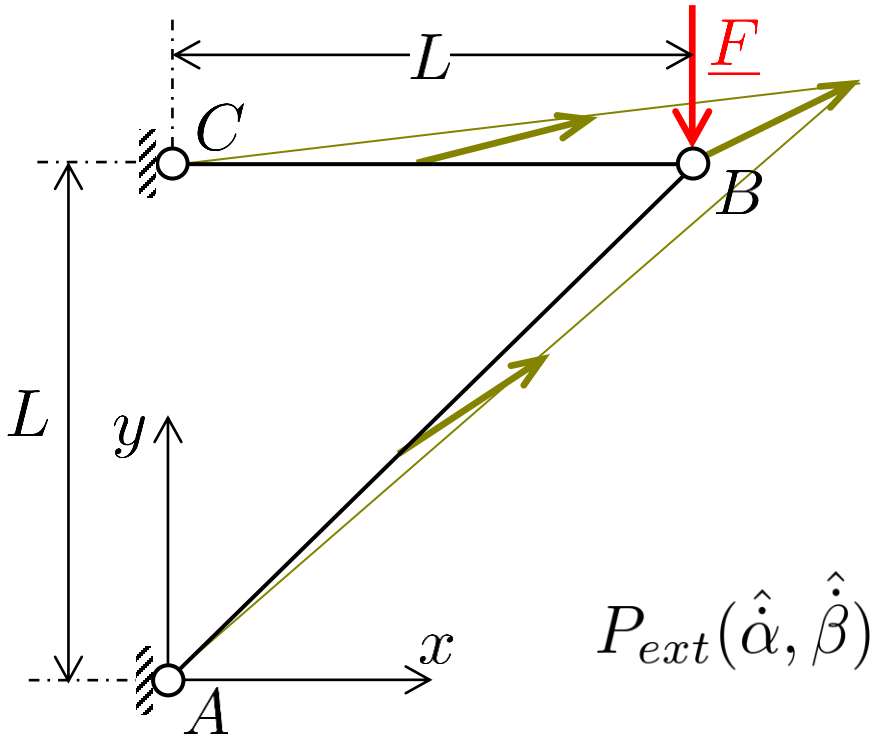
Exemple d'application (2)



$N_{AB} = ?\ ;\ N_{BC} = ?$

$$\begin{cases} \underline{\hat{\dot{\xi}}}(A) = \underline{\hat{\dot{\xi}}}(C) = \underline{0}\ ;\ \underline{\hat{\dot{\xi}}}(B) = \hat{\dot{\alpha}}\underline{e}_x + \hat{\dot{\beta}}\underline{e}_y \\ \\ \underline{\hat{\dot{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

Exemple d'application (2)

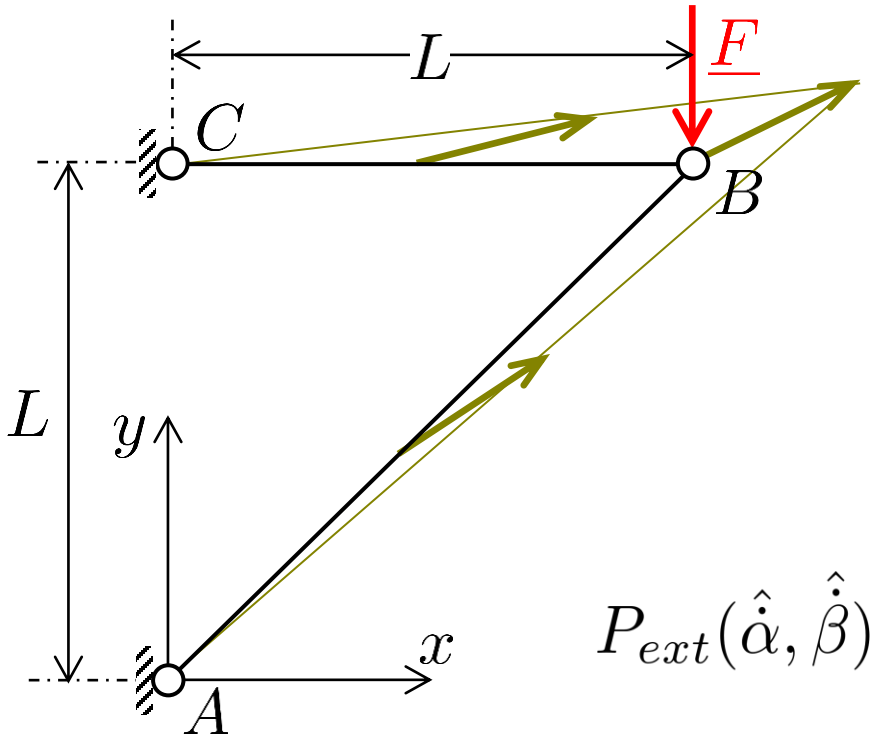


$N_{AB} = ? \; ; \; N_{BC} = ?$

$$\begin{cases} \underline{\dot{\hat{\xi}}}(A) = \underline{\dot{\hat{\xi}}}(C) = \underline{0} \; ; \; \underline{\dot{\hat{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \\ \underline{\dot{\hat{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \underline{\dot{\hat{\xi}}}(A) + \underline{R}_C \cdot \underline{\dot{\hat{\xi}}}(C) - F\underline{e}_y \cdot \underline{\dot{\hat{\xi}}}(B)$$

Exemple d'application (2)

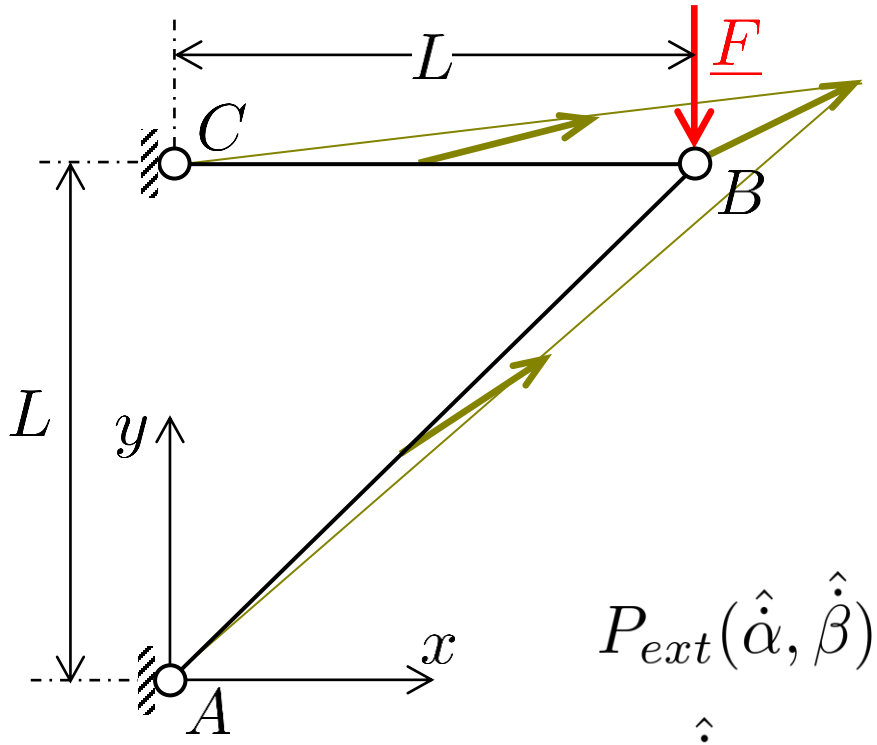


$N_{AB} = ? \; ; \; N_{BC} = ?$

$$\begin{cases} \underline{\dot{\hat{\xi}}}(A) = \underline{\dot{\hat{\xi}}}(C) = \underline{0} \; ; \; \underline{\dot{\hat{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \\ \underline{\dot{\hat{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \underline{\dot{\hat{\xi}}}(A) + \underline{R}_C \cdot \underline{\dot{\hat{\xi}}}(C) - F\underline{e}_y \cdot \underline{\dot{\hat{\xi}}}(B) = -F\dot{\hat{\beta}}$$

Exemple d'application (2)



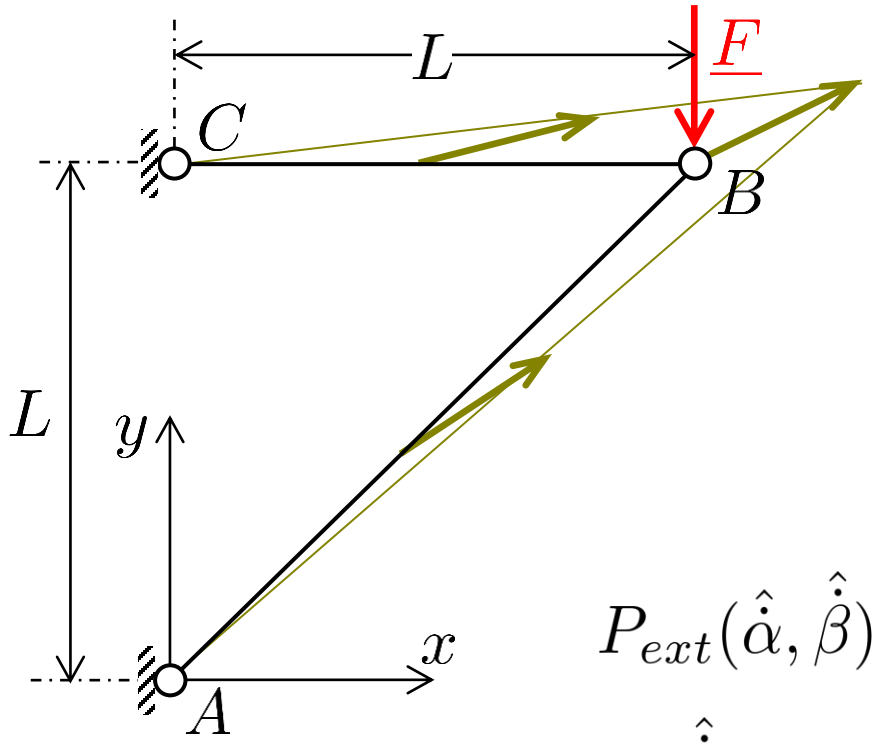
$N_{AB} = ? \; ; \; N_{BC} = ?$

$$\begin{cases} \dot{\hat{\underline{\xi}}}(A) = \dot{\hat{\underline{\xi}}}(C) = \underline{0} \; ; \; \dot{\hat{\underline{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \dot{\hat{\underline{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s$$

Exemple d'application (2)



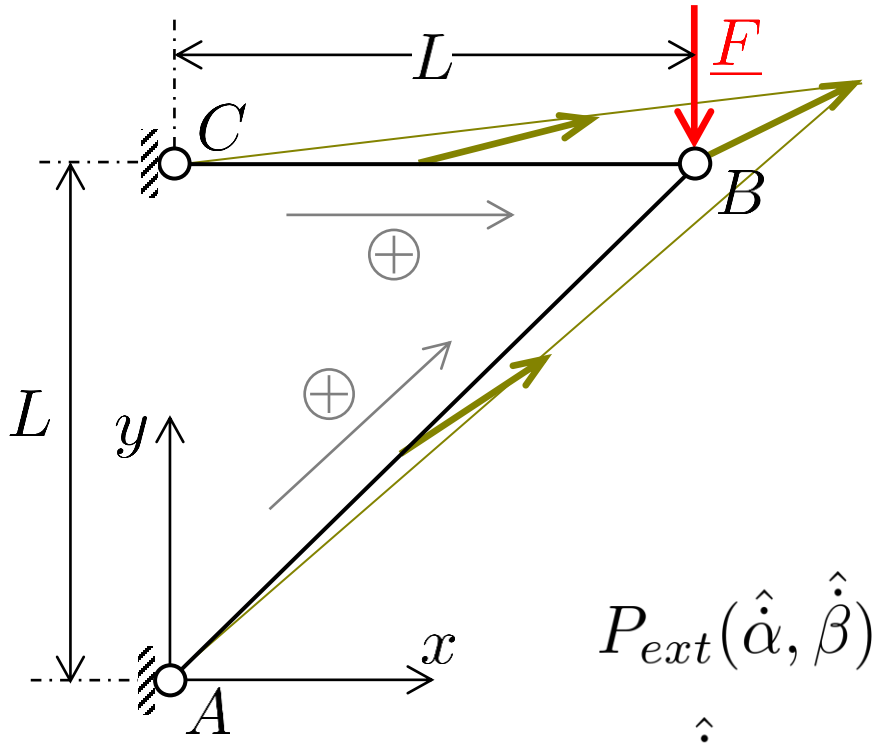
$N_{AB} = ? \; ; \; N_{BC} = ?$

$$\begin{cases} \underline{\dot{\hat{\xi}}}(A) = \underline{\dot{\hat{\xi}}}(C) = \underline{0} \; ; \; \underline{\dot{\hat{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \\ \underline{\dot{\hat{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \underline{\dot{\hat{\xi}}}(A) + \underline{R}_C \cdot \underline{\dot{\hat{\xi}}}(C) - F\underline{e}_y \cdot \underline{\dot{\hat{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s = -\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s - \int_C^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s$$

Exemple d'application (2)



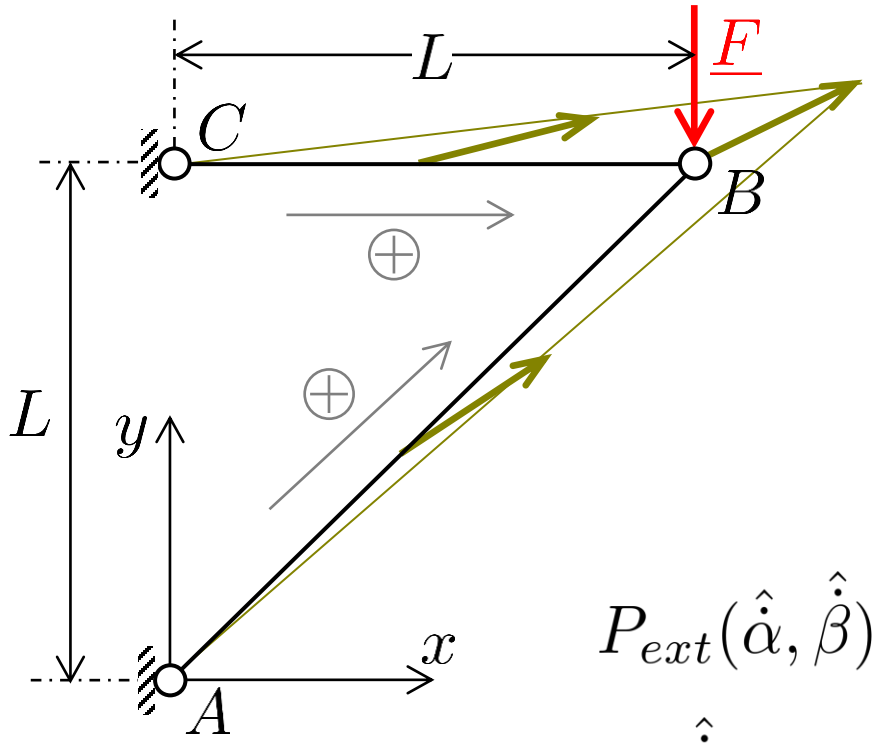
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$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \underline{\dot{\hat{\xi}}}(A) + \underline{R}_C \cdot \underline{\dot{\hat{\xi}}}(C) - F\underline{e}_y \cdot \underline{\dot{\hat{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s = -\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s - \int_C^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s$$

Exemple d'application (2)



$N_{AB} = ? \; ; \; N_{BC} = ?$

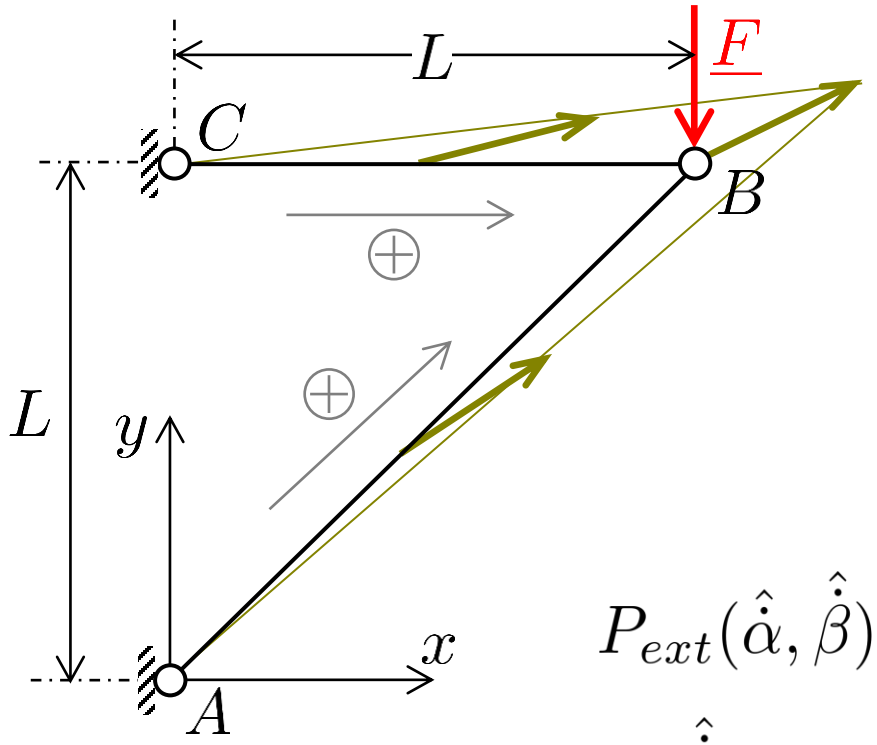
$$\begin{cases} \dot{\hat{\underline{\xi}}}(A) = \dot{\hat{\underline{\xi}}}(C) = \underline{0} \; ; \; \dot{\hat{\underline{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \dot{\hat{\underline{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s = -\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s - \int_C^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s$$

$$\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} = N_{AB} \left(\dot{\hat{\xi}}_t(B) - \dot{\hat{\xi}}_t(A) \right)$$

Exemple d'application (2)



$N_{AB} = ? \; ; \; N_{BC} = ?$

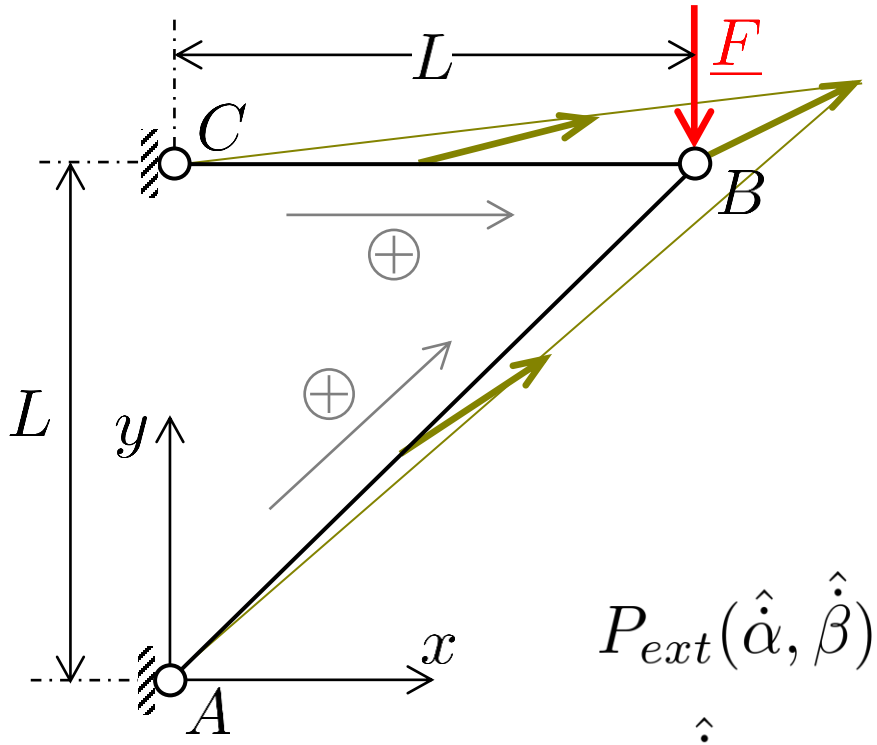
$$\begin{cases} \dot{\hat{\underline{\xi}}}(A) = \dot{\hat{\underline{\xi}}}(C) = \underline{0} \; ; \; \dot{\hat{\underline{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \dot{\hat{\underline{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s = -\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s - \int_C^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s$$

$$\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} = N_{AB} \left(\dot{\hat{\xi}}_t(B) - \dot{\hat{\xi}}_t(A) \right) = N_{AB} \dot{\hat{\underline{\xi}}}(B) \cdot \underline{t}_{AB}$$

Exemple d'application (2)



$N_{AB} = ?\ ;\ N_{BC} = ?$

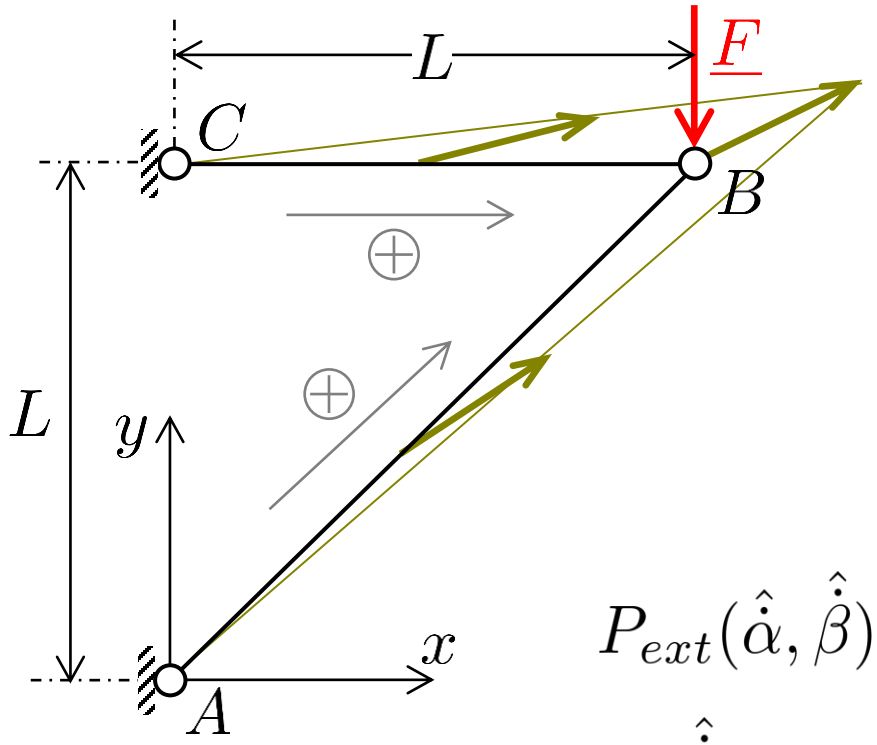
$$\begin{cases} \dot{\hat{\underline{\xi}}}(A) = \dot{\hat{\underline{\xi}}}(C) = \underline{0}\ ;\ \dot{\hat{\underline{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \dot{\hat{\underline{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\,\mathrm{d}s = -\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\,\mathrm{d}s - \int_C^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\,\mathrm{d}s$$

$$\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} = N_{AB}\left(\dot{\hat{\xi}}_t(B) - \dot{\hat{\xi}}_t(A)\right) = N_{AB}\dot{\hat{\underline{\xi}}}(B) \cdot \underline{t}_{AB} = N_{AB}(\dot{\hat{\alpha}} + \dot{\hat{\beta}})/\sqrt{2}$$

Exemple d'application (2)



$N_{AB} = ? \; ; \; N_{BC} = ?$

$$\begin{cases} \dot{\underline{\xi}}(A) = \dot{\underline{\xi}}(C) = \underline{0} \; ; \; \dot{\underline{\xi}}(B) = \dot{\alpha}\underline{e}_x + \dot{\beta}\underline{e}_y \\ \dot{\underline{\xi}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

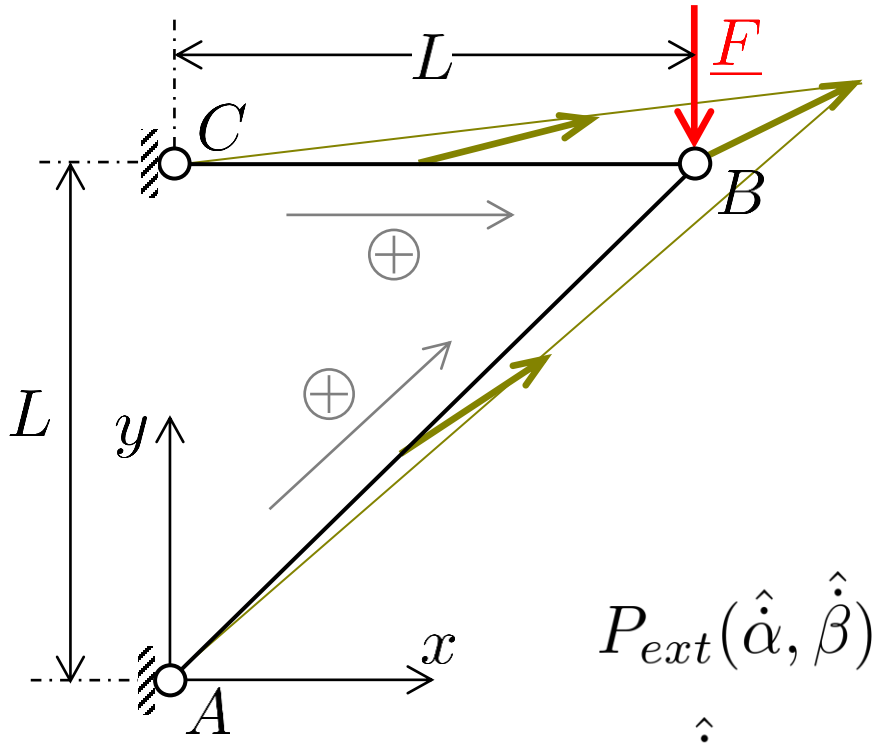
$$P_{ext}(\dot{\alpha},\dot{\beta}) = \underline{R}_A \cdot \dot{\underline{\xi}}(A) + \underline{R}_C \cdot \dot{\underline{\xi}}(C) - F\underline{e}_y \cdot \dot{\underline{\xi}}(B) = -F\dot{\beta}$$

$$P_{int}(\dot{\alpha},\dot{\beta}) = -\int_{struc} N\frac{\mathrm{d}\dot{\xi}_t}{\mathrm{d}s}\,\mathrm{d}s = -\int_A^B N\frac{\mathrm{d}\dot{\xi}_t}{\mathrm{d}s}\,\mathrm{d}s - \int_C^B N\frac{\mathrm{d}\dot{\xi}_t}{\mathrm{d}s}\,\mathrm{d}s$$

$$\int_A^B N\frac{\mathrm{d}\dot{\xi}_t}{\mathrm{d}s} = N_{AB}\left(\dot{\xi}_t(B) - \dot{\xi}_t(A)\right) = N_{AB}\dot{\underline{\xi}}(B)\cdot\underline{t}_{AB} = N_{AB}(\dot{\alpha}+\dot{\beta})/\sqrt{2}$$

$$\int_C^B N\frac{\mathrm{d}\dot{\xi}_t}{\mathrm{d}s} = N_{CB}\left(\dot{\xi}_t(B) - \dot{\xi}_t(C)\right) = N_{CB}\dot{\underline{\xi}}(B)\cdot\underline{t}_{CB}$$

Exemple d'application (2)



$N_{AB} = ? \; ; \; N_{BC} = ?$

$$\begin{cases} \dot{\hat{\underline{\xi}}}(A) = \dot{\hat{\underline{\xi}}}(C) = \underline{0} \; ; \; \dot{\hat{\underline{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \dot{\hat{\underline{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

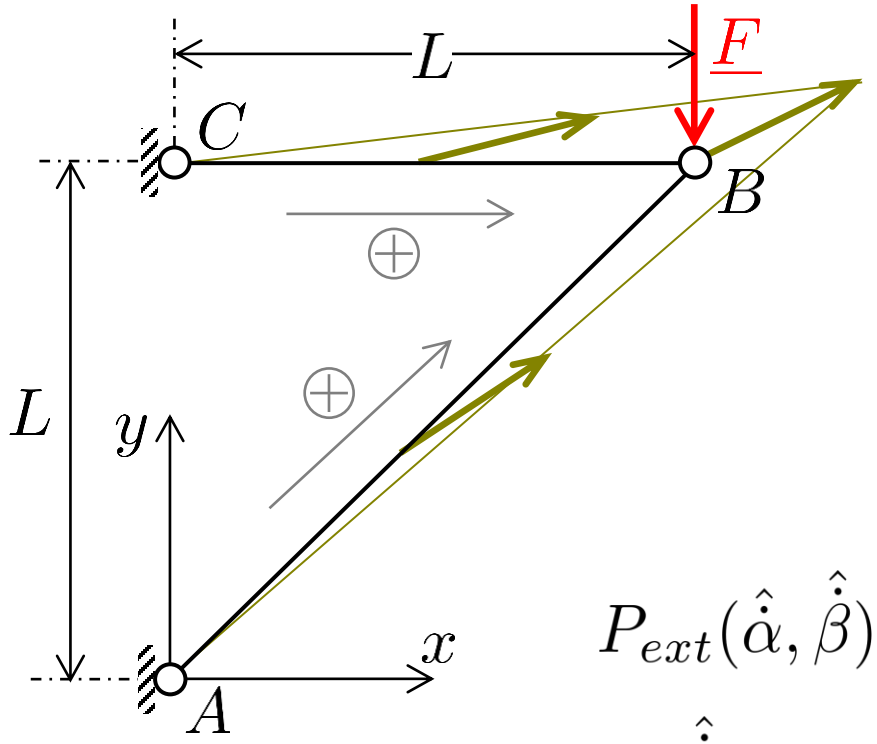
$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s = -\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s - \int_C^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s$$

$$\int_A^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} = N_{AB}\left(\dot{\hat{\xi}}_t(B) - \dot{\hat{\xi}}_t(A)\right) = N_{AB}\dot{\hat{\underline{\xi}}}(B) \cdot \underline{t}_{AB} = N_{AB}(\dot{\hat{\alpha}} + \dot{\hat{\beta}})/\sqrt{2}$$

$$\int_C^B N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} = N_{CB}\left(\dot{\hat{\xi}}_t(B) - \dot{\hat{\xi}}_t(C)\right) = N_{CB}\dot{\hat{\underline{\xi}}}(B) \cdot \underline{t}_{CB} = N_{CB}\dot{\hat{\alpha}}$$

Exemple d'application (2)



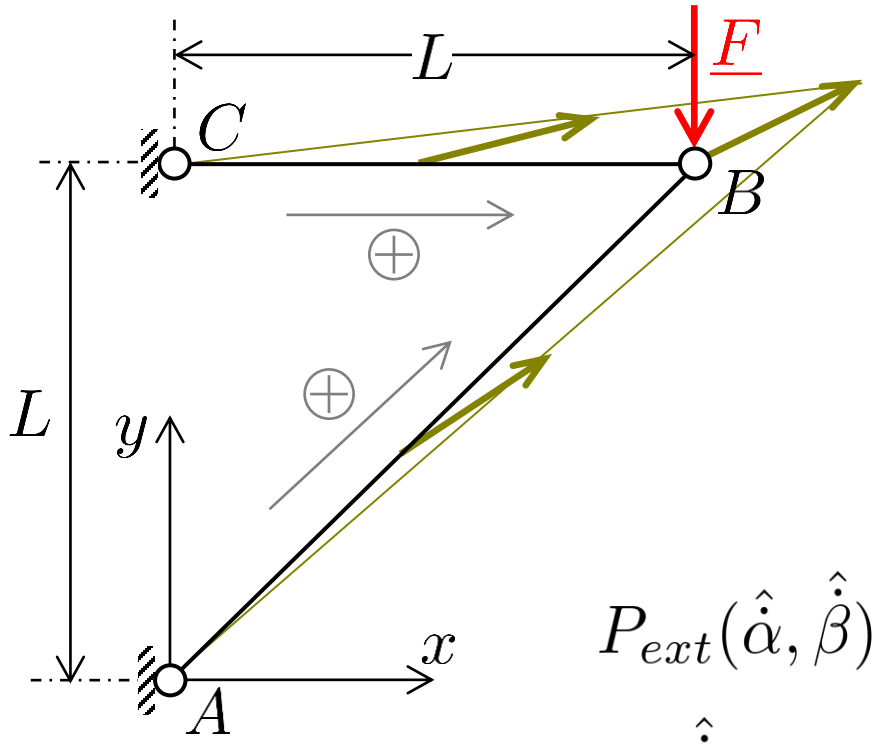
$N_{AB} = ?\ ;\ N_{BC} = ?$

$$\begin{cases} \dot{\hat{\underline{\xi}}}(A) = \dot{\hat{\underline{\xi}}}(C) = \underline{0}\ ;\ \dot{\hat{\underline{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \dot{\hat{\underline{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \dot{\hat{\underline{\xi}}}(A) + \underline{R}_C \cdot \dot{\hat{\underline{\xi}}}(C) - F\underline{e}_y \cdot \dot{\hat{\underline{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}\, \mathrm{d}s = -N_{AB}(\dot{\hat{\alpha}} + \dot{\hat{\beta}})/\sqrt{2} - N_{CB}\dot{\hat{\alpha}}$$

Exemple d'application (2)



$N_{AB} = ? \; ; \; N_{BC} = ?$

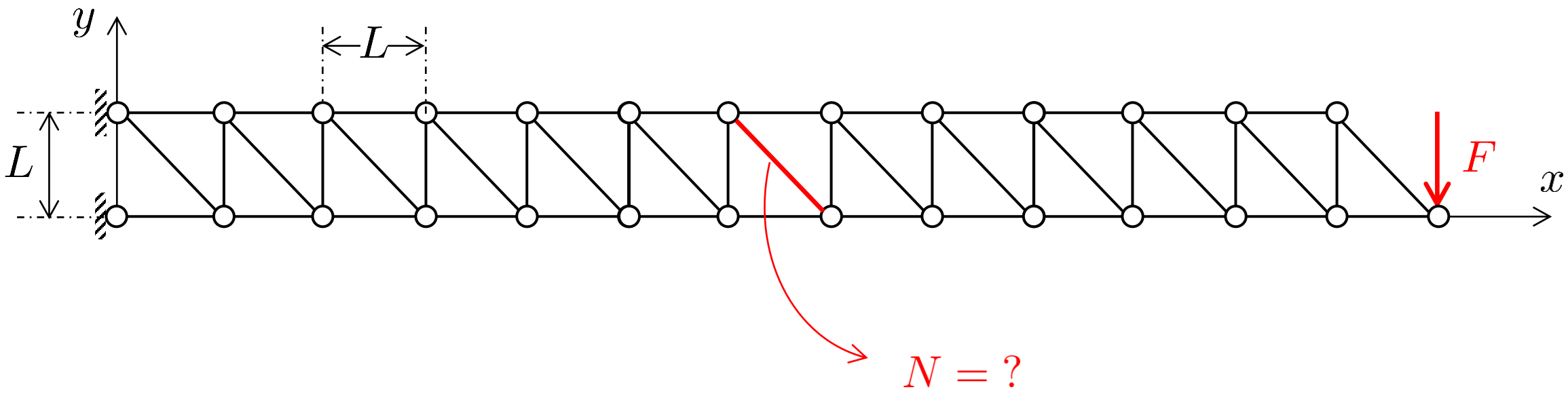
$$\begin{cases} \underline{\dot{\hat{\xi}}}(A) = \underline{\dot{\hat{\xi}}}(C) = \underline{0} \; ; \; \underline{\dot{\hat{\xi}}}(B) = \dot{\hat{\alpha}}\underline{e}_x + \dot{\hat{\beta}}\underline{e}_y \\ \underline{\dot{\hat{\xi}}} : \mathcal{C}_0 \text{ et } \mathcal{C}_1^{pm} \text{ sur } [AB] \text{ et } [CB] \end{cases}$$

$$P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = \underline{R}_A \cdot \underline{\dot{\hat{\xi}}}(A) + \underline{R}_C \cdot \underline{\dot{\hat{\xi}}}(C) - F\underline{e}_y \cdot \underline{\dot{\hat{\xi}}}(B) = -F\dot{\hat{\beta}}$$

$$P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = -\int_{struc} N \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s} \, \mathrm{d}s = -N_{AB}(\dot{\hat{\alpha}} + \dot{\hat{\beta}})/\sqrt{2} - N_{CB}\dot{\hat{\alpha}}$$

$$\forall (\dot{\hat{\alpha}}, \dot{\hat{\beta}}) : P_{ext}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) + P_{int}(\dot{\hat{\alpha}}, \dot{\hat{\beta}}) = 0 \implies \boxed{N_{AB} = -\sqrt{2}F \; ; \; N_{BC} = F}$$

Exercice d'application



Exercice d'application

